

M500 In-Car Video System User Guide, International Version

AUGUST 2024

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- To confirm troubleshooting results and analysis before taking action

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- The document title and part number
- The page number or title of the section with the error
- A description of the error

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Document History

Version	Description	Date
MN010314A01-D	Recording status update, Automatically Stop an Event, Configurable Live View camera default, Add event review to re-export and manually delete events, RFID update, Maintenance Mode Time Source Reset	August 2024
MN010314A01-C	RFID reader with RFID officer login, Peer Assisted Recording, Utilities error list, M5E camera, 5 camera sku, forced background, updated human form detection	May 2024
MN010314A01-B	Added screens with international date/time format	January 2023
MN010314A01-A	Initial release of international version	September 2023

Supplier's Declaration of Conformity

Per FCC CFR 47 Part 2 Section 2.1077(a)



Responsible Party

Name: Motorola Solutions, Inc.

Address: 415 East Exchange Parkway Allen TX 75013

Phone Number: 1-800-605-673

Hereby declares that the product:

Model Names: M500 DVR system includes the following devices:

- M500
- M5F
- M5D
- M5P
- M5R

These devices conform to the following regulations:

FCC Part 15, subpart B, section 15.109 and section 15.109(g)

Class B Digital Device

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:

1. This device may not cause harmful interference, and
2. This device must accept any interference received, including interference that may cause undesired operation.

Notice:

This equipment has been tested and found to comply with the limits for a Class B digital device, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation.

If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.

Notice to Users (FCC and Industry Canada)

This device complies with Part 15 of the FCC rules and Industry Canada license-exempt RSS's per the following conditions:

- This device may not cause harmful interference.
- This device must accept any interference received, including interference that may cause undesired operation.
- Changes or modifications made to this device, not expressly approved by Motorola Solutions, could void the authority of the user to operate this equipment.

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Chapter 1

Introduction

Welcome to the Motorola Solutions M500 In-Car Video System User Guide, International Version. This guide is designed to walk you through the basics of using your M500 to collect video and audio evidence.



NOTE: This user guide documents M500 system features through firmware version 24.3.1.

About This Document

The *M500 In-Car Video System User Guide, International Version* provides instructions for using the M500 system during your shift, including:

- Powering on ([Power On the M500 on page 15](#)) and off ([Power Off the M500 on page 52](#))
- Logging in ([Logging In from the Home Menu on page 16](#)) and out ([Logging Out from the Home Menu on page 51](#))
- Creating ([Start Events on page 21](#)) and tagging ([Event Tagging on page 28](#)) recorded events
- Reviewing events and other video ([Event and Video Review on page 33](#))
- Creating Record-After-the-Fact® (RATF) events ([Creating an RATF Event on page 42](#))
- Uploading events to the agency evidence management system ([Event Upload Over a Network on page 46](#))



NOTE: See [M500 Throughout a Shift on page 15](#) for more information on using the system during your shift.

This user guide includes information about the basic components and operation of the M500 in-car system, including:

- DVR ([DVR on page 59](#))
- Display ([Display on page 60](#))
 - Live View ([Live View on page 62](#))
 - Home menu ([Home on page 64](#))
- Cameras ([Cameras on page 102](#))
- Microphones ([Microphones on page 106](#))



NOTE: See [About the M500 In-Car Video System on page 56](#) for more information on the M500 components and their operation.

This user guide also includes instructions for some common administrator tasks ([Administrator Functions on page 109](#)).

The images in this document are representative of what you could see on your screen. They are meant to serve as a guide.



NOTE: This document covers the basic use of the M500 in-car video system. It is not a comprehensive manual for every possible action or situation you could experience when using the M500. If you have a question about the M500 that is not covered in this document, contact your Motorola Solutions representative.

Related Documents and Information

For subjects related to your M500 system that are not covered by the M500 In-Car Video System User Guide, International Version, see the following documents:

- *M500 In-Car Video System Installation Guide, International Version*
- *M500 Installation Overview and Quick Start Guide (EU)*
- *M500 Installation Overview and Quick Start Guide (UK)*

The M500 In-Car Video System

The M500 in-car video system captures, processes, stores, and uploads video and audio evidence.



The M500 system can include:

- DVR ([DVR on page 59](#))
- Display ([Display on page 60](#))
- Up to 5 cameras, including:
 - Front camera
 - Cabin camera with an integrated microphone
 - Rear camera
 - Up to two exterior camerasSee [Cameras on page 102](#) for more information.
- Two wired cabin microphones ([Stand-Alone Cabin Microphones on page 106](#))

The M500 DVR and Display function as the central component of the system.

- The DVR processes and stores evidence captured through the system cameras and microphones
- The Display allows you to interact with the DVR through Control Panel buttons and touchscreen icons and menus

The M500 connects to your agency evidence management system for provisioning and for uploading evidence.



NOTE: For more information, see [M500 Throughout a Shift on page 15](#) and [About the M500 In-Car Video System on page 56](#).

Chapter 2

M500 Throughout a Shift

The M500 DVR and Display function as the central component of the M500 in-car video system. You use the M500 to capture, process, and store video and audio evidence collected through connected cameras and microphones. Your M500 connects to your agency evidence management system for provisioning and for uploading its evidence.

You use the M500 throughout your shift. This section gives you instructions for common tasks you need to perform on the M500:

- At the start of a shift
- During a shift
- At the end of a shift



NOTE: Some M500 functions are beyond the normal scope of daily usage. Access to these administrator functions requires an administrator or supervisor password. See [Administrator Functions on page 109](#) for more information.

Start of a Shift

- Powering on ([Power On the M500 on page 15](#))
- Logging in ([Logging In from the Home Menu on page 16](#))

During a Shift

- Setting officer preferences ([Officer Preferences on page 19](#))
- Muting microphones ([Muting Microphones on page 19](#))
- Starting events ([Start Events on page 21](#))
- Stopping events ([Stop Events on page 25](#))
- Creating events in Covert Mode ([Entering Covert Mode on page 24](#))
- Tagging events ([Event Tagging on page 28](#))
- Reviewing video ([Event and Video Review on page 33](#))
- Creating Record-After-the-Fact® (RATF) events ([Creating an RATF Event on page 42](#))

End of a Shift

- Verifying event upload from M500 ([Event Upload Over a Network on page 46](#))
- Logging out ([Logging Out from the Home Menu on page 51](#))
- Powering off ([Power Off the M500 on page 52](#))

Power On the M500

You can power on the M500 DVR two ways:

- With the ignition [Powering On with the Ignition on page 16](#)
- With the **On** button [Powering On with the On Button on page 16](#)

Powering On with the Ignition

Your agency administrator typically configures the M500 to power on with your vehicle ignition.

Procedure:

To power on the M500, start your vehicle.

Result: The M500 boots up. When it has finished, the Display Screen shows the Live View streaming from the front camera.

Powering On with the On Button

You can use the **On** button on the Display Control Panel to power on the M500.

Procedure:

To power on the M500, press and hold the **On** button for 2 seconds.



Result:



IMPORTANT: If your M500 is configured to power on with the vehicle ignition, avoid pressing the **On** button on the Display Control Panel.

The M500 boots up. When it has finished, the Display Screen shows the Live View streaming from the default configured camera.

Log In to the M500

You can log in to the M500 three ways:

- From the Home menu [Logging In from the Home Menu on page 16](#)
- By touching the officer name on Live View [Logging In from Live View on page 17](#)
- By tapping your RFID reader [Logging in with RFID on page 18](#)

Logging In from the Home Menu

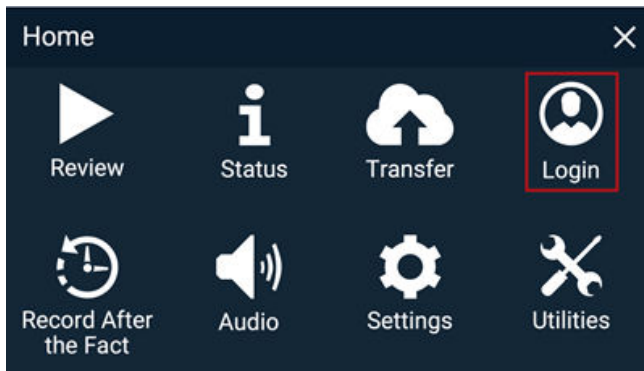
You log in to the M500 at the beginning of each shift to ensure your name is associated with the events recorded in your vehicle.

Procedure:

1. From Live View, press **Home** on the Display Control Panel.

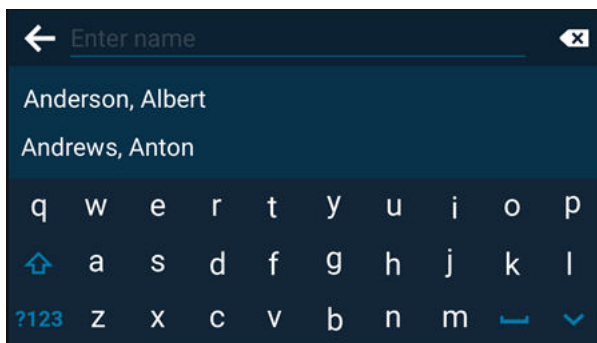


The **Home** menu appears.



2. Touch **Login**.

The login screen appears.



3. Start entering your name using the keyboard.



NOTE: Depending on your agency settings, you may also be able to use your badge number.

The system shows you a list of names containing the letters you entered.



NOTE: Touch the down arrow  to collapse the keyboard and see the whole list of officers.

4. Touch your name when it appears on the list.

The M500 logs you in.

Logging In from Live View

You log in to the M500 at the beginning of each shift to ensure your name is associated with the events recorded in your vehicle.

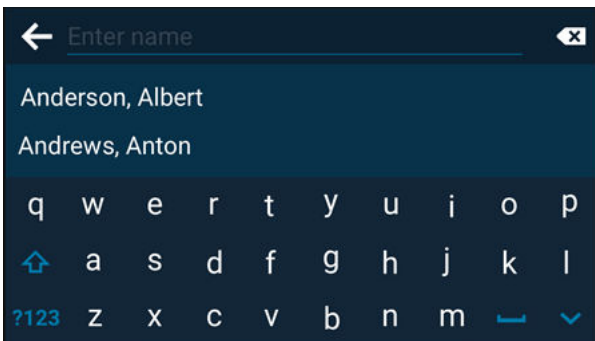
Procedure:

1. From Live View, touch the officer name in the top right corner of the Display Screen.



NOTE: If there is a recorded event in progress, you cannot log in.

The login screen appears.



2. Start entering your name using the keyboard.



NOTE: Depending on your agency settings, you may also be able to use your badge number.

The system shows you a list of names containing the letters you entered.



NOTE: Touch the down arrow  to collapse the keyboard and see the whole list of officers.

3. Touch your name when it appears on the list.

The M500 logs you in.

Logging in with RFID

If you have a Radio-Frequency Identification (RFID) reader, you can log in to the M500 by tapping your ID on the RFID reader.



NOTE: See [RFID Reader on page 107](#) for more information.

Procedure:

Tap your ID on the RFID reader.

The M500 logs you in.



TIP: If there is another officer logged in, you can log out the current user and retap your ID on the RFID reader to log in.



NOTE: For information on logging out of the M500, see [Log Out of the M500 on page 51](#).

Officer Preferences

The M500 tracks certain settings that you make after you log in. The system associates these setting preferences with your login name in the agency evidence management system.

When you log in to an M500 again, your preferences are set as they were during your last logged-in shift.

 **NOTE:** Your preferences are associated with your login name, so they are applied to any agency M500 you log in to.

To set your officer preferences, navigate to these pages and follow the instructions:

Display

- Display Control Panel button brightness ([Using Button Brightness on page 102](#))
- Display Screen brightness ([Using Screen Brightness on page 101](#))
- Screen saver options ([Screensaver on page 79](#))

Audio

- Wired microphone input levels ([Mic Input on page 72](#))
- Wired microphone mute status ([Muting Microphones on page 19](#))
- System tone and review player volumes ([Volume on page 72](#))

Camera

- Cameras LEDs state ([Disabling the Camera LEDs on page 77](#))
- Recording state of system cameras ([Camera Views on page 96](#))

Automatic License Plate Recognition (ALPR)

- License plate detection indicators ([Enabling ALPR on page 78](#))

Muting Microphones

The M500 in-car video system uses two types of microphones:

- Stand-alone cabin microphone
- Integrated microphone in the M5P camera

For more information on the microphones in the M500 system, see [Microphones on page 106](#).

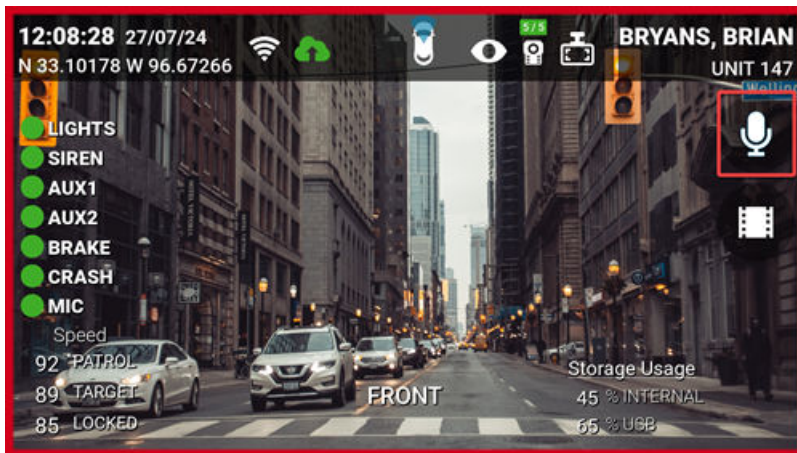
You can mute the microphones from the Live View, if you have permissions.

 **NOTE:** Microphone muting permissions are configured by your administrator in your agency evidence management system.

The Live View shows the microphone mute status with an icon:

-  (unmuted): One or more of the system microphones are unmuted
-  (muted): All the system microphones are muted

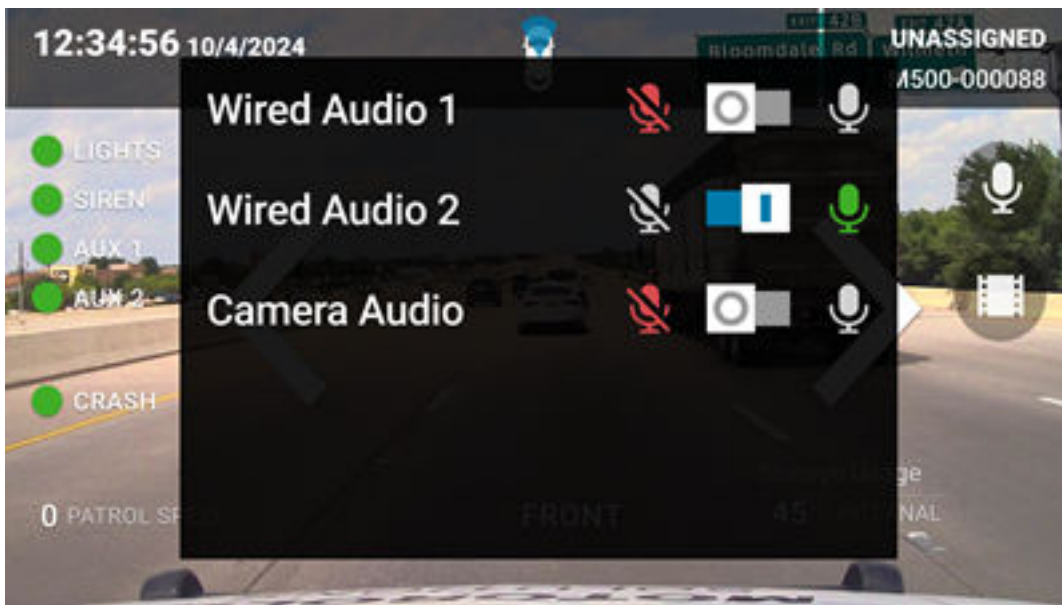
See [Live View on page 62](#) for more information about the icons and information on the Live View.



 **NOTE:** If you do not have permissions to mute any of the microphones, the microphone status icon is not present.

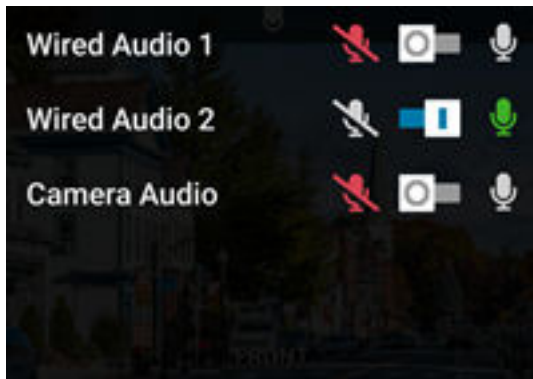
Procedure:

1. To change the status of individual microphones, from the Live View, touch the microphone status icon. The microphone status screen opens showing the mute status of each microphone in the M500 system.



 **NOTE:** You may not have all of these microphones on your list. The M500 only shows the microphones on the status screen list that it has detected in the system.

2. Touch the toggle switch for the microphone you want to mute (or unmute). The microphone you selected is muted (or unmuted).



 **NOTE:** The muted icon  turns red when you mute a microphone.

3. Touch outside the microphone status screen to return to the Live View.

The microphone status icon on the Live View reflects the change you made on the microphone status screen, as appropriate.

Result:



IMPORTANT: Microphones are unmuted automatically when a recorded event starts, by default. You can re-mute a microphone during an event, if needed. Any microphones that were automatically unmuted for an event are automatically re-muted when the event is stopped.

The wired microphone mute status you set is saved to the agency evidence management system as part of your officer preferences.

Each time you log in to an M500, your set preferences are restored.

For more information, see:

[Audio Evidence on page 57](#)

[Start Events on page 21](#)

[Stopping an Event Manually on page 26](#)

[Entering Covert Mode on page 24](#)

[Officer Preferences on page 19](#)

Start Events

On the M500, you can start a recorded event:

- Automatically ([Start an Event Automatically on page 22](#))
Your system administrator typically sets up the M500 to start a recorded event automatically when triggered by auto-start inputs. The active inputs show on the Live View.
- Manually ([Starting an Event Manually on page 24](#))
You can start an event manually using the **Record** button on the Display Control Panel.

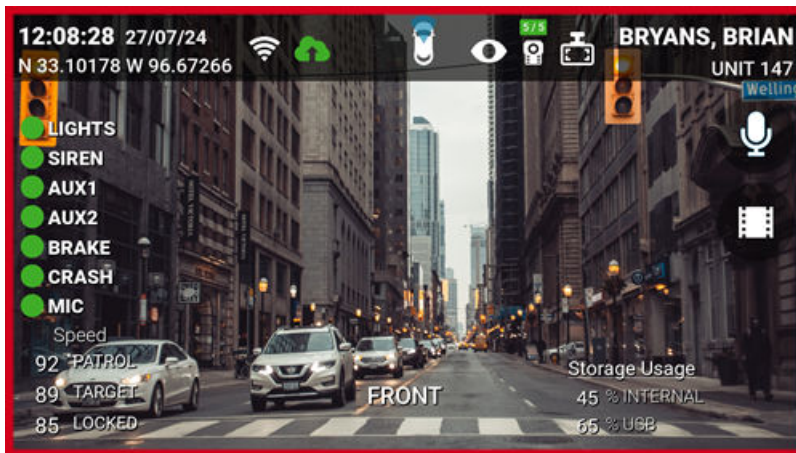


IMPORTANT: Depending on your configuration and agency evidence management system, the M500 may prompt you to log in before an event can be started.



NOTE: A recorded event (or event) is a unique, protected segment of recorded video, audio, and metadata ([Evidence on page 56](#)).

When an event starts on the M500, the **Record** button lights up and remains on while the event is in progress. Red bars appear at the top and bottom of the screen and remain while the event is in progress.



When a non-Covert Mode event starts, all microphones immediately unmute to start recording the audio for the event. Cameras remain in the state they were in before the event started.

See [Muting Microphones on page 19](#) for information about muting and unmuting microphones. See [Cameras Button on page 96](#) for information about camera states.

If you want to start recording an event without anyone in the vehicle being aware they are being recorded, you can use Covert Mode. When a covert event starts, all microphones unmute to start recording audio. All cameras are restarted to record video for the covert event.

See [Entering Covert Mode on page 24](#) for instructions.



WARNING: The M500 system will not start new events if its internal storage drive is full. In this case, you must upload existing events to your agency evidence management system before you can start a new event ([Event Upload Over a Network on page 46](#)).

For more information, see:

[Stopping an Event Manually on page 26](#)

[Recording Reminder Alert on page 24](#)

Start an Event Automatically

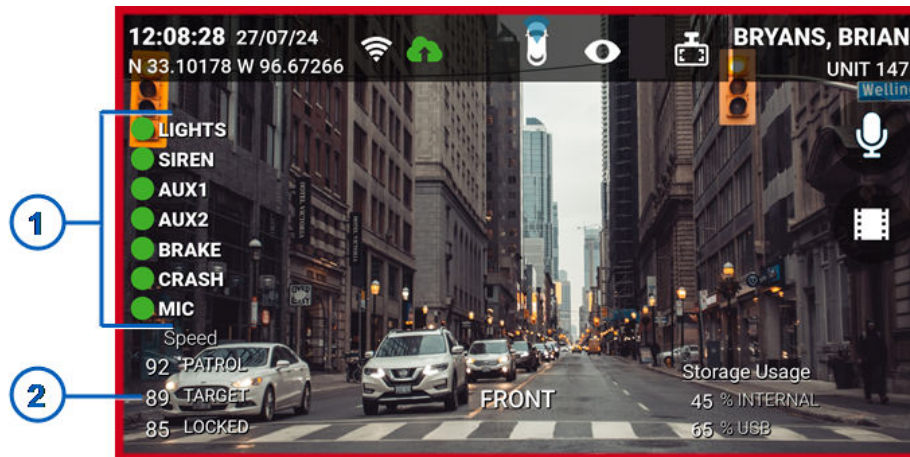
Your system administrator typically sets up the M500 to start a recorded event automatically when prompted by certain external inputs or a vehicle crash.

Depending on your configuration, a recorded event may start automatically if your M500 is a part of a peer recording group. The M500 can be configured to receive notifications from other peer devices and can start an event automatically when a notification is received. See [Peer Assisted Recording on page 108](#) for more information.



IMPORTANT: Depending on your configuration and agency evidence management system, the M500 may prompt you to log in before an event can be started.

The Live View shows which inputs (or other) are active.



1	Active external inputs and crash indicator
2	Speed indicator

 **NOTE:** The list of active external inputs is an example.

External Inputs

These external inputs can be configured to start an event automatically:

- **Lights**
- **Siren**
- **Aux1**
- **Aux2**
- **Patrol speed**

The external inputs are wired up to the M500 system by the system installer. The **Aux1** and **Aux2** inputs are determined by your agency. Some examples of auxiliary inputs:

- Dog kennel door opening
- Vehicle door opening
- Removal of a firearm from its mounting in the vehicle

When you activate any of the configured inputs or exceed the threshold for excessive speed, M500 starts a recorded event automatically.

Your system administrator configures the external inputs and speed threshold as auto-start inputs in the agency evidence management system.

Crash

The M500 uses an algorithm with an accelerometer to measure the G-force on the system. If the M500 experiences a sudden extreme change in G-force, the system registers a crash and a recorded event starts automatically. Your system administrator enables **Crash** as an event auto-start input in the agency evidence management system.

 **NOTE:** If an event is started from the **Crash** auto-start input, the system automatically adds two minutes of pre-event video to the event. See [Pre-event on page 58](#) for more information.

Starting an Event Manually

You can start a recorded event manually using the **Record** button on the Display Control Panel.

Procedure:

To manually start an event, press **Record** on the Display Control Panel.



The **Record** button lights up to show the event has started.



IMPORTANT: Depending on your configuration and agency evidence management system, the M500 may prompt you to log in before an event can be started.

Result:

For more information, see [Stopping an Event Manually on page 26](#).

Recording Reminder Alert

Your system administrator may set up your M500 to remind you periodically that it is still recording an event. When this recording reminder is enabled, the M500 beeps at regular intervals during a recorded event. The interval can be from 1 to 60 minutes.

The recording reminder does not beep during Covert Mode ([Entering Covert Mode on page 24](#)).

Your administrator sets up recording reminder alerts in the agency evidence management system.

Covert Mode

Covert Mode lets you start recording an event without anyone in the vehicle being aware they are being recorded. The system gives no indication that an event has been started and appears to power off. Your system administrator sets up Covert Mode in the M500 device configuration in the agency evidence management system.

Entering Covert Mode

Procedure:

To place the M500 in Covert Mode, on the Display Control Panel, press and hold the **On** button for 3 seconds.



The power down tones sound, the Display Screen and the Display Control Panel button LEDs go dark. Camera LEDs turn off. All touch input on the screen is ignored. All buttons are ignored except **On** and **Stop**.

At the same time, the M500 starts an event. Deactivated microphones are restarted for the event. Backgrounded cameras are forced on for the event.



NOTE: See [Background Recording on page 99](#) for more information.

 **NOTE:** An event started with Covert Mode does not include pre-event.

Exiting Covert Mode

Procedure:

To exit Covert Mode, press the **On** button or the **Stop** button.

The exiting Covert Mode tone sounds and Live View appears on the Display Screen showing that an event is in progress ([Start Events on page 21](#)). The Display Control Panel buttons and the camera LEDs light up. All cameras and microphones remain active.

The M500 continues recording the event until you stop it.

See [Stopping an Event Manually on page 26](#) for instructions.

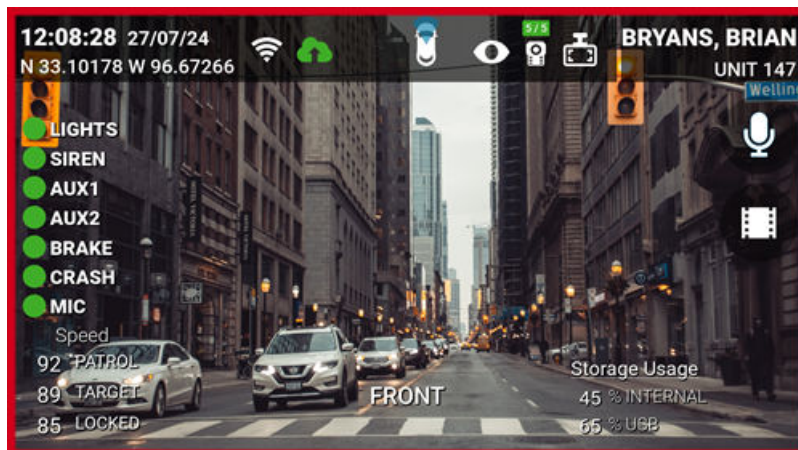
After the event is stopped, the cameras return to their previous configuration based on your officer preferences. See [Officer Preferences on page 19](#) for more information.

Stop Events

On the M500, you can stop a recorded event:

- Automatically ([Stop an Event Automatically on page 26](#))
Your system administrator can set up the M500 to stop a recorded event automatically when triggered by auto-stop inputs. The active inputs show on the Live View.
- Manually ([Stopping an Event Manually on page 26](#))
You can stop an event manually using the **Stop** button on the Display Control Panel.
Your administrator can configure the M500 to stop a recorded event automatically if the event reaches a time limit.

 **NOTE:** See [Maximum Recorded Event Time on page 27](#) for information.



 **NOTE:** A recorded event (or event) is a unique, protected segment of recorded video, audio, and metadata ([Evidence on page 56](#)).

Stop an Event Automatically

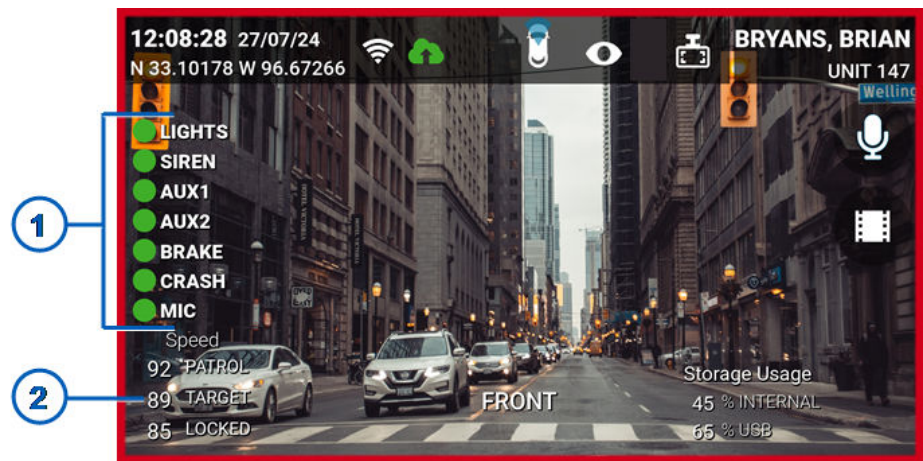
Your system administrator can set up the M500 to stop a recorded event automatically when prompted by auto-stop inputs. The active inputs show on the Live View.

Depending on your configuration, an event can be stopped automatically when prompted by the following external inputs:

- Lights
- Siren
- AUX 1
- AUX 2

When an event is stopped automatically, the **Record** button light goes out. On the Live View, the red bars disappear. The **Event Tags** screen appears for you to categorize the event.

The Live View shows which inputs (or other) are active.



1	Active external inputs and crash indicator
2	Speed indicator

 **NOTE:** The list of active external inputs is an example.

Stopping an Event Manually

You stop a recorded event with the **Stop** button on the Display Control Panel.

 **NOTE:** Whether the event was started automatically ([Start an Event Automatically on page 22](#)) or manually ([Starting an Event Manually on page 24](#)), you stop the event by pressing the **Stop** button.

Procedure:

To stop a recorded event, press **Stop** on the Display Control Panel.



The **Record** button light goes out. On the Live View, the red bars disappear. The **Event Tags** screen appears for you to categorize the event.

 **NOTE:** Depending on how your administrator has configured your device, you may need to press the **Stop** button twice to stop a recorded event.

Result:

See [Event Tagging on page 28](#) for information about event categories and tags.

 **NOTE:** Your administrator can configure the M500 to stop a recorded event automatically if the event reaches a time limit. See [Maximum Recorded Event Time on page 27](#) for information.

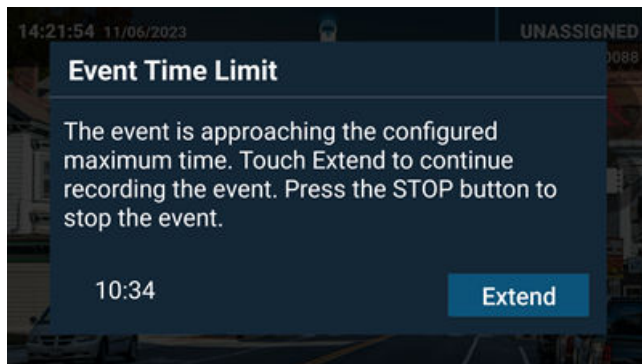
Maximum Recorded Event Time

The Maximum Recorded Event Time feature on the M500 allows the system to automatically stop a recorded event when it reaches a configured duration. Your administrator can configure the Maximum Recorded Event Time feature in the agency evidence management system.

When enabled, the system automatically stops an event when it reaches the configured duration unless you intervene using the **Event Time Limit** dialog box.

Event Time Limit Warning

Two minutes before it reaches the time limit, the M500 warns you it is going to stop the event.



If you do nothing, the **Event Time Limit** dialog box remains on the screen for the two-minute warning period, counting down the time before the event is stopped.

When the timer expires, the event is stopped, the dialog box disappears, and the **Event Tags** screen opens ([Event Tagging on page 28](#)).

 **NOTE:** If the system is in Covert Mode when the time limit is reached, the 2-minute warning is suspended until the system exits Covert Mode. When that happens, the **Event Time Limit** dialog box appears and the 2-minute countdown starts. See [Entering Covert Mode on page 24](#) for information about Covert Mode.

Extending the Event

Procedure:

To continue recording the event, touch **Extend** on the dialog box.

The dialog box disappears and the system continues the event for another Maximum Recorded Event Time period.

Stopping the Event

Procedure:

To stop the event immediately, press **Stop** on the Display Control Panel.

The event stops and the **Event Tags** screen opens.

Result:

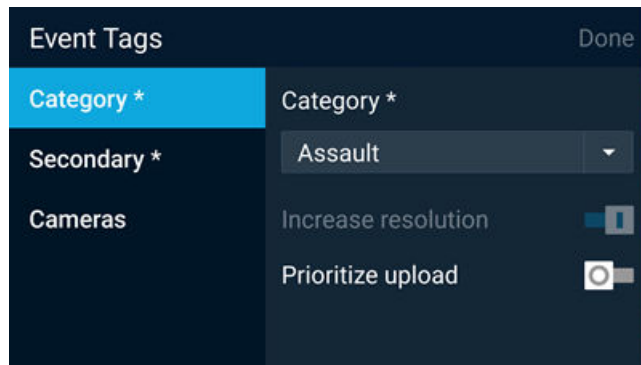
For more information, see [Stopping an Event Manually on page 26](#).

Event Tagging

Event tags are information prompts that you answer about the recorded event after you stop the event. The agency administrator sets up event tags in the M500 device configuration in the agency evidence management system.

See [Tagging an Event on page 31](#) for instructions on how to tag your recorded event.

When you press the M500 **Stop** button to stop a recorded event, the Display Screen shows you the **Event Tags** screen.



There are two types of event tags:

- **Category** ([Category on page 28](#))
- **Secondary** ([Secondary Event Tags on page 29](#))

Category and **Secondary** event tags can be required ([Required Event Tags on page 30](#)). An asterisk next to the tag type means at least one of the event tags on that tab is required. The agency administrator indicates whether a tag is required in the M500 device configuration.

The M500 records multiple camera streams. Depending on the configuration and your assigned permissions, you can choose whether a camera uploads its streams to your agency evidence management system with the rest of the recorded event.



NOTE: See [Event Tags Cameras on page 30](#) and [Cameras on page 102](#) for more information.

Category

The **Category** event tag is used to classify a recorded event. This event tag prompts you to categorize recorded events using a list of agency-determined categories, for example, traffic warning, traffic citation, or accident.

Event categories are either critical or routine. Tagging an event with a critical category makes the event a critical event ([Critical Categories on page 29](#)).

The **Category** tag is usually required ([Required Event Tags on page 30](#)).

Event categories are used for:

- Determining whether M500 event video is saved and uploaded using the critical or routine video quality
- Determining the event upload order (priority events upload first)
- Searching for evidence in the agency evidence management system
- Applying retention rules to events stored in the database



NOTE: A recorded event could be improperly purged from evidence storage if you set an event category value incorrectly. If needed, your system administrator can update the tag later in your agency evidence management system.

Critical Categories

Each camera connected to the M500 can be configured to record at least two streams. Each stream can be set to a different video quality, either critical or routine. Critical events are saved with a higher quality video stream.

A recorded event is designated as critical or routine depending on the category you apply to it. When you apply a critical category, the event is saved using the higher quality video stream.

Your agency administrator configures the video quality for each camera and stream, and determines which event categories correspond to critical or routine video quality.

Increase Resolution

If you have permissions, you can save a routine-tagged event with the critical quality video stream using the **Increase resolution** toggle switch. The toggle switch is only active when you select a routine category from the **Category** drop-down list.



NOTE: You can only increase the video quality.

Prioritize Upload

If you have permissions, you can save an event and prioritize its upload using the **Prioritize upload** toggle switch. The event that is prioritized is moved to the front of the upload queue, ahead of all events. The toggle switch is active when you select any category from the **Category** drop-down list.

Secondary Event Tags

Secondary event tags are used to identify anything other than **Category** that your agency wants you to document about a recorded event.

Event Tags		Done
Category *	VIN	
Secondary *	4Y1SL65848Z411439	
Cameras	Vehicle Make *	
	Honda	▼
	State	

Secondary tags can be drop-down selection lists or text entry fields. A drop-down list tag prompts you to select an answer from a list of options, for example, **Jurisdiction**. A text entry field tag prompts you to enter an alphanumeric or numeric value in a field, for example, **Incident** or **Gang Affiliation**.

Secondary tags can be required or optional.

Required Event Tags

An asterisk next to the tag type means at least one of the event tags on that tab is required. An asterisk next to the event tag means an answer is required for that individual tag.

You must answer all required tags, **Category** and **Secondary**, before you can move on from the **Event Tags** screen. **Done** is enabled after all required tags are answered.



NOTE: If another recorded event starts (manually or automatically) while you are in the middle of tagging an event, the M500 aborts the tagging and starts the new event. The untagged event can be tagged later in your agency evidence management system.

The agency administrator indicates whether a tag is required in the M500 device configuration in the agency evidence management system.

Event Tags Cameras

The **Event Tags Cameras** screen is used to identify cameras that are in a background recording state and whether their views are either included or excluded by default in an event upload, depending on your configuration.

Event Tags		Done
Category *	Camera	Include in upload
Secondary *	Front (wide)	☒
Cameras	Cabin	☒
	<i>Included based on AI detection</i>	
	Rear	☐



NOTE: Depending on your configuration, you may not be able to change the state of a camera if AI human form detection is enabled during an event. See [AI Human Form Detection Override on page 99](#) for more information.

Depending on your configuration, a camera view set to a background recording state is either included or excluded by default. If you have permissions, you can include or exclude the camera view in an event upload by selecting or clearing the checkbox next to the camera. See [Camera States on page 98](#) and [Background Recording on page 99](#) for more information.

By default, cameras are included in an event upload if human form detection is enabled during an event. Depending on your configuration, you may not be able to change the state of the camera. See [AI Human Form Detection Override on page 99](#) for more information.



NOTE: If you have permissions, you can change the state of your camera. See [Camera States on page 98](#) for more information.

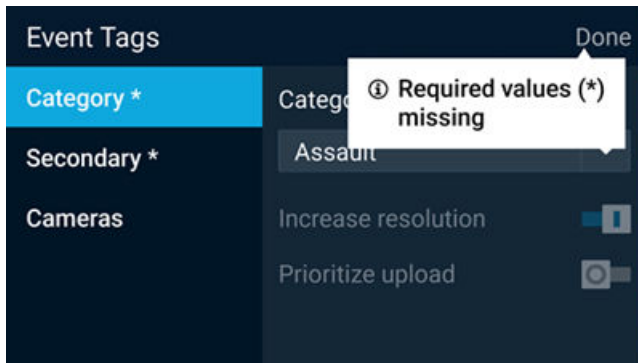
Tagging an Event

Procedure:

1. To tag a recorded event on the M500, press **Stop** on the Display Control Panel to stop the event.

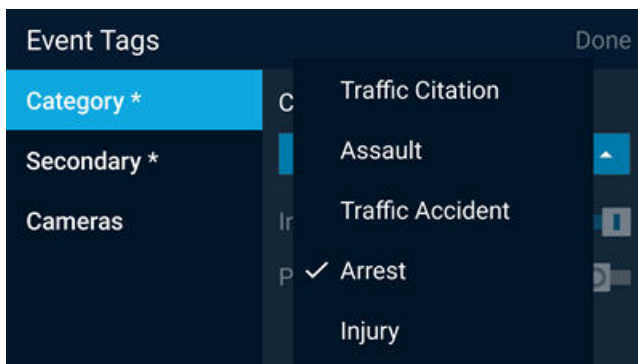


The **Event Tags** screen appears on the Display Screen.

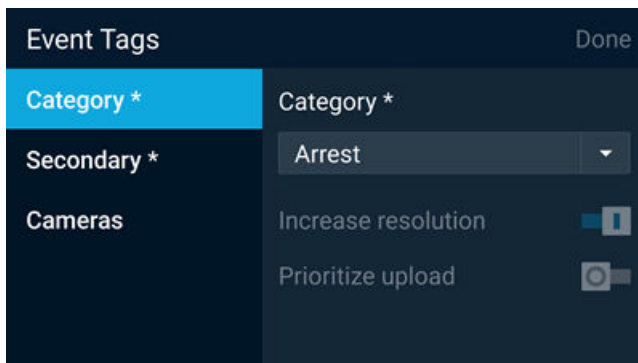


 **NOTE:** Complete all required values before touching **Done**.

2. Touch the drop-down arrow on the **Category** field and select the category to apply to the event you just stopped.



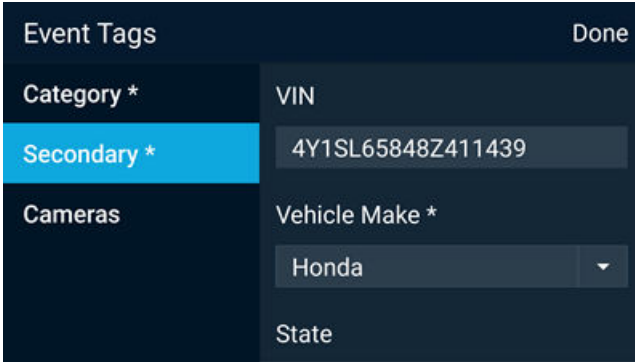
The category you selected is entered into the **Category** field.



3. If you selected a routine tag but want to save the event with a higher quality video resolution, touch the **Increase resolution** toggle switch to enable it ([Increase Resolution on page 29](#)).

 **NOTE:** You can only increase the resolution of routine tags. You cannot decrease the resolution of critical tags.

4. If you want an event moved to the front of the upload queue, touch the **Prioritize upload** toggle switch to enable it ([Prioritize Upload on page 29](#)).
5. Touch **Secondary**.
The **Secondary Event Tags** screen appears.
6. Select or enter the secondary tag answers.



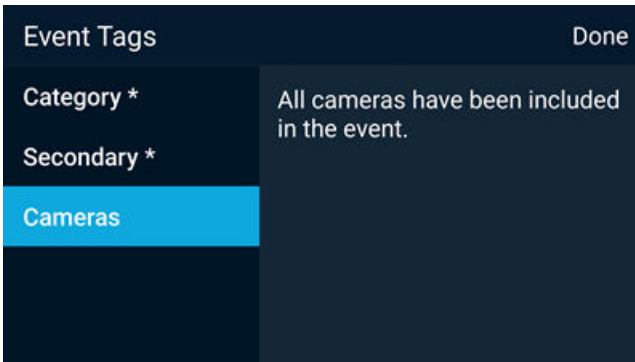
Event Tags		Done
Category *	VIN	
Secondary *	4Y1SL65848Z411439	
Cameras	Vehicle Make *	Honda
	State	

 **NOTE:** The secondary tags shown in this image are examples. Your agency secondary tags may be different.

7. Touch **Cameras**.

The **Event Tags Cameras** screen appears.

If your agency configuration does not support camera background state, then the following screen appears. Go to step 9.



Event Tags		Done
Category *	All cameras have been included in the event.	
Secondary *		
Cameras		

8. Select the cameras in a background recording state that you want to be included in the event upload.

Event Tags		Done
Category *	Camera	Include in upload
Secondary *	Front (wide)	☒
Cameras	Cabin	☒
	<i>Included based on AI detection</i>	
	Rear	☐

 **NOTE:** By default, cameras are included in an event upload if human form detection is enabled during an event. Depending on your configuration, you may not be able to change the state of the camera. See [Background Recording on page 99](#) for more information.

 **IMPORTANT:** By default, cameras are included in an event upload if human form detection is enabled during an event. Depending on your configuration, you may not be able to override the AI decision to include a stream in the upload. See [AI Human Form Detection Override on page 99](#) for more information.

9. Verify that you have answered all required tags ([Required Event Tags on page 30](#)).

10. Touch **Done**.

The tags are applied to the event and the previous screen appears on the Display Screen.

For more information, see:

- [Event Tagging on page 28](#)
- [Camera Views on page 96](#)

Event and Video Review

You can review events and video on the M500, if you have permissions:

- By event ([Review by Event on page 33](#))
- By time ([Review by Time on page 39](#))

Your administrator configures review permissions in the agency evidence management system.

 **IMPORTANT:** **Review** is not available while the vehicle is moving.

Review by Event

When you choose to review by event, the M500 shows you a list of the events present on the DVR.

← Review				×
Time		Category	Duration	
Thursday 11 April 2024				
13:57	2/2	motorist-assist	00:17:28	RATF
13:40	1/2	motorist-assist	00:17:45	RATF
Wednesday 10 April 2024				
16:12		traffic-warning	00:01:26	

See [Reviewing by Event on page 37](#) for instructions how to review an event on your M500.

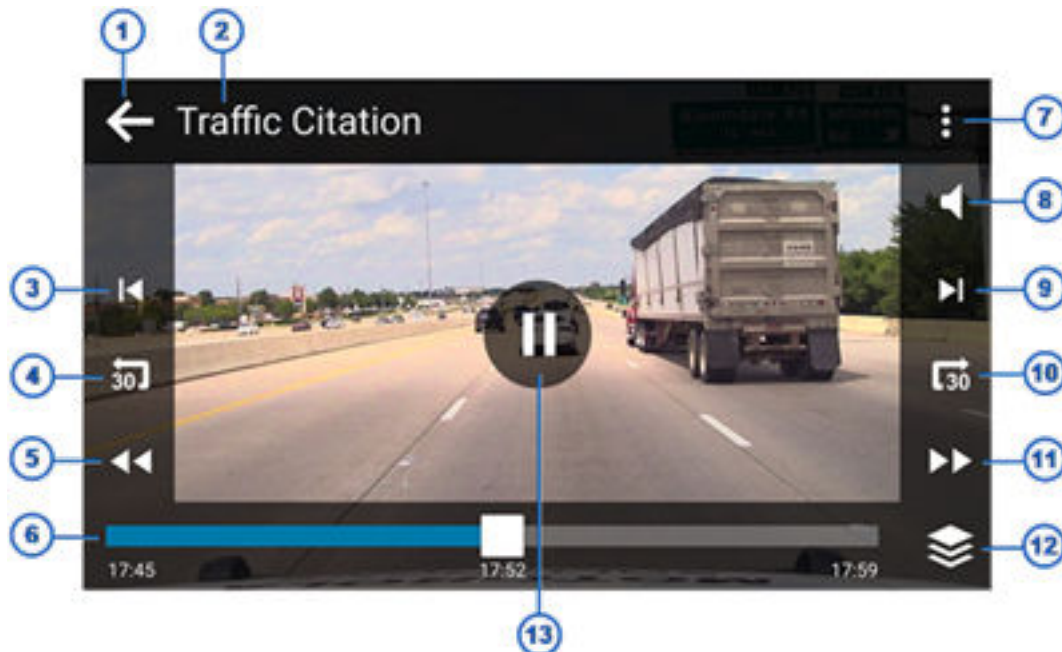
The **Review** event list screen shows basic information about each event on the M500, including:

- Event date
- Start time
- Category
- Duration
- Whether the event is a Record-After-the-Fact® (RATF) event ([Record-After-the-Fact Events on page 42](#))
- Split event indicator ([Split Event Indicator on page 36](#))

When you select an event from the list, depending on your video review permissions, the **Review** player opens playing back the event you selected ([Event Player on page 34](#)).

Event Player

When you select an event to review from the event list, the **Review** player opens playing back the event you selected.



NOTE: Your administrator configures review permissions in the agency evidence management system.

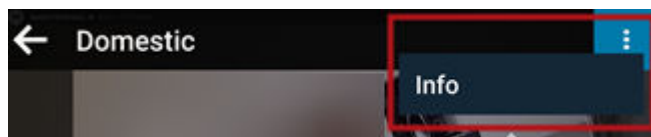
1	Return to Events list (Review by Event on page 33)
2	Event category (Category on page 28)
3	Go to the start of the event
4	Step backward 30 seconds in the event
5	Fast rewind the event; touching the arrows again increases the speed (2x, 4x, 8x)
6	Playback progress bar and indicator; dragging the indicator moves the playback to that location in the event
7	Menu of more information and/or actions depending on permissions (Event Information on page 35)
8	Playback volume control
9	Go to the end of the event
10	Step forward 30 seconds in the event
11	Fast forward the event; touching the arrows again increases the speed (2x, 4x, 8x)
12	View event metadata (Metadata Overlay in Review on page 40)
13	Pause/Play control

If the event has been split into smaller events, a split event indicator displays in the top left corner of the event player. See [Split Event Indicator on page 36](#) for more information.

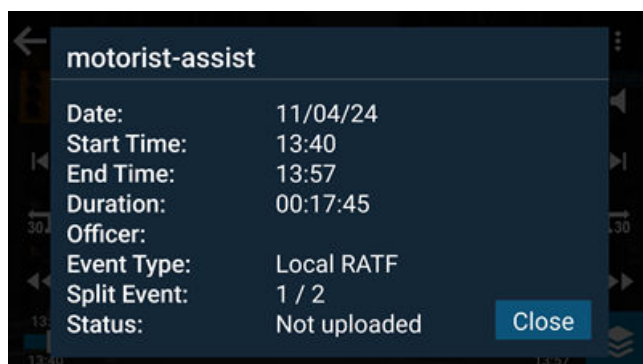
Event Information

You can find information about an event through the menu on the **Review** player.

On the top right of the **Review** player, touch the menu icon, then touch **Info**.



The event information dialog box opens on top of the **Review** player.



The information dialog box provides the following details about the event the **Review** player is playing:

- Assigned event category
- Event date
- Event start and end times
- Event duration

- Officer who recorded the event
- Event type
 - **Local:** Event recorded locally on the M500
 - **Local RATF:** Record-After-the-Fact® (RATF) event created locally on the M500 ([Creating an RATF Event on page 42](#))
 - **Remote RATF:** RATF event requested remotely from the agency evidence management system
- Split event ([Split Event Indicator on page 36](#))
- Upload status
 - **Uploaded:** Event was successfully uploaded to the agency evidence management system
 - **Not uploaded:** Event has not yet been uploaded to the agency evidence management system

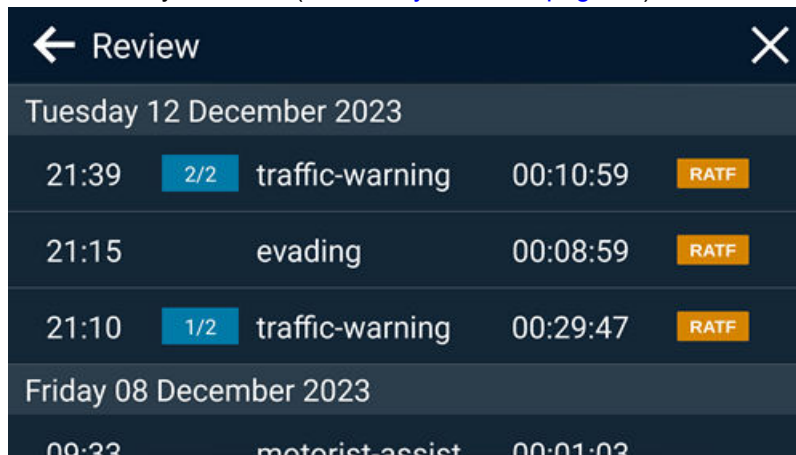
Split Event Indicator

If the system determines that an event is too large for efficient upload to the agency evidence management system (in duration, size, or number of files), the system splits the event into smaller events.

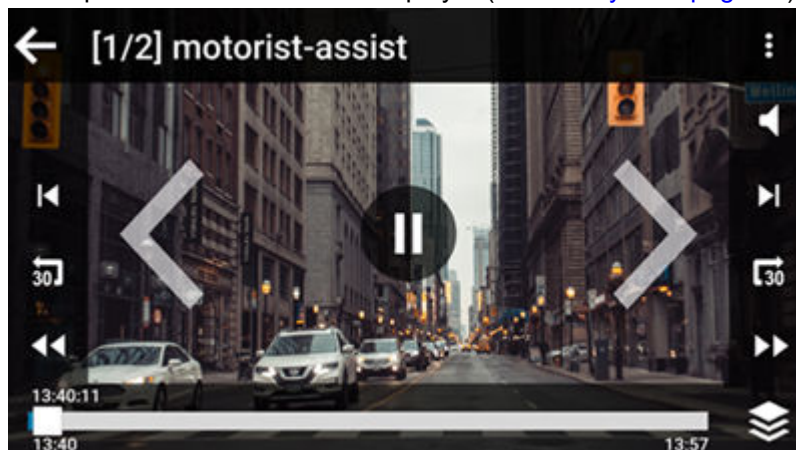
These smaller events are assigned the same category and are linked together when imported into the agency evidence management system. The smaller events are marked with a split event indicator. For example, if an event has been split into two events, the indicators show [1/2] and [2/2].

The split event indicator shows on the following screens:

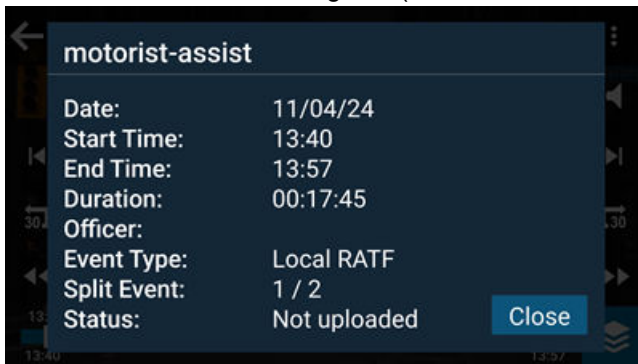
- The review by event list ([Review by Event on page 33](#))



- The top left corner of the **Review** player ([Event Player on page 34](#)):



- The event information dialog box ([Event Information on page 35](#))



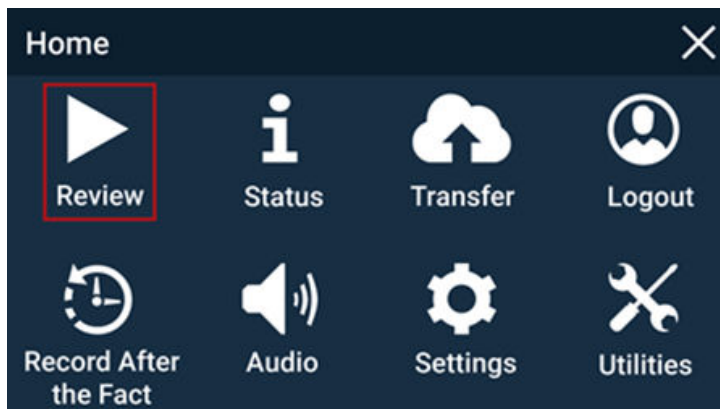
Reviewing by Event

Procedure:

1. From Live View, press **Home** on the Display Control Panel.



The **Home** screen opens.



2. Touch **Review**.

The **Review by Events** screen appears.

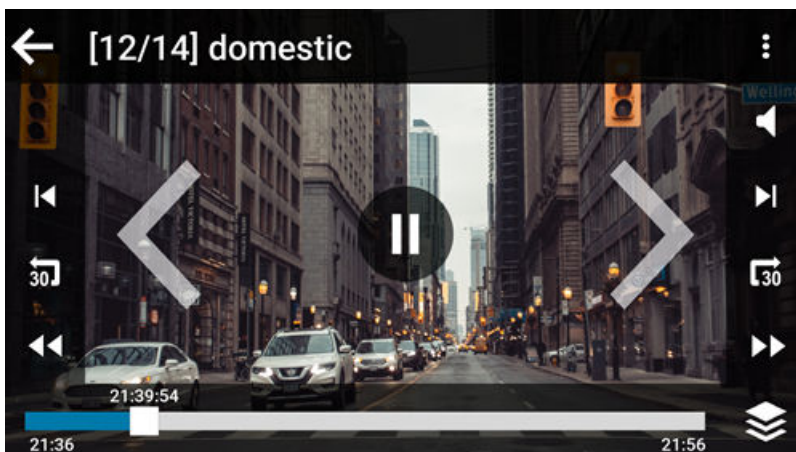
← Review					×
Time		Category	Duration		
Thursday 11 April 2024					
13:57	2/2	motorist-assist	00:17:28	RATF	
13:40	1/2	motorist-assist	00:17:45	RATF	
Wednesday 10 April 2024					
16:12		traffic-warning	00:01:26		

3. Touch to select the event you want to review.



NOTE: You may need to enter your supervisor password to review the event you selected.

The **Review** player opens showing the event you selected.



NOTE: If the video was recorded by a camera on background, **Background video** appears on the screen. See [Background Recording on page 99](#) for more information.

4. Use the player controls to review the event ([Review by Event on page 33](#)).
5. Touch the arrow on the top left to return to the **Events** list and select another event to review, or press **Home** on the Display Control Panel to return to Live View.

Result:

For more information, see:

[Review by Event on page 33](#)

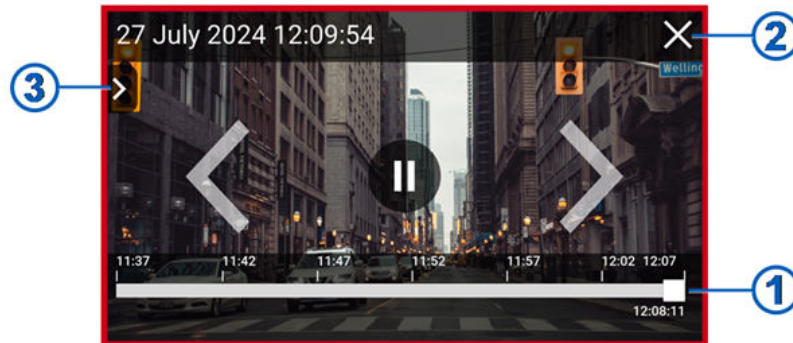
[Event Player on page 34](#)

[Event Information on page 35](#)

Review by Time

When you choose to review by time, a player opens on the Live View screen showing you near real-time video. You can move the playback indicator back in time along the playback progress bar to review video recorded in the near past.

See [Reviewing by Time on page 39](#) for instructions how to review the video by time on your M500.



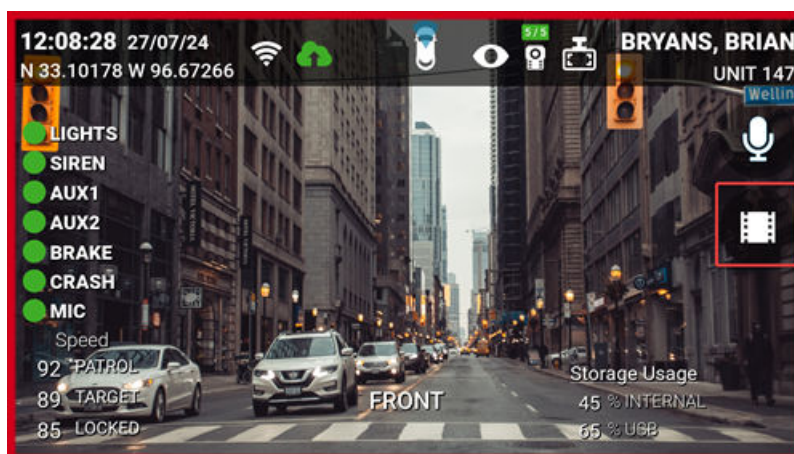
1	Playback progress bar and indicator; dragging the indicator moves the playback to that moment in time
2	Close the near real-time player and return to Live View
3	View event metadata (Metadata Overlay in Review on page 40)

 **NOTE:** You can also review video by time when you create a Record-After-the-Fact® (RATF) event. See [Record-After-the-Fact Events on page 42](#) for information.

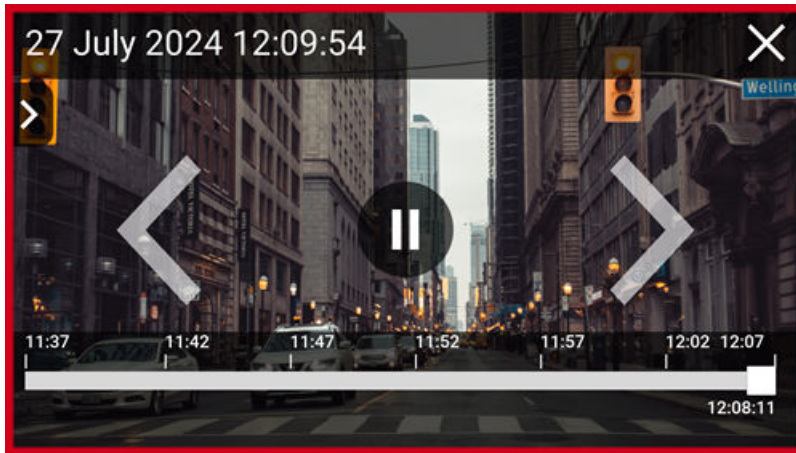
Reviewing by Time

Procedure:

1. On the Live View, touch the review by time filmstrip icon.



The icons and text on the Live View disappear and the near real-time player controls appear ([Review by Time on page 39](#)). The playback indicator is as close to real-time as the system can provide.



TIP: Touch the screen to see the right and left on-screen arrows. Touch an arrow to move to the stream from another camera in the system.

2. Move the playback indicator back in time along the playback progress bar to review video recorded in the near past.



NOTE: You may need to enter your supervisor password to review video from before you logged in.

3. Touch the arrow to review metadata associated with the near real-time video.



NOTE: For more information, see [Metadata Overlay in Review on page 40](#) and [Metadata on page 58](#).

4. Touch the **X** to close the near real-time player and return to Live View.

Result:

For more information, see

[Event and Video Review on page 33](#)

[Record-After-the-Fact Events on page 42](#)

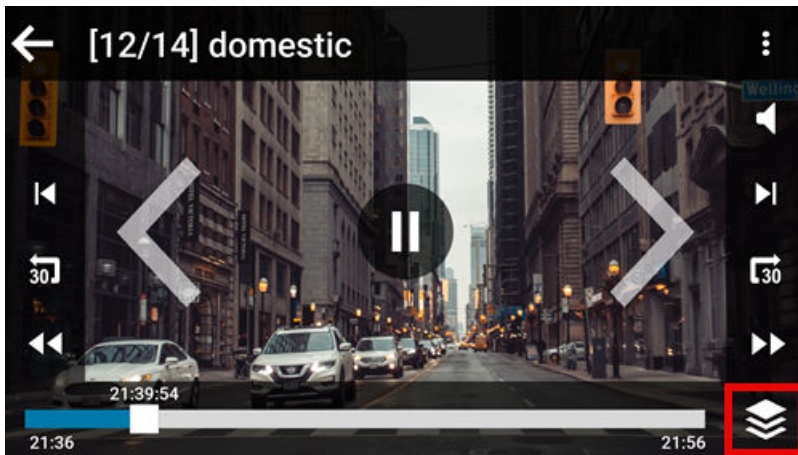
Metadata Overlay in Review

You can view the following metadata related to an event:

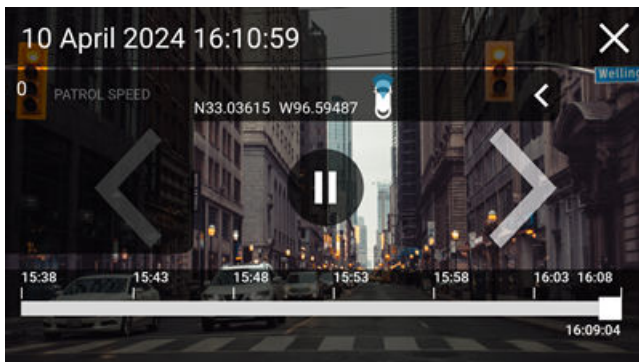
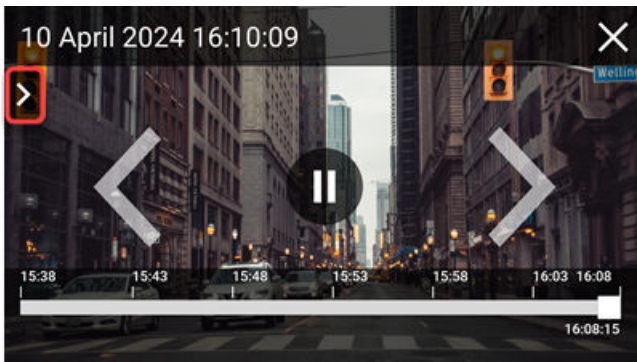
- Patrol speed
- GPS coordinates
- Camera view indicator

You can view metadata on the following screens:

- **Event Player** ([Event Player on page 34](#))
When you review an event by time, you can touch the overlay icon in the bottom right corner of the screen to view metadata related to the event:



- **Review by Time** ([Review by Time on page 39](#))
When you review an event by time, you can touch the arrow icon in the upper left corner of the screen to view metadata related to the event:



 **TIP:** Touch the arrow icon to hide the metadata.

- **Record After the Fact (RATF)** ([Record-After-the-Fact Events on page 42](#))
When you review an RATF event, you can touch the arrow icon in the upper left corner of the screen to view metadata related to the RATF event:



Record-After-the-Fact Events

The Record-After-the-Fact® (RATF) feature allows you to go back and save video (and associated data) that was not originally part of a recorded event.

 **IMPORTANT:** The presence of any data on an M500 from a given time period depends on many factors, for example, its configuration and how heavily your agency used that particular M500. Consult with your Motorola Solutions representative for more information.

Your administrator can configure the M500 to capture video (and associated data) all the time or only during recorded events. Even if the system is capturing video all the time (RATF is enabled), only the recorded events (the protected segments between the start and stop) are saved and uploaded to your agency evidence management system.

 **NOTE:** In your agency evidence management system, your administrator can configure how far in the past you can set the start time for an RATF event.

You can view a subset of system metadata during RATF Preview. See [Metadata Overlay in Review on page 40](#) for more information.

See [Evidence on page 56](#) for more information.

If you need to save some of the video (or associated data) that was originally not part of a recorded event, and RATF is enabled, you can create an RATF event. The video (and associated data) you include in the RATF event becomes protected when you create the event. This protected RATF event is uploaded to your agency evidence management system with other events.

You must have the correct permissions to create an RATF event.

 **NOTE:** You cannot create an RATF event while the M500 is recording an event. The RATF icon is disabled until you stop the event.

 **IMPORTANT:** RATF is not available while the vehicle is moving.

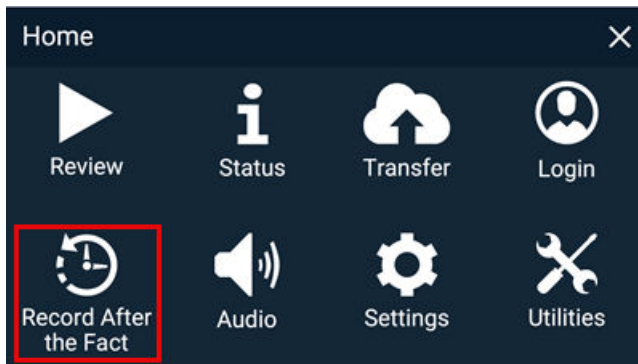
Creating an RATF Event

Procedure:

1. From the Live View, press **Home** on the Display Control Panel.



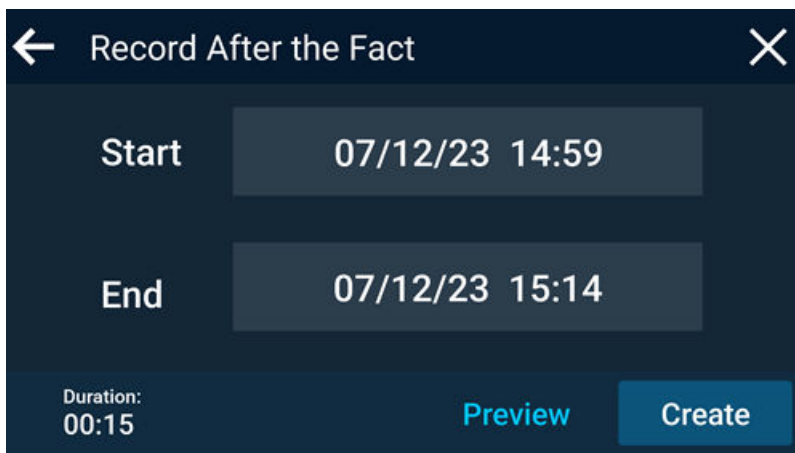
The **Home** menu appears.



 **NOTE:** If your administrator has not enabled RATF in the M500 configuration, the **Record After the Fact** icon is not active on the **Home** menu.

2. Touch **Record After the Fact**.

The **Record After the Fact** screen appears.



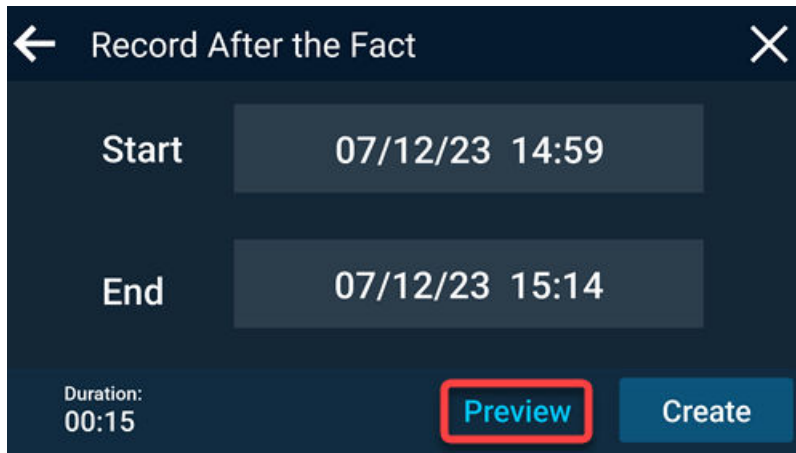
3. Touch the **Start** field to enter the start time of your RATF event.

The **Start** screen appears.

4. Touch a field (**Date dd, mm, or yy** or **Time h or m**) then use the number pad to enter the **Start** date and time.
5. Touch **Done**.

The **Record After the Fact** screen reappears.

6. Touch the **End** field to enter the end time of your RATF event.
The **RATF End Time** screen appears.
7. Touch a field (**Date dd, mm, or yy** or **Time h or m**) then use the number pad to enter the **End** date and time.
8. Touch **Done**.
The **Record After the Fact** screen reappears.



← Record After the Fact ×

Start 07/12/23 14:59

End 07/12/23 15:14

Duration: 00:15

Preview Create

9. Touch **Preview**.

If you do not want to preview the RATF event, go to step 11.

The **Record After the Fact** preview screen appears.

 **NOTE:** If the video was recorded by a camera on background, **Background video** appears on the screen. See [Background Recording on page 99](#) for more information.



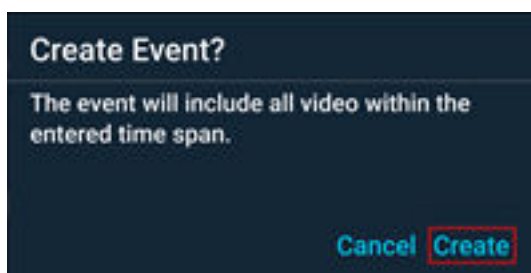
 **NOTE:** Touch the screen to see the right and left on-screen arrows. Touch an arrow to move to the stream from another camera in the system.

 **TIP:** You can also review metadata on the RATF preview screen. See [Metadata Overlay in Review on page 40](#) for more information.

10. Move the playback indicator along the playback progress bar to preview the RATF event.

11. Touch **Create**.

The **Create Event?** confirmation dialog box opens.



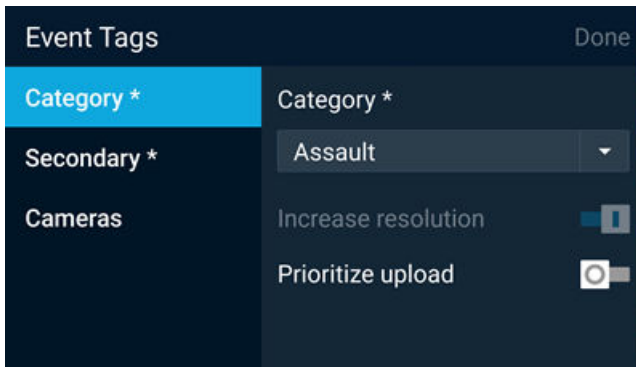
Create Event?

The event will include all video within the entered time span.

Cancel Create

12. Touch **Create**.

The **Event Tags** screen appears.



13. Tag the RATF event.

See [Tagging an Event on page 31](#) for instructions.

Result:



NOTE: RATF events can also be requested remotely through the agency evidence management system. See your agency evidence management system documentation for more information.

Event Upload

At the end of your shift, you verify the events you recorded during your shift are uploading automatically or take steps to upload them manually.

Recorded events can be uploaded:

- Automatically over a Wi-Fi® or LTE network connection with the agency evidence management system ([Event Upload Over a Network on page 46](#))
- Manually using the M500 USB drive inserted into a computer with a network connection to the agency evidence management system ([Event Upload Using a USB Drive on page 48](#))

Event Upload Over a Network

If your agency connects to its M500 in-car video systems through a wireless network (using Wi-Fi® and/or LTE), your M500 typically uploads events automatically to the agency evidence management system.

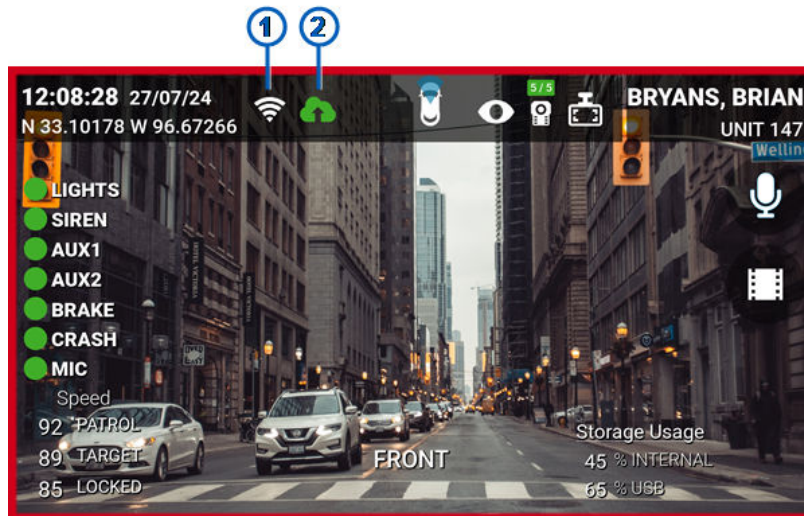
Recorded events upload automatically from your M500 if your vehicle:

- Has a wireless radio and is in range of your agency Wi-Fi network ([Wi-Fi Upload on page 47](#))
Or
- Has a Sierra Wireless® or other LTE appliance installed ([LTE Upload on page 47](#)) and is in a cellular service area



NOTE: The M500 must be configured to connect to the agency evidence management location.

You can see the Wi-Fi signal strength and event transfer status on the Live View, when applicable.



1	Wi-Fi signal strength (used for MikroTik Groove only)
2	Event transfer status White : Connected to agency evidence management, Wi-Fi or LTE Green : Actively uploading events, Wi-Fi or LTE

The M500 can continue to upload recorded events after the ignition is turned off, if configured to do so. See [Shutdown Timers on page 53](#) for more information.

Wi-Fi Upload

The M500 automatically begins to upload the recorded events on its hard drive to your agency evidence management system when you are in range of the agency access points (Wi-Fi hotspots).

 **NOTE:** M500 must be configured to connect to the agency evidence management location.

Before you leave your vehicle at the end of a shift, verify that your M500 is uploading its events:

- Check the Wi-Fi signal strength on Live View to verify connection to your agency network
- Check the event transfer status on the Live View to verify it is green  (actively uploading events)
- If you want more detailed information about the transfer status, check the **Transfer Status** screen ([Transfer Status on page 69](#))

LTE Upload

The M500 automatically uploads recorded events to your agency evidence management system any time you have cellular service, if your vehicle has a Sierra Wireless or other LTE appliance installed.

 **NOTE:** The M500 system supports two Sierra Wireless routers: the AirLink® RV55 Industrial LTE-A Pro Router and the AirLink® MP70 LTE Router.

The event transfer status icon on the Live View indicates your LTE connection and upload status, when you are using LTE for event transfer:

- White cloud icon  indicates a good LTE connection to the agency evidence management location

- Green cloud icon  indicates events are actively uploading over the LTE connection

 **NOTE:** If your agency uploads events over an LTE connection, your M500 may be configured to only upload critical events over LTE. If this is the case, new firmware versions are not downloaded from your agency evidence management system over an LTE connection. For more information about critical events, see [Critical Categories on page 29](#).

See [Status on page 65](#) for more detailed information about the LTE connection and upload status.

Event Upload Using a USB Drive

You can manually upload recorded events from your M500 to the agency evidence management system using the USB drive.

 **IMPORTANT:** You must upload events manually if your agency does not connect to its M500 in-car video systems through a wireless network (using Wi-Fi® and/or LTE).

When the M500 records events, the events are automatically saved to the DVR hard drive (SSD) and the USB drive ([DVR on page 59](#)).

To manually upload events at the end of each shift, your M500 USB drive must be inserted into a computer in the agency with a connection to the evidence management system. It is recommended that the same M500 USB drive be brought back to the M500 to confirm that the events were successfully uploaded ([Uploading Events Using a USB Drive on page 49](#)).

 **IMPORTANT:** Only use the USB drives shipped with the M500 to transfer evidence. Other USB devices are not approved and may not function properly.

Events on the USB drive that have not yet been confirmed as uploaded show the USB icon  on the **Transfer Event List** screen ([Transfer Event List on page 70](#)).



The M500 confirms a successful upload when the same USB drive is inserted back into the M500.

Events successfully uploaded from the USB drive are shown on the **Transfer Event List** screen with a cloud icon .

For information and instructions how to upload events from a USB drive through the USB upload application in the agency evidence management system, see your evidence management system documentation.

Uploading Events Using a USB Drive



IMPORTANT: Only use USB drives shipped with the M500 to transfer evidence. Other USB devices are not approved and may not function properly.

Procedure:

1. On the **Transfer Event List**, verify that recorded events are present on the USB drive ([Accessing Transfer on page 68](#)).
2. Safely eject the USB drive:
 - Using the **Stop** button on the Display Control Panel ([Quick-Ejecting the USB Drive by Using Stop on page 50](#))
Or
 - Using the **Utilities** screen ([Ejecting the USB Drive Using Utilities on page 49](#))
3. Unlock the DVR door panel and remove the USB drive from the M500.
4. Take the USB drive into the agency (or another authorized location) and insert it into the computer designated for USB upload to the agency evidence management system.
5. Follow the instructions for uploading events from an M500 USB drive through the USB upload application.

For instructions, see your agency evidence management system documentation.



NOTE: If there are any firmware or configuration updates available in your agency evidence management system, they are automatically loaded on the USB drive while it is inserted into the computer.

6. After all the events have uploaded, eject the USB drive from the computer.
7. Take the USB drive back to the M500 it came from and reinsert it into the DVR.



NOTE: If there are any updates loaded on the USB drive, they are staged on the M500 and then applied during the next reboot. See [Automatic Firmware Upgrade on page 115](#) and [Automatic Configuration Deployment on page 109](#) for more information.

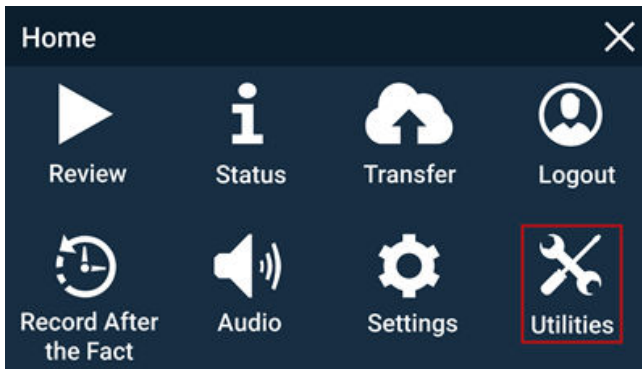
The M500 system confirms that the events on the USB drive were successfully uploaded to the agency evidence management system and marks the events as unprotected.

8. Close and lock the DVR door panel.
9. On the **Transfer Event List** screen, verify the events are showing the cloud icon ([Transfer Event List on page 70](#)).

Ejecting the USB Drive Using Utilities

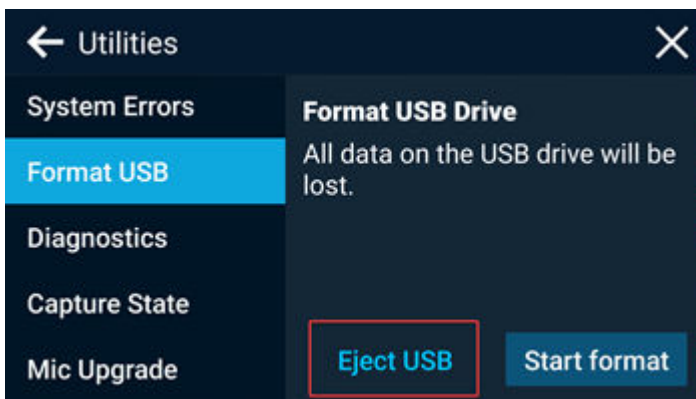
Procedure:

1. From the Live View, press **Home** on the Display Control Panel.
The **Home** screen appears.



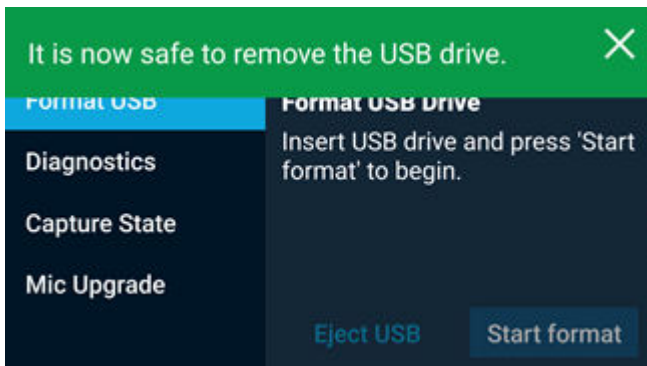
2. Touch **Utilities**.

The **Format USB Drive** screen appears.



3. Touch **Eject USB**.

The **It is now safe...** message appears.



4. Remove the USB drive from the M500.

Result:

For more information, see [Quick-Ejecting the USB Drive by Using Stop](#) on page 50.

Quick-Ejecting the USB Drive by Using Stop

You can use the **Stop** button to quickly eject the USB drive from the M500.

Procedure:

1. On the Display Control Panel, press and hold **Stop** for 3 seconds.

The **It is now safe to remove the USB drive** message appears at the top of the screen and the USB drive is ejected.

 **IMPORTANT:** If the USB drive is busy when you try to quick-eject it, the message may not appear for a few seconds. Remove the USB drive only after the message appears.

2. Unlock the DVR door panel and remove the USB drive from the M500 ([DVR on page 59](#)).

Result:

 **NOTE:** If the M500 is recording an event when you try to quick-eject the USB drive, pressing **Stop** stops the event. When the **Event Tags** screen appears, tag the event, then try again to quick-eject the USB drive ([Event Tagging on page 28](#)).

For more information, see [Ejecting the USB Drive Using Utilities on page 49](#).

Log Out of the M500

You can log out of the M500 two ways:

- From the Home menu [Logging Out from the Home Menu on page 51](#)
- By touching the officer name on Live View [Logging Out with Live View on page 52](#)

Depending on your configuration, the M500 may log you out at the end of the shutdown process.

Logging Out from the Home Menu

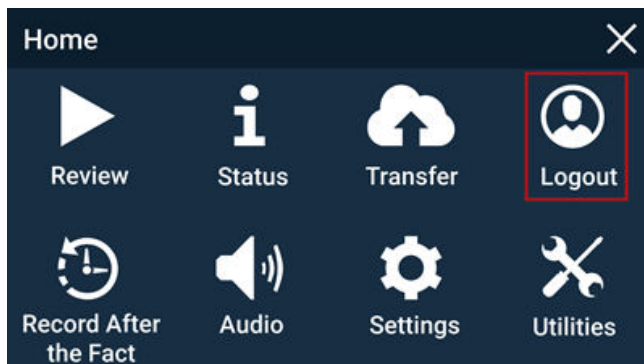
You log out of the M500 at the end of each shift to ensure your name is not associated with events that were made after your shift ended.

Procedure:

1. From Live View, press **Home** on the Display Control Panel.



The **Home** menu appears.



2. Touch **Logout** on the **Home** menu.

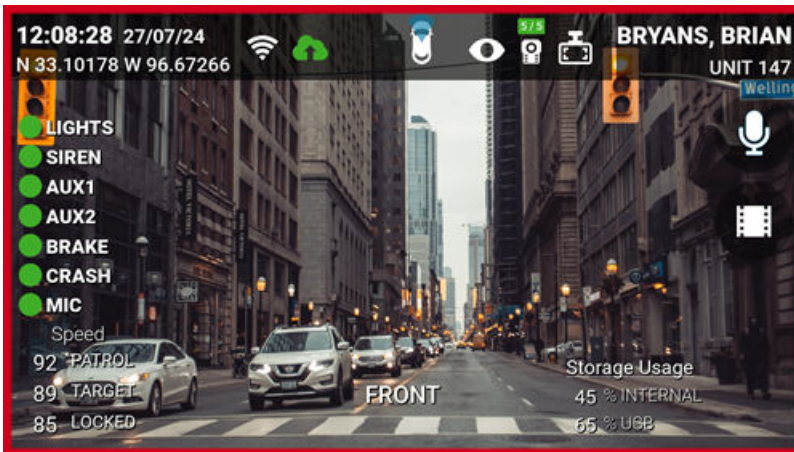
The M500 logs you out.

Logging Out with Live View

You log out of the M500 at the end of each shift to ensure your name is not associated with events that were made after your shift ended.

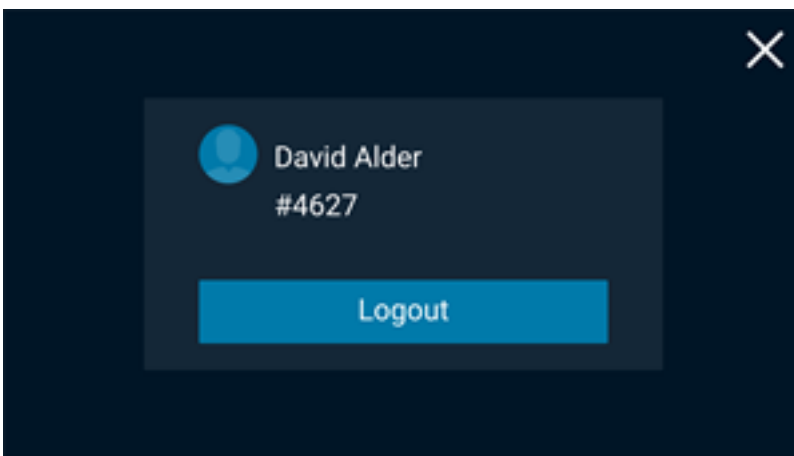
Procedure:

1. From Live View, touch the officer name.



NOTE: If there is a recorded event in progress, you cannot log out.

2. Touch **Logout**.



The M500 logs you out.

Power Off the M500

At the end of your shift, you can power off the M500 DVR two ways:

- With the ignition ([Powering Off with the Ignition on page 53](#))
- With the **Off** button ([Powering Off with the Off Button on page 53](#))

Depending on your configuration, the M500 may log you out at the end of the shutdown process.

Powering Off with the Ignition

Your agency administrator typically configures the M500 to power off with your vehicle ignition.

Procedure:

To power off the M500, turn off your vehicle.

The M500 starts the power-off process.

 **NOTE:** Depending on your configuration, the M500 may log you out at the end of the shutdown process.

Result:

If your agency configures M500 to use shutdown timers, the M500 can continue to upload recorded events after it starts its power-off process.

See [Shutdown Timers on page 53](#) for more information.

Powering Off with the Off Button

You can use the **Off** button on the Display Control Panel to power off the M500.

 **IMPORTANT:** If your M500 is configured to power off with the vehicle ignition, avoid pressing the **Off** button on the Display Control Panel.

Procedure:

To power off the M500, press **Off**.



The M500 powers off.

 **NOTE:** Depending on your configuration, the M500 may log you out at the end of the shutdown process.

Result:

 **NOTE:** When you power off the M500 using the **Off** button, shutdown timers are not applicable.

Shutdown Timers

Shutdown timers determine how long the M500 waits before shutting down after you turn off the vehicle ignition. The timers are configured by your administrator in the agency evidence management system.

 **NOTE:** To use the M500 shutdown timers, your M500 must be configured to shut down when the ignition turns off.

The shutdown timers allow you to turn off your vehicle for a period of time during your shift without completely shutting down the M500. They also allow the M500 to continue uploading events for a period of time at the end of your shift, if needed.

Depending on the length of time your administrator configures for the timers, the M500 can start shutting down immediately when you turn off the vehicle ignition (0 minutes for each timer) or start shutting down after a delay.

There are two shutdown timers that run in the following order, if configured by your agency administrator:

- [Ignition Shutdown Timer on page 54](#)
- [Wireless Transfer Shutdown Timer on page 55](#)



IMPORTANT: The M500 is on and fully functional while the Ignition Shutdown Timer counts down. The M500 stays ready to start a recorded event while the Wireless Transfer Shutdown Timer counts down.

For example, your administrator sets the Ignition Shutdown Timer for 15 minutes and the Wireless Transfer Shutdown Timer for 1 hour.

When you turn off the vehicle ignition, the Ignition Shutdown Timer shows a 15-minute timer on the Live View that starts counting down. When the timer on the Live View reaches 0, the Display Screen goes dark and the Display Control Panel button LEDs start pulsing.

The Wireless Transfer Shutdown Timer starts its countdown. The screen remains dark and the buttons continue to pulse for the configured hour, or until the last event finishes uploading, whichever comes first. When the Wireless Transfer Shutdown Timer expires, the M500 begins the final shutdown process, then powers off.



NOTE: Depending on your configuration, the M500 may log you out at the end of the shutdown process.

You can cancel the shutdown timers at any point until the final shutdown process begins ([Shutdown Timer Cancellation on page 55](#)). You can skip the remaining time on the shutdown timers and start the final shutdown process at any point ([Immediate Shutdown on page 55](#)).

For more information, see:

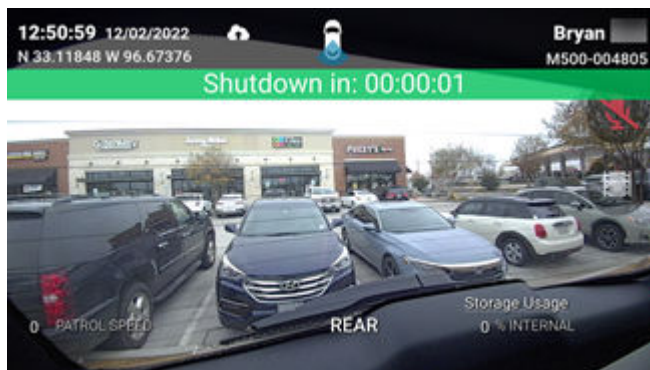
[Power Off the M500 on page 52](#)

[Event Upload Over a Network on page 46](#)

Ignition Shutdown Timer

The Ignition Shutdown Timer starts counting down when you turn off the vehicle ignition.

This timer starts a countdown clock, visible on the Display Screen.



The timer starts at the time configured by your administrator. When the timer reaches 0, the Wireless Transfer Shutdown Timer starts counting down ([Wireless Transfer Shutdown Timer on page 55](#)).



IMPORTANT: The M500 is on and fully functional while the Ignition Shutdown Timer counts down.

For more information, see [Shutdown Timers on page 53](#).

Wireless Transfer Shutdown Timer

The Wireless Transfer Shutdown Timer starts counting down when the Ignition Shutdown Timer reaches 0.

See [Ignition Shutdown Timer on page 54](#).

When the Wireless Transfer Shutdown Timer starts, the Display Screen goes dark and the Display Control Panel button LEDs start to pulse. When the timer expires, the M500 starts shutting down.



IMPORTANT: The M500 stays ready to start a recorded event while the Wireless Transfer Shutdown Timer counts down.

The Wireless Transfer Shutdown Timer expires early (before it reaches 0) in the following scenarios:

- No events remain to be uploaded when the Wireless Transfer Shutdown Timer starts
In this case, the Wireless Transfer Shutdown Timer expires immediately.
- The M500 system is not connected to the agency evidence management system when the Wireless Transfer Shutdown Timer starts
In this case, the Wireless Transfer Shutdown Timer expires immediately.
- All events finish uploading while the Wireless Transfer Shutdown Timer is counting down
In this case, the Wireless Transfer Shutdown Timer expires as soon as the last event finishes uploading.



NOTE: Depending on your configuration, the M500 may log you out at the end of the shutdown process.

For more information, see [Shutdown Timers on page 53](#).

Shutdown Timer Cancellation

You can cancel the shutdown timers (and the M500 shutdown) at any time during their countdowns by performing one of the following actions:

- Turning the ignition on
- Pressing **On** on the Control Panel
- Starting an event manually or automatically

The M500 shutdown cannot be canceled after the timers have both reached 0 and the final shutdown process has started.

Immediate Shutdown

You can skip the remaining time on the shutdown timers and start the M500 shutdown process immediately by pressing **Off** on the Control Panel.

Chapter 3

About the M500 In-Car Video System

The M500 in-car video system captures, processes, stores, and uploads video and audio evidence.

The M500 system can include:

- M500 DVR ([DVR on page 59](#))
- M500 Display ([Display on page 60](#))
- Up to 5 cameras, including:
 - M5F front camera ([M5F Front Camera on page 103](#))
 - M5P cabin camera ([M5P Camera on page 104](#)) with an integrated microphone ([M5P Camera Integrated Microphone on page 106](#))
 - M5R rear camera ([M5R Camera on page 104](#))
 - 2 M5E exterior cameras ([M5E Camera on page 105](#))
- 2 wired cabin microphones ([Stand-Alone Cabin Microphones on page 106](#))



NOTE: Future cameras and microphones will be added to the system as they are developed.

The M500 DVR and Display function as the central component of the M500 system:

- The DVR processes and stores evidence captured through the connected cameras and microphones
- The Display allows you to interact with the DVR through Control Panel buttons and touchscreen icons and menus

The M500 connects to your agency evidence management system for provisioning and for uploading evidence. The connection can be over a network or using a USB drive. See ([USB Drive on page 59](#)) for more information.



IMPORTANT: Motorola Solutions recommends that you provision the M500 using your agency evidence management software before first use.

Each camera connected to the M500 can be configured to record at least two streams of video. Each stream can be set to a different video quality, higher quality for critical events, lower quality for routine events.

The M500 is typically configured to capture and store video whenever it is powered on. Only protected evidence segments (recorded events) are uploaded to the agency evidence management system.

See [Evidence on page 56](#) for more information about recorded events.

The M500 system overwrites the oldest unprotected area in storage with any newly captured video or recorded events. This process continues until the storage is filled with events (protected segments). Then you must upload events to your evidence management system to free up storage space before you can continue to use the system.

Evidence

The M500 in-car video system captures, processes, stores, and uploads evidence in the form of recorded events.

A recorded event (or event) is a unique, protected segment of recorded:

- Video ([Video Evidence on page 57](#))
- Audio ([Audio Evidence on page 57](#))

- Metadata ([Metadata on page 58](#))

When the M500 starts and then stops a recorded event, the system protects the video/audio/metadata segment between the start and the stop. The protected segment is the recorded event.



NOTE: For information on starting events manually and automatically, see [Start Events on page 21](#). For information on stopping events, see [Stopping an Event Manually on page 26](#).

Depending on the configuration applied to the M500, the system may be capturing video all the time or only during a recorded event. If the system is capturing video all the time (Record-After-the-Fact® (RATF) or pre-event is enabled), only the protected segment (between the start and stop) is a recorded event.

Only recorded events (protected segments) are uploaded to your evidence management system.

Unprotected video (non-event video) can become protected if it is:

- Added to an event as pre-event video ([Pre-event on page 58](#))
- Included in an RATF event ([RATF on page 59](#))

Video Evidence

The M500 system can capture video from four types of physical cameras:

- M5F front camera ([M5F Front Camera on page 103](#))



NOTE: The physical front camera is configured as two separate camera views, **Front** and **Front (wide)**.

- M5P cabin camera ([M5P Camera on page 104](#))
- M5R rear camera ([M5R Camera on page 104](#))
- M5E exterior camera ([M5E Camera on page 105](#))

Each camera view (**Front**, **Front (wide)**, **Cabin**, **Rear**, **Left**, or **Right**) can be configured to record two streams of video, a primary and a secondary stream. Each stream is compressed using h.264 high-profile compression.



NOTE: Future cameras will be added to the system as they are developed.

The primary stream video quality for any camera view can be configured as:

- High definition (HD), 1080p, at a frame rate of 5, 10, 15, or 30 frames per second, or
- HD, 720p, at a frame rate of 5, 10, 15, or 30 frames per second, or
- Standard definition (SD), 480p, at a frame rate of 5, 10, 15, or 30 frames per second

The secondary stream video quality for any camera view can be configured as:

- HD, 720p, at a frame rate of 5, 10, 15, or 30 frames per second, or
- SD, 480p, at a frame rate of 5, 10, 15, or 30 frames per second

Audio Evidence

The M500 in-car video system records audio using the following types of microphones:

- Three wired, in-vehicle microphones:
 - Two stand alone
 - One built in to the M5P camera



NOTE: Future microphones will be added to the system as they are developed.

The in-vehicle microphones can be configured to:

- Continuously record audio, or
- Only record audio during recorded events, or
- Never record audio

All microphones can be configured to allow muting.



NOTE: Starting a recorded event automatically unmutes any muted microphones.

Metadata

Metadata is the information (text and icons) that can be overlaid on the video. Different subsets of the metadata are viewable depending on the system configuration and where in the system you are reviewing the video.

The M500 includes the following metadata, depending on its configuration:

- Department name
- Officer name
- Date and time
- Vehicle ID
- Microphone on or muted
- GPS location
- Patrol speed
Excessive speed can also act as an event trigger.
- Event triggers when active:
 - Auxiliary 1
 - Auxiliary 2
 - Crash
 - Lights
 - Siren
- Brakes when active

Pre-event

Video recorded before the event start is not protected unless it is added to the event as pre-event video. An M500 recorded event includes the pre-event video if the pre-event feature is enabled in the M500 configuration.

The pre-event feature causes the M500 to continuously record then save video for a period of time before an event start. The pre-event video is added before the start of the recorded event.

For example, a recorded event starts at 2:15 PM. You stop the event at 2:35 PM. After the system adds a configured pre-event video time of 1 minute, the recorded event includes video recorded from 2:14 PM to 2:35 PM.

M500 pre-event video time can be none up to 10 minutes. Your administrator configures the pre-event feature in the agency evidence management system.

Audio is not typically included with pre-event video. The M500 only begins to record audio when a recorded event starts, unless configured otherwise by your administrator.

 **NOTE:** Your administrator can configure the M500 (using the agency evidence management system) to include audio whenever the system is capturing video (recorded events, pre-event, and RATF).

RATF

When the RATF feature is enabled on the M500, the system captures and stores video whenever it is powered on. The segments between event starts and stops are protected as recorded events. The rest of the video is unprotected and can be recorded over. Only recorded events are uploaded to your evidence management software.

RATF video typically consists of video that was not originally part of a recorded event. You can use the M500 RATF feature to review video in M500 storage and designate a segment as protected, if you have permissions. This protected RATF video is then uploaded with other events to evidence management.

For more information, see [Record-After-the-Fact Events on page 42](#).

 **NOTE:** RATF video can include parts of or whole recorded events depending on the time-frame of the video that you designate.

 **NOTE:** In your agency evidence management system, your administrator can configure how far in the past you can set the start time for an RATF event.

Audio is not typically included in the saved unprotected (non-event) video when RATF is enabled. The M500 only begins to record audio when a recorded event starts unless configured otherwise by your administrator. This means that audio is not typically included with RATF video that was not originally part of a recorded event.

 **NOTE:** Your administrator can configure the M500 (using the agency evidence management system) to include audio whenever the system is capturing video (recorded events, pre-event, and RATF).

DVR

The M500 DVR is a computer that processes and stores evidence captured through connected cameras and microphones.



The DVR contains two devices for processing and storing evidence:

- 1 TB solid-state drive (SSD)
- 128 GB removable USB drive ([USB Drive on page 59](#))

The Radio-Frequency Identification (RFID) reader connects to the USB port on the back of the M500 DVR. See [RFID Reader on page 107](#) for more information.

 **IMPORTANT:** The USB port on the back of the M500 DVR can only be used for the RFID reader.

USB Drive

The USB drive is accessible through a locked door panel on the front of the DVR. When you unlock the door panel, a small arm extends outward holding the USB drive. All door panel keys are identical. Keys are typically kept by a supervisor.

When the M500 records events, the events are automatically saved to the SSD and the USB drives. The USB drive functions as a backup storage device for the SSD.



NOTE: The M500 remains fully functional when no USB drive is present.

You can use the USB drive in the M500 to:

- Function as backup storage to the main M500 hard drive (SSD)
- Transfer recorded events to the agency evidence management system if a network link is unavailable ([Event Upload Using a USB Drive on page 48](#))
- Provision the system ([Manual Configuration Deployment on page 110](#))
- Upgrade firmware ([Manual Firmware Upgrade on page 116](#))
- Collect a system state capture ([Generating a State Capture \(Utilities\) on page 87](#))



WARNING: Only remove a USB drive from an M500 when it is safe to do so. See [Ejecting the USB Drive Using Utilities on page 49](#) for instructions how to safely remove the USB drive.

If the USB drive is inserted in the M500, it is backing up recorded events that have not yet been successfully uploaded to the agency evidence management system. The system manages the space on the USB drive to ensure that events that were uploaded and confirmed using a network connection to the agency are removed from drive.

The USB drive has less capacity than the SSD. It typically does not have the capacity to perform a full backup of all the events on the system.

If the USB drive fills with events that have not been uploaded and confirmed, you must take the filled drive into the agency to upload events. The USB drive cannot be used for anything else until the events have been successfully uploaded to the agency evidence management system. You can replace the filled drive with a drive that has available capacity to continue to store new events.

Display

The M500 Display has a high-resolution (1920 x 1080), responsive, multi-touch Screen. Its Control Panel has buttons that serve as dedicated hard controls for some of the M500 main functions. The icons and menus on the Display Screen and the buttons on the Display Control Panel allow you to interact with the DVR computer.

The M500 Display lets you:

- View a live feed from any camera in the system ([Live View on page 62](#))
- Create recorded events ([Start Events on page 21](#)) and tag them ([Event Tagging on page 28](#))
- Review recorded events ([Reviewing by Event on page 37](#)) and other video stored on the system ([Reviewing by Time on page 39](#))
- Control and monitor M500 functions ([Home on page 64](#))



1	Display Screen showing Live View while recording an event
2	Display Control Panel

NOTE: Your Display Screen may change to an M500 logo when your vehicle begins to move. You cannot interact with any Display Screen functions when this happens. Your administrator configures this behavior in the agency evidence management system.

The Display Screen has two main starting points:

- **Live View:** Shows a live video stream from one of the cameras in the M500 system, overlaid with information and icons indicating the current status of the system ([Live View on page 62](#))
- **Home** menu: Functions as the main menu screen for the M500, providing access to M500 features and system information ([Home on page 64](#))

The Display Control Panel has seven buttons:

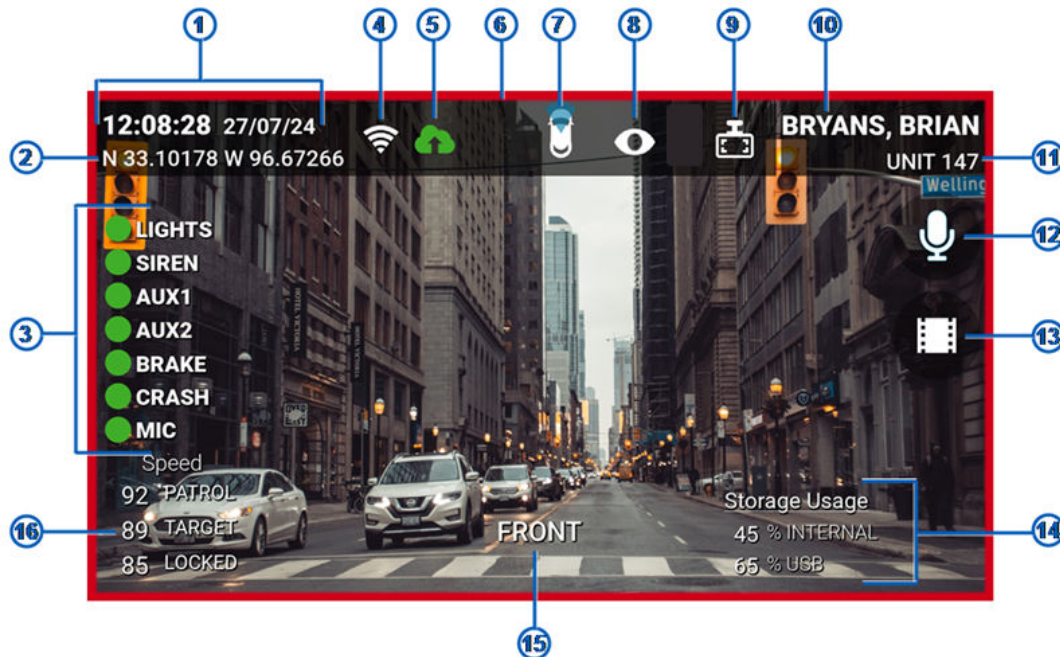
- **On:** Powers on the DVR ([Power On the M500 on page 15](#)) and activates/deactivates Covert Mode ([Covert Mode on page 24](#))
- **Off:** Powers off the DVR ([Power Off the M500 on page 52](#))
- **Home:** Switches between **Home** menu ([Home on page 64](#)) and Live View ([Live View on page 62](#))
- **Record:** Starts a recorded event ([Starting an Event Manually on page 24](#))
- **Stop:** Stops a recorded event ([Stopping an Event Manually on page 26](#))
- **Cameras:** Shows the views from the system cameras and allows you to turn off recording from an individual camera view, if you have permissions ([Cameras Button on page 96](#))
- **Display:** Shows the display options including brightness, **Dark Mode**, and **Hide Info** settings ([Display Button on page 100](#))

Live View

Live View is the first screen you see after you power on the M500 and it has fully booted up. It shows a live video stream from one of the cameras in the M500 in-car video system.

Initially, Live View shows the live stream coming from the camera your administrator has set as your default camera. You can touch the right or left on-screen arrows to move to the live stream from another camera in the system.

Live View typically shows you text and icons laid over the view of the live stream. The text and icons give you information about the current state of the system and let you access some features with a touch.



1	Current time and date
2	GPS coordinates of current location
3	Active external inputs (External Inputs on page 23)
4	Wi-Fi signal strength  location (Event Upload Over a Network on page 46), present when Wi-Fi is available (MikroTik Groove only)
5	Event transfer status (Event Upload Over a Network on page 46) White  : Connected to agency evidence management, Wi-Fi or LTE Green  : Actively uploading events, Wi-Fi or LTE
6	Active event recording indicators (red bars) (Start Events on page 21)  NOTE: The red bars appear on all screens when an event is in progress.
7	Camera view indicator (icon)
8	Live streaming icon (Live Stream on page 63)  NOTE: If there are events to upload, the upload is paused during live streaming.

9	Pending update icon (Automatic Configuration Deployment on page 109 and/or Automatic Firmware Upgrade on page 115)
10	Logged-in officer name (Logging In from the Home Menu on page 16)  NOTE: You can log in or log out of the M500 by touching the officer name on Live View. See Logging In from Live View on page 17 or Logging Out with Live View on page 52 for more information.
11	Vehicle ID
12	Microphone status (muted or unmuted) (Muting Microphones on page 19)
13	Near real-time review player access (Reviewing by Time on page 39)
14	Used storage status for internal hard drive and USB drive
15	Camera view indicator (text)
16	Patrol speed calculated using GPS

You can hide the information on the Live View if you want to see the live stream unobstructed. See [Display Button on page 100](#) for information about hiding on-screen text.

 **NOTE:** Your administrator can configure which text and icons your agency wants to show on the Live View.

Live Stream

The system can live stream directly to your agency evidence management system depending on which evidence management system you use. Dispatch can watch and listen to real-time events from your device.

The Live Streaming icon  shows on the Display Screen when your device is live streaming to your agency. Depending on the configuration, your device may make an audible notification when a live stream is initiated from your agency. See [Live View on page 62](#) for more information.

 **NOTE:** You cannot pause or stop a live stream on your device.

If you do not have an active connection to the agency evidence management system, your system does not live stream.

 **NOTE:** If there are events to upload, the upload is paused during live streaming.

License Plate Detection

Your administrator can enable the automatic license plate recognition (ALPR) feature in your evidence management system. When it is enabled, you see active license plate detection on the Live View.

 **NOTE:** When using the ALPR feature, enable **Hide Info** to turn off all on-screen text except plate detection ([Enabling Hide Info on page 102](#)).



NOTE: The example image blurs the license plates and reproduced text for privacy.

The system outlines the license plates in blue. The plate number is reproduced in white text below the blue box.

You can disable the license plate indicators if you have permissions. See [Enabling ALPR on page 78](#) for information.



NOTE: Contact your Motorola Solutions representative for more information about using ALPR with the M500 system.

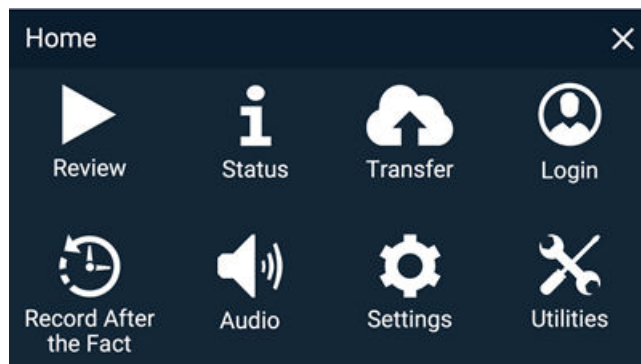
Home

The **Home** menu is the main menu screen for the M500. The icons give you access to M500 functions and system information.

You can access the **Home** menu from Live View by pressing **Home** on the Display Control Panel.



NOTE: To return to Live View from the **Home** menu, press **Home** on the Display Control Panel.



NOTE: If an event is in progress, red bars appear at the top and bottom of the screen.

The **Home** menu icons give you access to the following functions:

- **Review:** Opens the **Review by Event** feature ([Reviewing by Event on page 37](#))
- **Status:** Shows current user, system, and network status information ([Status on page 65](#))
- **Transfer:** Shows the status of event transfer to the evidence management location ([Transfer on page 68](#))

- **Login/Logout:** Allows officers to log in and associate their names with events recorded during their shifts ([Logging In from the Home Menu on page 16](#)) and log out after their shifts ([Logging Out from the Home Menu on page 51](#))
- **Record After the Fact:** Opens the Record-After-the-Fact® (RATF) feature to allow RATF event creation ([Record-After-the-Fact Events on page 42](#))
- **Audio:** Allows officers to adjust the volume of tones, review, and live audio, as well as the input level of wired (in-car) microphones ([Audio on page 71](#))
- **Settings:** Allows access to some system settings ([Settings on page 73](#))
- **Utilities:** Allows access to system tools ([Utilities on page 80](#))

For more information, see [Display on page 60](#).

Status

The **Status** icon on the **Home** menu gives you access to information about the current:

- Logged-in user ([User on page 66](#))
- Network connection to the evidence management system (MikroTik Groove only) ([Network on page 66](#))
- System firmware versions and connected external inputs ([System on page 67](#))
- Associated body-worn cameras (**Group Devices**)



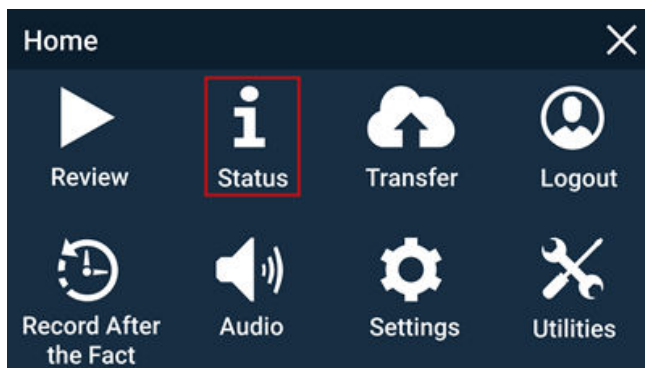
NOTE: The **Group Devices** screen is only applicable to North American customers with integrated systems.

Accessing Status

Procedure:

1. From Live View, press **Home** on the Display Control Panel.

The **Home** menu appears.



2. Touch **Status**.

The **User Status** screen opens.

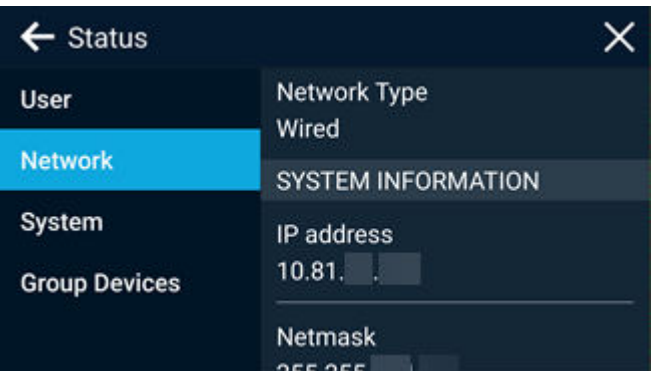
User

The **Status User** screen shows information about the currently logged-in user.

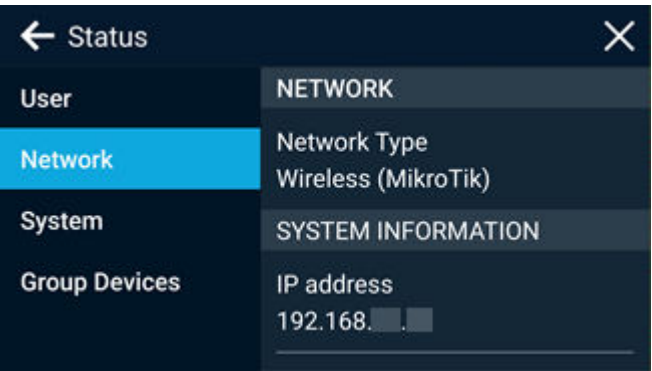


Network

The **Status Network** screen shows information about the network connections between the in-vehicle network and the agency evidence management system.



Full information is only available when a MikroTik Groove radio is used in the system.



The **Status Network** screen shows the IP address and other information for:

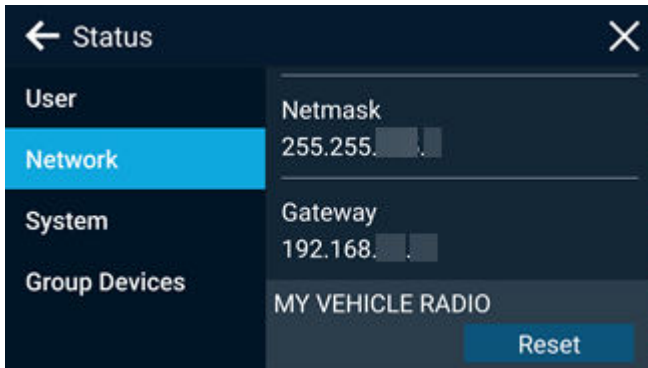
- M500
- **Integrated System** including the **SPS Type** and **SPS DHCP Server**, if applicable



NOTE: The **Integrated System** screen is only applicable to North American customers with integrated systems.

- Body-worn cameras in the system with their device names and IP addresses
- Radio (MikroTik Groove) that provides the Wi-Fi connection to the agency access point

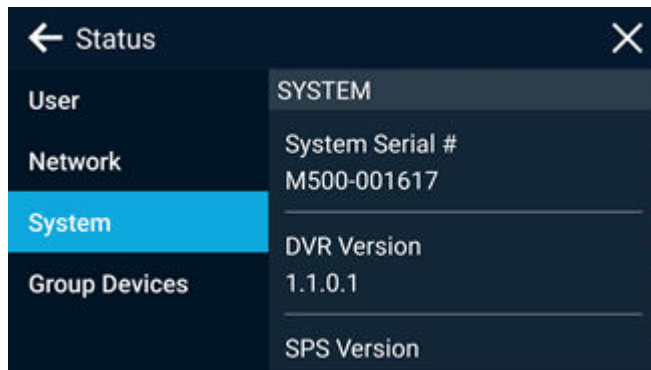
Touching **Reset** in the **My Vehicle Radio** section of the screen causes the system to reboot the MikroTik Groove radio.



CAUTION: You should only reboot the radio when instructed to by Motorola Solutions Technical Services.

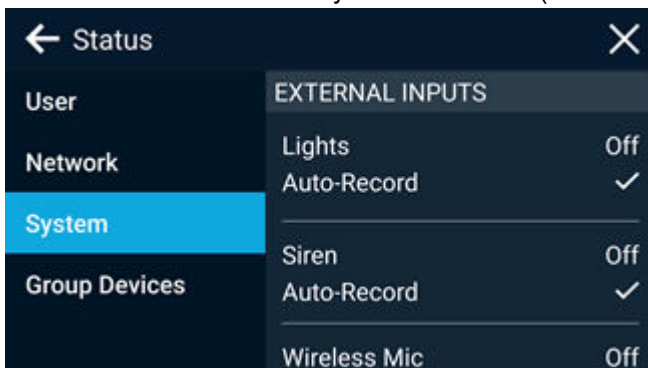
System

The **Status System** screen shows information about the M500 system, its peripherals, its external input connections, and its internal components.

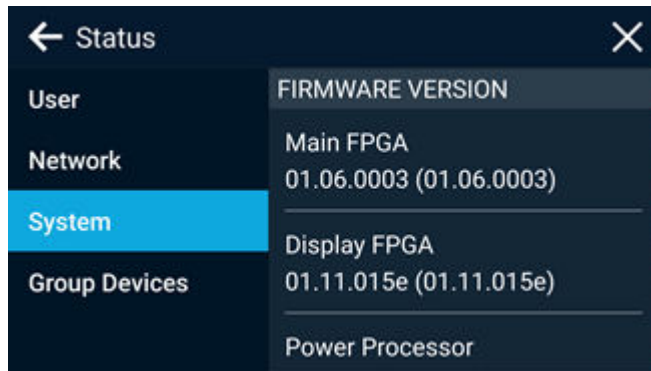


The information includes:

- Overall system versions for the DVR, Smart Power Switch (if present)(if present)
- GPS information including whether GPS has a lock
- List of connected external inputs ([External Inputs on page 23](#)) and whether they are set up to start a recorded event automatically when activated (**Auto-Record**)



- Firmware versions for the DVR internal components and connected cameras
In the **Firmware Version** section, the current version is shown first, the expected version is shown in parentheses.



If the current and expected versions do not match, that component may need to be upgraded. Contact your system administrator for instructions.

Transfer

The **Transfer** icon on the **Home** menu gives you access to the following information

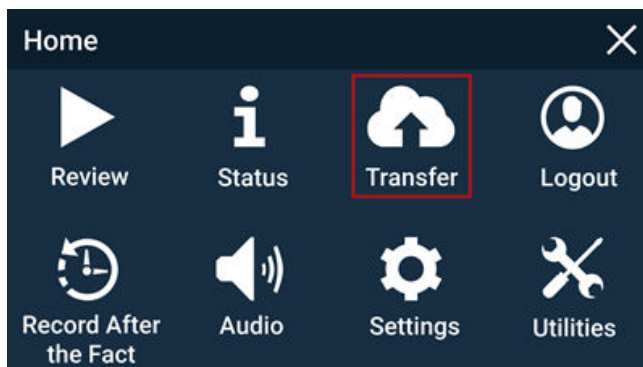
- **Transfer Status**
Provides a list of the recorded events that are uploading or waiting to upload from M500 to the agency evidence management system ([Transfer Status on page 69](#))
- **Transfer Event List**
Provides a list of the individual events that are in the process of uploading or waiting to upload ([Transfer Event List on page 70](#))

Accessing Transfer

Procedure:

1. From the Live View, touch **Home** on the Display Control Panel.

The **Home** menu appears.

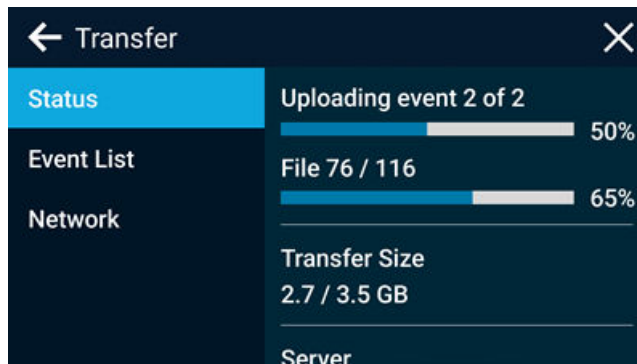


2. Touch **Transfer**.

The **Transfer Status** screen appears.

Transfer Status

The **Transfer Status** screen shows detailed information about the event upload status and Wi-Fi or LTE connection to the evidence management system.

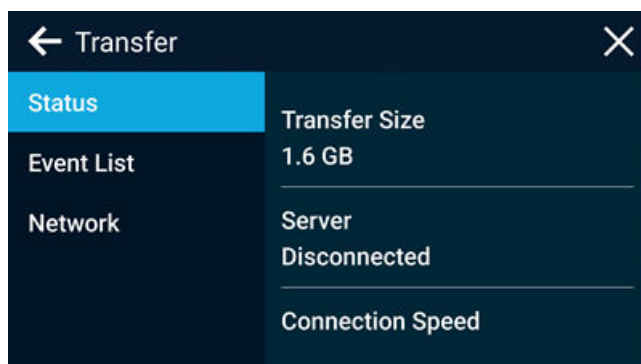


The **Transfer Status** screen gives you the following information:

- **Uploading event x of n:** Indicates which event is uploading (out of the total number of events to upload) and the percentage finished of the entire upload package
- **File x/n:** Indicates which file is in process (out of the total number of files in the uploading event) and the percentage finished of the entire event
- **Transfer Size:** Indicates how much of the upload package has been uploaded out of the entire upload package
- **Server:** Shows the status of the connection (Connected, Disconnected) to the evidence management system (using Wi-Fi or LTE)
- **Connection Speed:** Indicates the upload speed of the Wi-Fi or LTE connection to the evidence management system

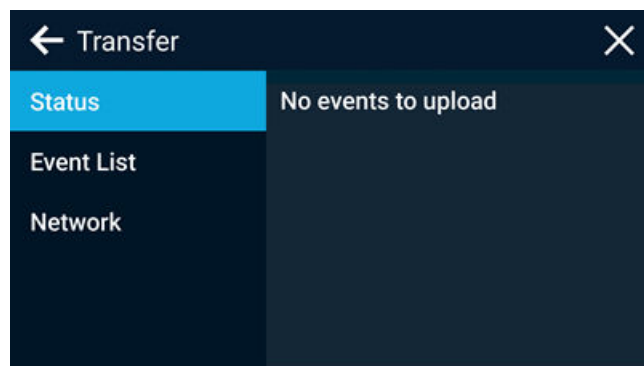
Disconnected

If the status of the **Server** connection is **Disconnected** and there are events on the M500 to upload, the **Transfer Status** screen shows the number of events waiting to be uploaded and the size of the upload package.



No Events to Upload

If there are no events on the M500 to upload, the **Transfer Status** screen shows **No events to upload**.



Transfer Event List

The **Transfer Event List** screen shows a list of the events that have been uploaded, are uploading currently, or are waiting to upload to the agency evidence management system.



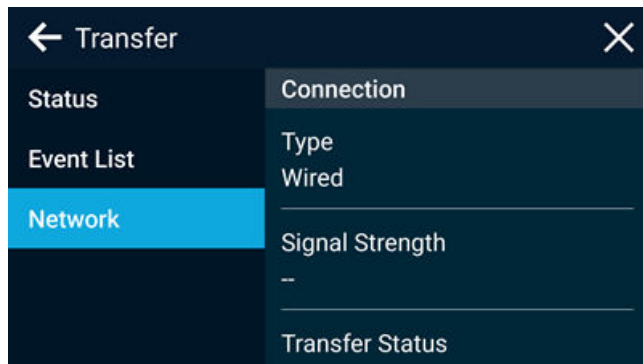
 **NOTE:** If the system determines that an event is too large for efficient upload to the agency evidence management system (in duration, size, or number of files), the system splits the event into smaller events. These smaller events are assigned the same category and are linked together. See [Split Event Indicator on page 36](#) for more information.

The **Transfer Event List** screen gives you the following information about the listed events:

- **Date heading:** Indicates the date the events were recorded
- **Category:** Shows the category assigned to the event
- **Time:** Shows the start time of the event
- **Duration, in parentheses:** Shows the length of the event in hours:minutes:seconds
- : Indicates the event has been uploaded to the agency evidence management system
- : Indicates the event has been offloaded to the USB drive, but has not yet been verified as uploaded to the agency evidence management system
- **Critical** or routine: Bold text indicates the event is critical; normal text indicates the event is routine

Transfer Network

The **Transfer Network** screen shows the network and the connection to the agency evidence management system.



The **Transfer Network** screen gives you the following information about the network connection:

- **Connection**
 - **Type:** Indicates the type of network the device is connected to (**Wi-Fi**, **LTE**, **Wired**, or -- when not connected)
 - **Signal Strength:** Indicates the signal strength only if the device is connected to Wi-Fi (**Weak**, **Fair**, **Good**, **Excellent**)
 - **Transfer Status:** Indicates the status of the system connection to the agency evidence management system (**Connected** if connected to the agency evidence management system, **Uploading** if uploading evidence, or **Disconnected** if not connected)
- **Upload Server:** Depending on your device configuration, this section indicates whether the Primary Domain Name or Secondary Domain Name is connected
- **DNS Server:** Shows the IP addresses associated with your agency primary and secondary DNS servers

Audio

The **Audio** icon on the **Home** menu gives you access to:

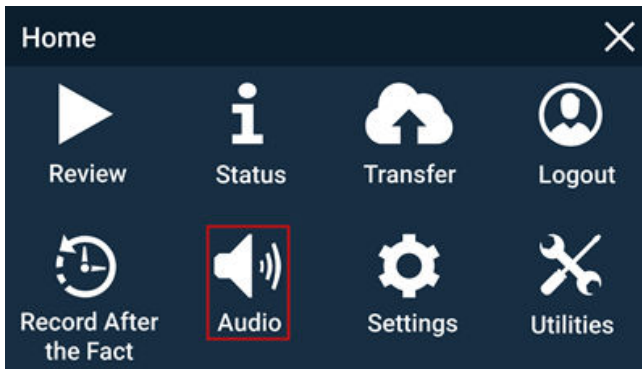
- Volume controls for M500 tones including event and video review
- Input level controls for the wired, in-vehicle microphones in the M500 system ([Mic Input on page 72](#))

For more information, see [Muting Microphones on page 19](#).

Accessing Audio

Procedure:

1. From the Live View, touch **Home** on the Display Control Panel.
The **Home** menu appears.



2. Touch **Audio**.

The **Audio Volume** screen opens.

Volume

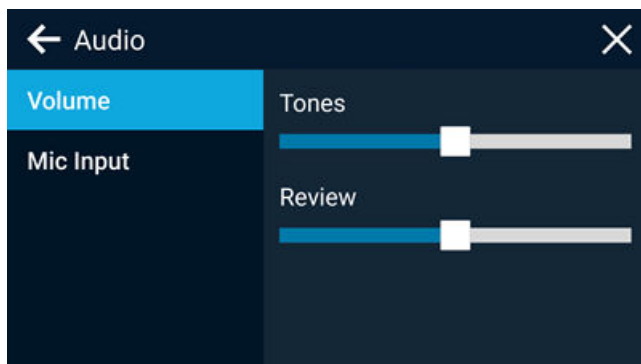
The **Audio Volume** screen provides volume controls for:

- **Tones**
This slider controls the volume for all of the system tones.
- **Review**
This slider controls the volume for each individual or all of the players:
 - Event review player ([Review by Event on page 33](#))
 - Near real-time video review player ([Review by Time on page 39](#))
 - Record-After-the-Fact® preview player ([Record-After-the-Fact Events on page 42](#))



NOTE: You can also adjust the player volume on each individual **Review** player.

You adjust the volume by dragging the slider. Dragging the slider to the right increases the volume.



The volume levels you set for **Tones** and **Review** are saved to the agency evidence management system as part of your officer preferences. Each time you log in to an M500, your set preferences are restored.

See [Officer Preferences on page 19](#) for information about all your officer preference settings.

Mic Input

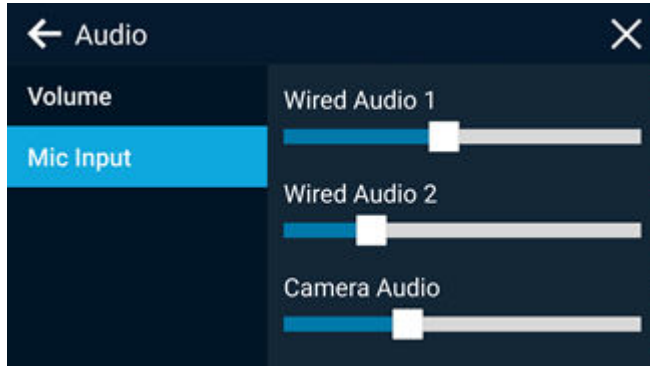
The M500 in-car video system uses three wired, in-vehicle microphones: two stand-alone cabin microphones and one integrated microphone in the M5P camera.

See [Microphones on page 106](#) for more information on the microphones in the M500 system.

The **Audio Mic Input** screen provides input level controls for the wired (in-vehicle) microphones in the M500 system.

You access the microphone input level by touching **Mic Input** on the **Audio Volume** screen.

You adjust the microphone input level by dragging the slider. Dragging the slider to the right increases the input level.



 **NOTE:** The **Camera Audio** microphone input is only available when the M5P is connected to the cabin camera port on the back of the M500.

See [Checking the Input Levels on page 73](#) for help setting the input levels.

 **NOTE:** Adjust the **Camera Audio** microphone input separately from the **Wired Audio** microphone inputs. The audio characteristics of the stand-alone cabin microphones (**Wired Audio 1** and **2**) are different than the integrated M5P camera microphone (**Camera Audio**).

The microphone input levels you set are saved to the agency evidence management system as part of your officer preferences. Each time you log in to an M500, your set preferences are restored.

See [Officer Preferences on page 19](#) for information about all your officer preference settings.

Checking the Input Levels

You can use the **Review by Time** feature to check your microphone input levels.

Procedure:

1. Start a recorded event ([Starting an Event Manually on page 24](#)) that includes audio input to your in-vehicle microphones.
2. Open the **Review by Time** player ([Review by Time on page 39](#)).
3. Move the player back in time a few seconds.
4. Listen to the audio level.
5. Open the **Audio Mic Input** screen ([Mic Input on page 72](#)).
6. Use the controls on the **Audio Mic Input** screen to adjust the levels as needed.
7. Re-open the **Review by Time** player.
8. Move the playback to a time after you adjusted the input levels.
9. Repeat steps 4 through 8 as needed.
10. Stop the recorded event.

Settings

The **Settings** icon on the **Home** menu gives you access to some system settings including:

- Time zone, date, and time ([Date & Time on page 74](#))
- Camera power and recording LEDs enable/disable ([Disabling the Camera LEDs on page 77](#))
- Automatic License Plate Recognition (ALPR) bounding boxes enable/disable ([Enabling ALPR on page 78](#))



NOTE: The ALPR screen will only be present if the feature has been enabled by your administrator in the agency evidence management system.

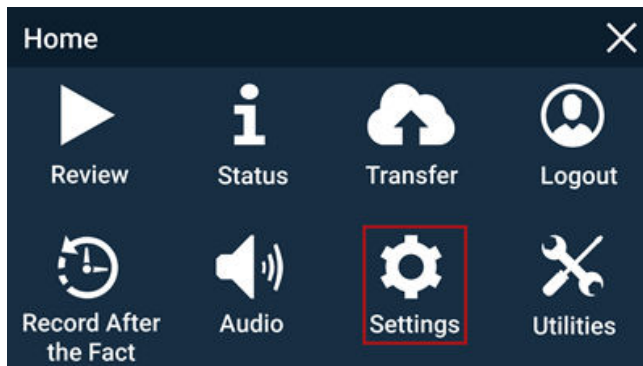
- Display options including brightness, **Dark Mode**, **Hide Info**, and **Screensaver** ([Display Settings on page 78](#))

Accessing the System Settings

Procedure:

1. From the Live View, press **Home** on the Display Control Panel.

The **Home** menu appears.

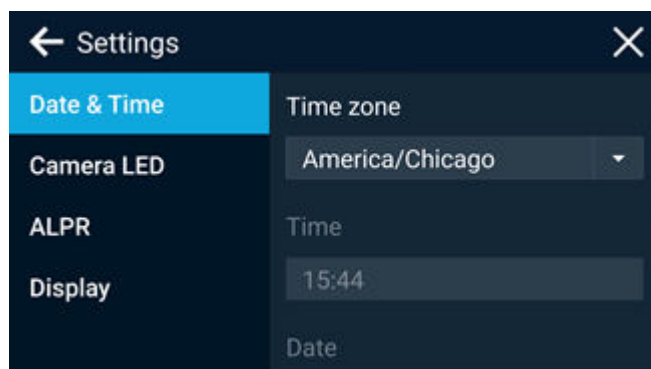


2. Touch **Settings**.

The **Settings** screen opens.

Date & Time

The **Settings Date & Time** screen shows you the time zone where your M500 is operating, and the current time and date.

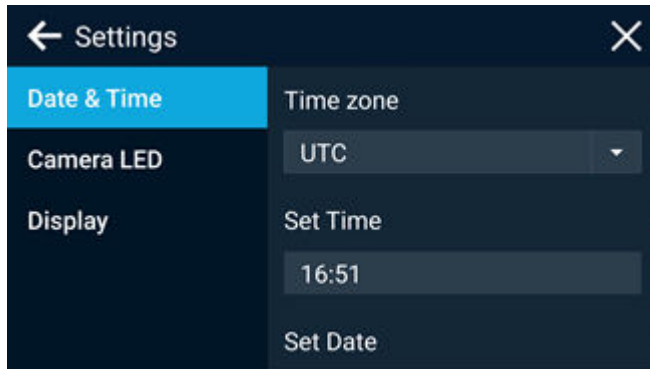


If you have permissions, you can update the **Time zone** if needed ([Setting the Time Zone on page 75](#)).

Typically, the **Time zone** is set from the configuration ([Configuration Deployment on page 109](#)), and the **Time** and **Date** are set automatically through communication with the agency evidence management system or by the device GPS.

If the M500 has not yet connected with the agency evidence management system or detected a GPS signal, you can manually set the **Time** and **Date** if you have permissions. The **Set Time** and **Set Date** fields are active if you are allowed to manually set them.

 **NOTE:** This scenario can happen at initial installation or if the cell battery that preserves the time while the system is powered down dies.



The date and time you set manually is overridden as soon as the M500 receives a date and time through its connection with the agency evidence management system or detects a GPS signal.

Setting the Time Zone

Procedure:

To set the **Time Zone**, from the **Settings Date & Time** screen, touch the **Time zone** field and select the appropriate time zone from the drop-down list.



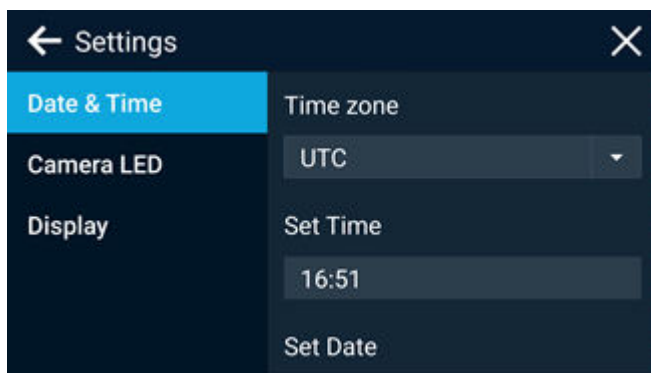
 **NOTE:** You can only adjust the **Time Zone** if you have permissions.

Setting the Time and Date

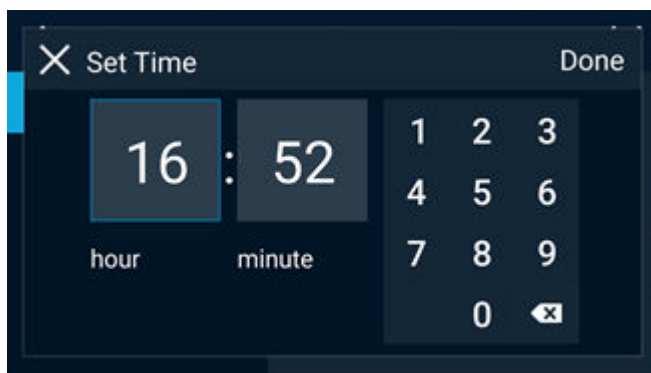
Procedure:

1. From the **Settings Date & Time** screen with active fields, touch the **Set Time** field.

 **NOTE:** You can only adjust the date and time if you have permissions.

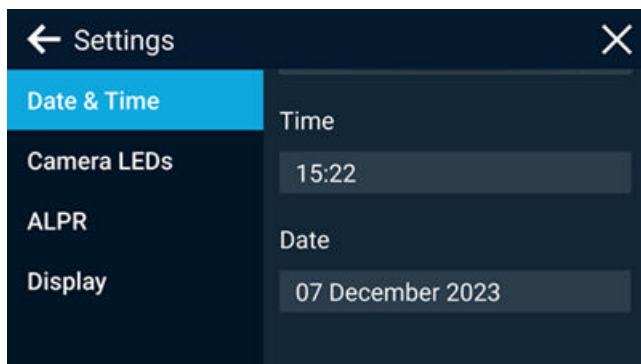


The **Set Time** dialog box appears.



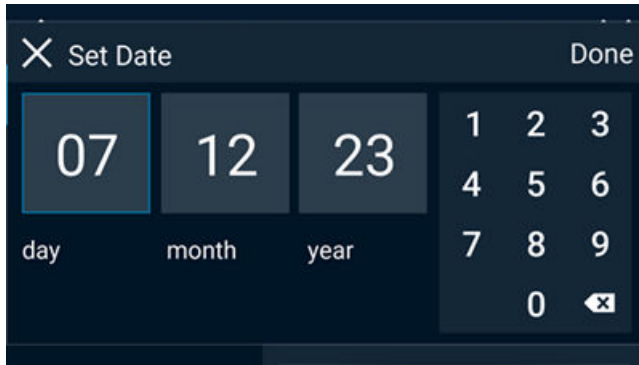
2. Touch a field (**hour** or **minute**) then use the number pad to enter the time.
3. Touch **Done**.

The **Settings Date & Time** screen reappears.



4. Touch the **Set Date** field.

The **Set Date** dialog box appears.



5. Touch a field (**day**, **month**, or **year**) then use the number pad to enter the date.
6. Touch **Done**.

Result:

 **NOTE:** The date and time you set manually is overridden as soon as the M500 receives a date and time from the agency evidence management system or finds a GPS signal.

Disabling the Camera LEDs

The **Settings Camera LED** screen allows you to disable the **Power LED** and **Record LED** on the following cameras:

- M5F ([M5F Front Camera on page 103](#))
- M5R ([M5R Camera on page 104](#))
- M5E ([M5E Camera on page 105](#))

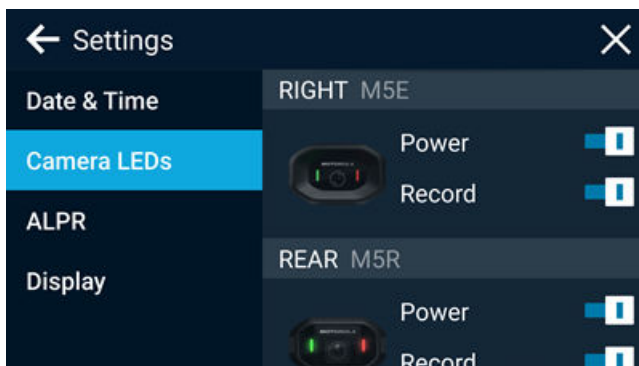
 **NOTE:** All LEDs are enabled by default.

 **NOTE:** All changes to camera LEDs are independent.

Procedure:

1. To disable (or re-enable) the camera LEDs, from the **Settings** screen, touch **Camera LEDs**.

The **Settings Camera LEDs** screen appears.



2. Touch the toggle switch for the LED you want to disable .

Result:

The state (enabled or disabled) you set for each camera LED is saved to the agency evidence management system as part of your officer preferences. Each time you log in to an M500, your set preferences are restored.

See [Officer Preferences on page 19](#) for information about all your officer preference settings.

Enabling ALPR

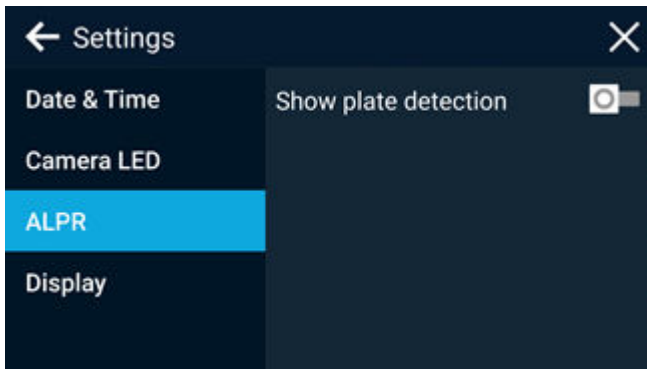
The **Settings ALPR** (automatic license plate recognition) screen allows you to enable the license plate detection indicators (blue boxes and white text). The indicators are disabled by default.

 **NOTE:** The **Settings ALPR** screen is present only when the feature has been enabled by your administrator in the agency evidence management system and you have permissions to enable/disable license plate detection indicators.

Procedure:

1. To enable (or re-disable) the plate detection indicators, from the **Settings** screen, touch **ALPR**.

The **Settings ALPR** screen appears.



2. Touch the **Show plate detection** toggle switch to enable  the plate detection indicators.

Result:

 **NOTE:** Turning off the **Show plate detection** toggle switch does not turn off the ALPR feature. It only removes the indicators from the Live View screen.

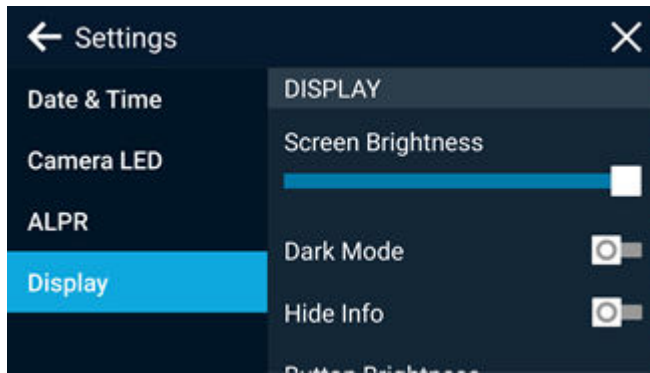
The state (disabled or enabled) you set for the license plate detection indicators is saved to the agency evidence management system as part of your officer preferences. Each time you log in to an M500, your set preferences are restored.

See [Officer Preferences on page 19](#) for information about all your officer preference settings.

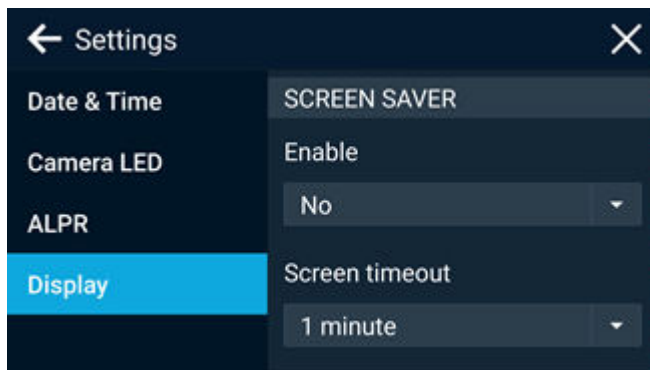
Display Settings

The **Settings Display** screen provides the same options that are available from the **Display** button on the Control Panel:

- **Screen Brightness** ([Using Screen Brightness on page 101](#))
- **Dark Mode** ([Enabling Dark Mode on page 101](#))
- **Hide Info** ([Enabling Hide Info on page 102](#))
- **Button Brightness** ([Using Button Brightness on page 102](#))



The screen also provides a section, when you scroll down, to enable a **Screen Saver** ([Screensaver on page 79](#)).



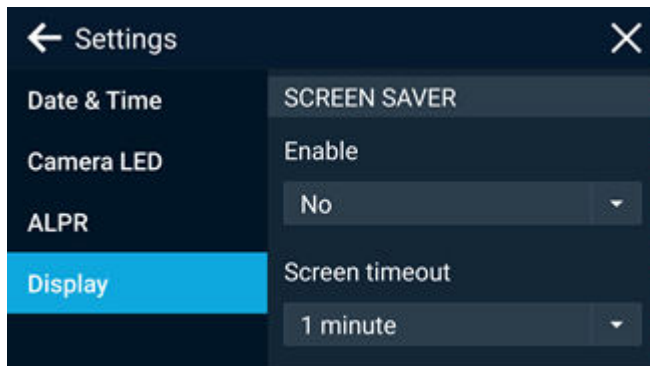
Screensaver

The **Settings Display** screen, **Screen Saver** section, allows you to enable a screen saver on the M500 Display Screen.

When the screen saver is enabled, after the set timeout period expires, the screen illumination turns off and the Control Panel buttons dim to their lowest visible setting.

If you interact with the display (touch the screen or press the buttons) during the timeout period, the **Screen timeout** timer resets. If you interact with the display while the screen saver is active, the screen saver is deactivated and the screen and buttons return to preference-level brightness. The timer resets.

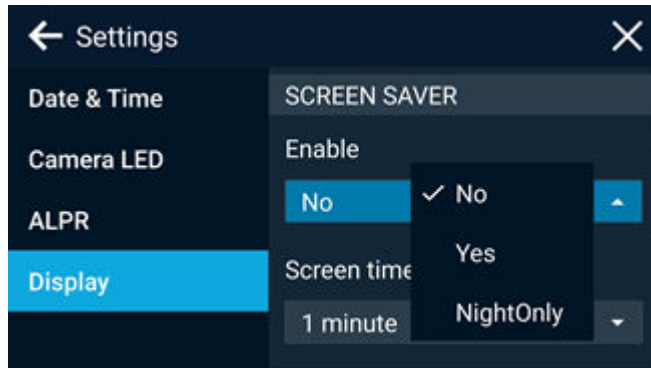
If any system messages pop up while the screen saver is active, the screen saver is deactivated (screen and buttons return to preference-level brightness). The timer resets when you dismiss the system message.



You can choose the following options for a screen saver:

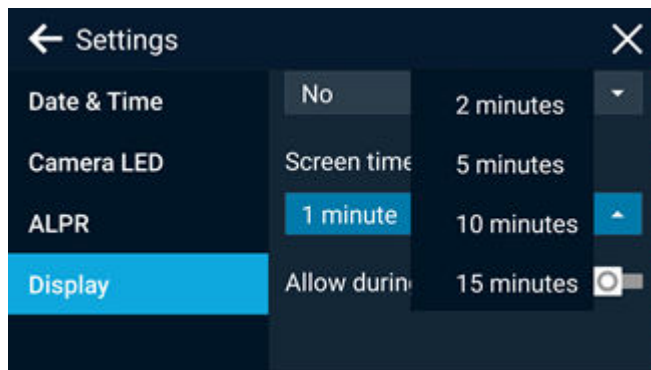
- **Enable:** Yes, No, or NightOnly

You can set the screen saver to be enabled or not, or to only be enabled during the time period 8 PM to 8 AM.



- **Screen timeout: 1, 2, 5, 10, or 15 minutes**

You can set the timer to activate the screen saver after 1, 2, 5, 10, or 15 minutes of inactivity (no screen touches, button presses, or system pop-up messages).



- **Allow during event: Disabled or enabled**

You can set the screen saver to activate (or not) when the system is in the middle of recording an event. The default is not allowed (disabled).

The options you set for a screen saver are saved to the agency evidence management system as part of your officer preferences. Each time you log in to an M500, your set preferences are restored.

See [Officer Preferences on page 19](#) for information about all your officer preference settings.

Utilities

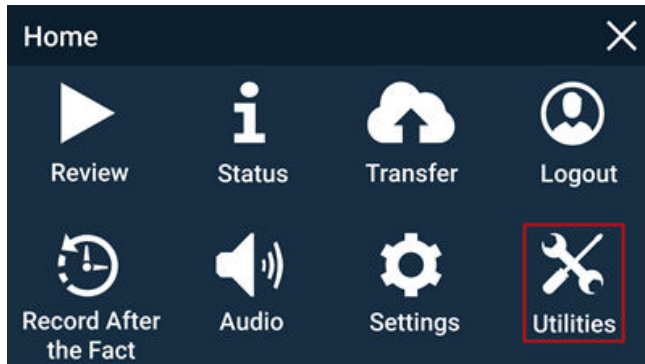
The **Utilities** icon on the **Home** menu gives you access to tools that can help you maintain and troubleshoot issues with your M500 system, including:

- Active system errors and their details ([System Errors on page 81](#))
- Reformatting the USB drive ([Formatting the USB Drive on page 83](#))
- System diagnostics information ([Diagnostics on page 84](#))
- Generating a state capture ([Generating a State Capture \(Utilities\) on page 87](#))
- Adjusting the cabin camera region of interest for detection ([Adjusting the Detection Area on page 89](#))
- Re-exporting events to the USB drive ([Re-Export Events to the USB Drive on page 91](#))
- Deleting events from the device ([Delete Events on page 93](#))

Accessing Utilities

Procedure:

1. From the Live View, press **Home** on the Display Control Panel.
The **Home** menu appears.



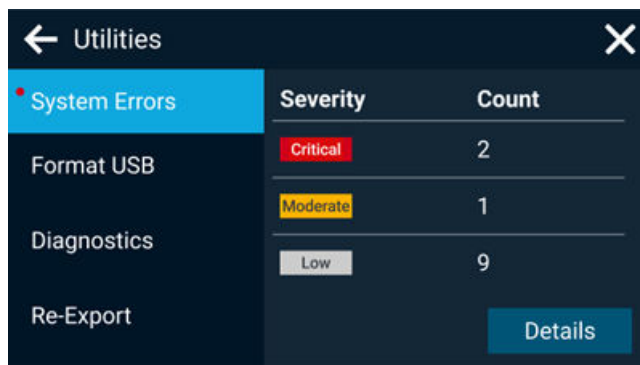
2. Touch **Utilities**.
The **Utilities** screen opens.

System Errors

The **Utilities System Errors** screen gives you a list of active error conditions in the system.

On the **Utilities System Errors** screen, the active system errors are organized into three categories based on severity of the error:

- **Critical**
- **Moderate**
- **Low**

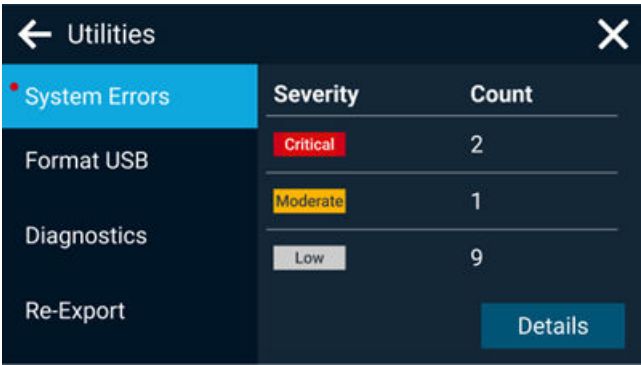


 **NOTE:** A red notification dot appears next to the **System Errors** tab when one or more errors are active in the system.

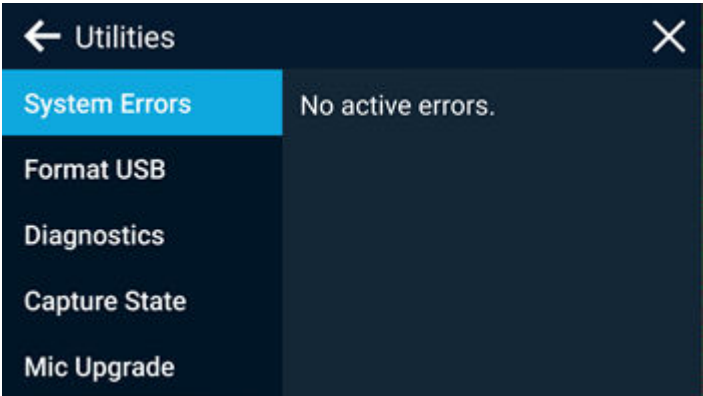
Accessing System Errors

Procedure:

1. From the **Utilities** screen, touch **System Errors**.
The **Utilities System Errors** screen appears.
If there are active errors in the system, the errors table appears.



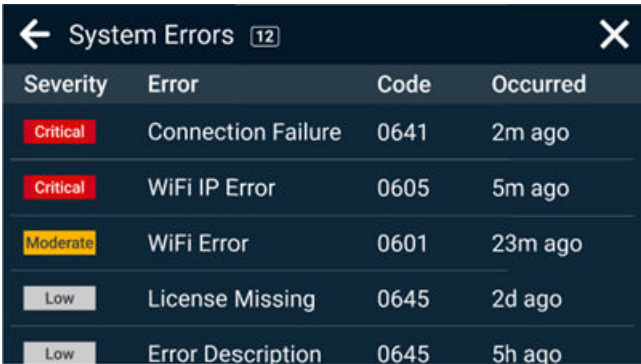
If there are no active errors in the system, the **No active errors** screen appears.



2. Touch the **Details** button.

The **System Errors** screen appears.

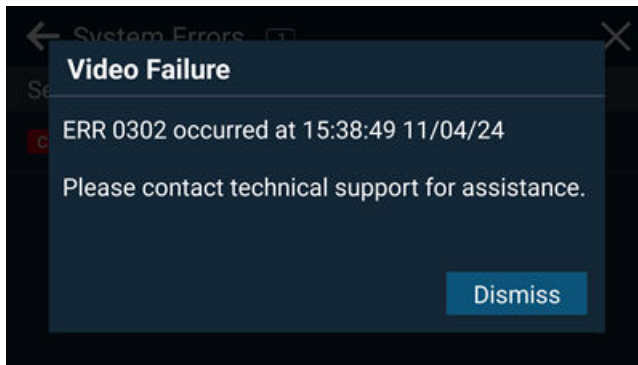
 **NOTE:** The total number of errors appears next to **System Errors**. The screen refreshes automatically.



 **NOTE:** The errors are organized by level of severity.

3. Touch the error to view details about that error.

The error detail dialog box appears.



 **NOTE:** The error details may give you additional information about the error and how to resolve it.

4. Touch **Dismiss** to return to the **System Errors** screen.

Formatting the USB Drive

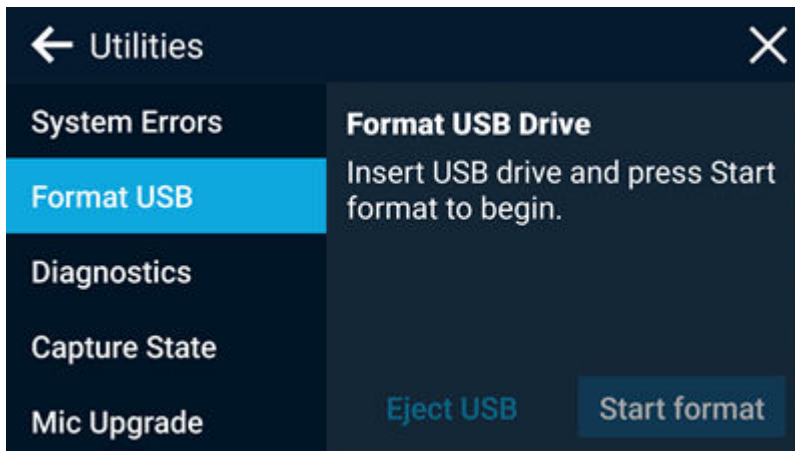
The **Utilities Format USB** screen allows you to reformat a USB drive.

 **WARNING:** Reformatting a USB drive causes all data on the drive to be lost.

Procedure:

1. From the **Utilities** screen, touch **Format USB**.

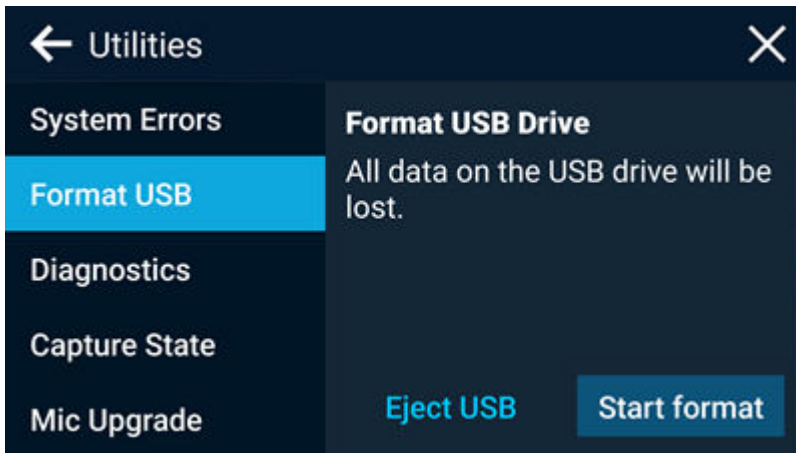
The **Utilities Format USB** screen appears.



2. Insert the USB drive you want to reformat.

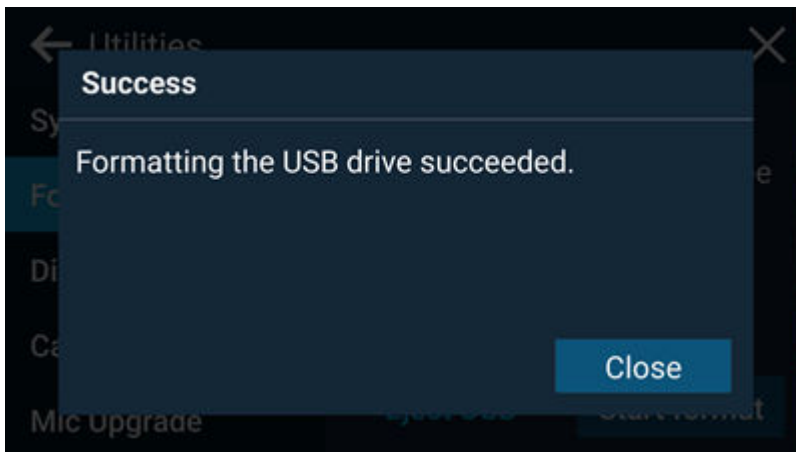
 **NOTE:** If you need to remove a USB drive before you insert one, touch **Eject USB** and remove it safely ([Ejecting the USB Drive Using Utilities on page 49](#)).

The **Format USB Drive** options become active.



3. Touch **Start format**.

When the system has finished reformatting the USB drive, the **Success** message appears.



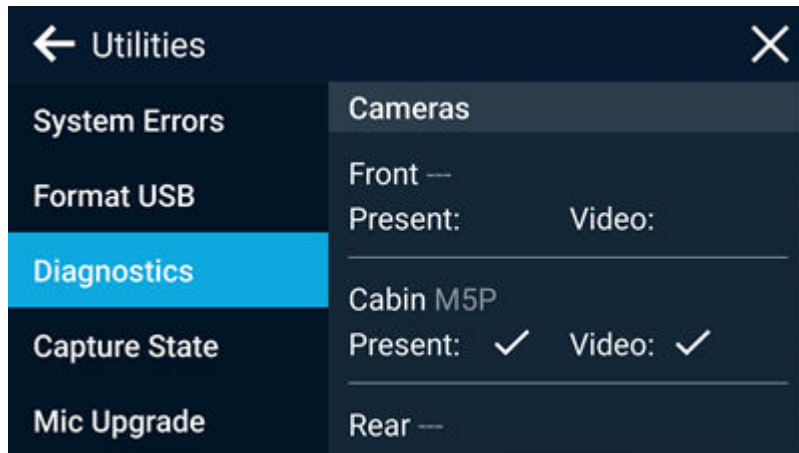
4. Touch **Close**.
5. Remove the reformatted USB drive from the M500.

Diagnostics

The **Utilities Diagnostics** screen gives you access to system diagnostics information including:

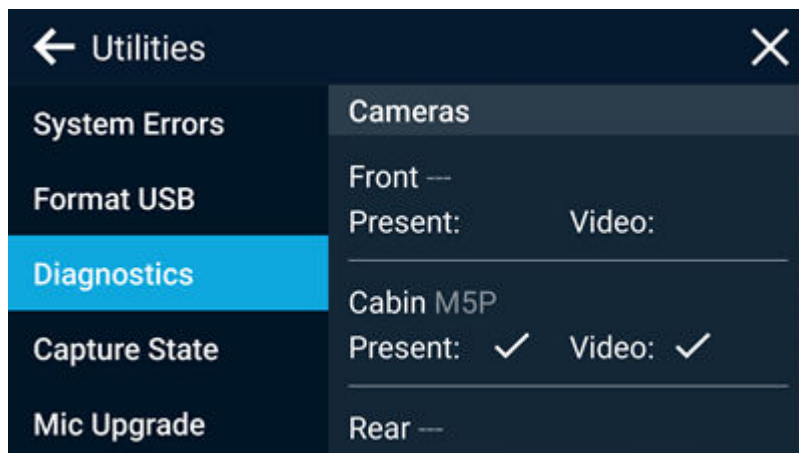
- System camera details ([Cameras on page 85](#))
- GPS connection details ([GPS on page 85](#))
- System temperatures ([Temperature on page 86](#))
- System voltages ([Voltage on page 86](#))

You can scroll down to see each section of the **Utilities Diagnostics** screen.



Cameras

The **Cameras** section of the **Utilities Diagnostics** screen shows every camera that can be in the M500 system. When a camera is present in the system, the model name (for example, **M5P**) appears next to the camera port (for example, **Cabin**) and a check mark appears.



If there are no check marks for a camera, no camera of that type is connected to the system or that camera has not been discovered by the system.

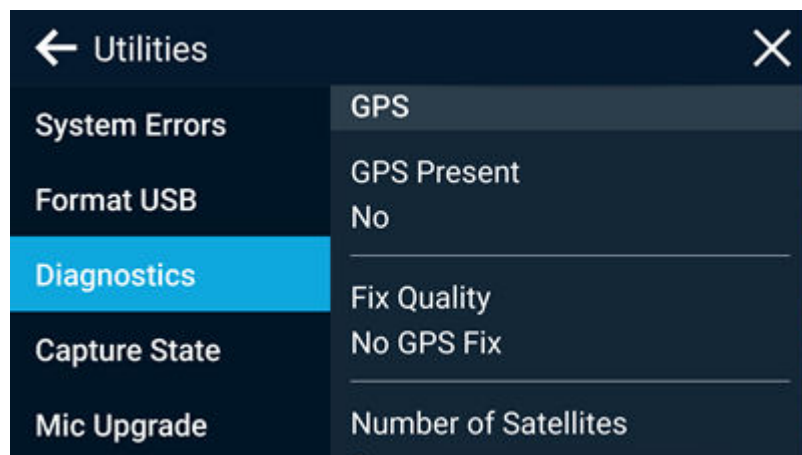


NOTE: If you believe a camera is connected to the system that is not showing as present, contact your system administrator.

GPS

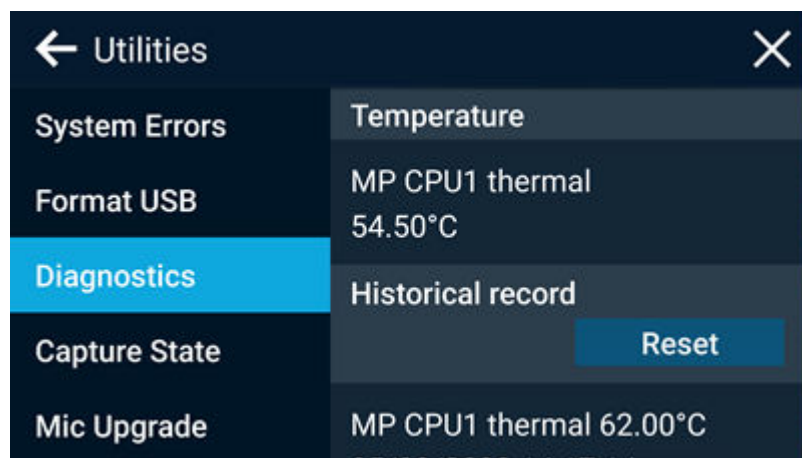
The **GPS** section of the **Utilities Diagnostics** screen shows:

- GPS presence (or not)
- Information about the quality of the current GPS fix including the number of satellites, if GPS is present
- Current GPS location of the vehicle, if GPS is present



Temperature

The **Temperature** section of the **Utilities Diagnostics** screen lists the current internal and historical high internal temperatures of the M500.

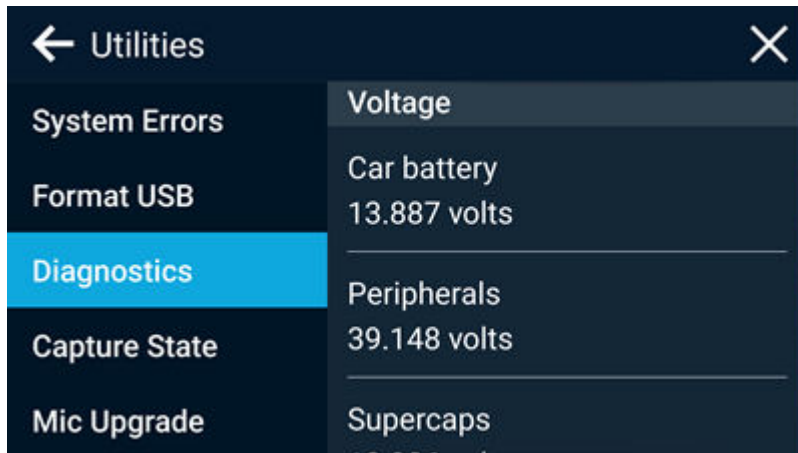


You can clear the **Historical record** temperature of the device by touching **Reset**.

Voltage

The **Voltage** section of the **Utilities Diagnostics** screen shows the current voltage for:

- **Car battery**
- System **Peripherals** (the display and system cameras)
- **Supercaps** (short-term backup in case of a sudden loss of power to the system)



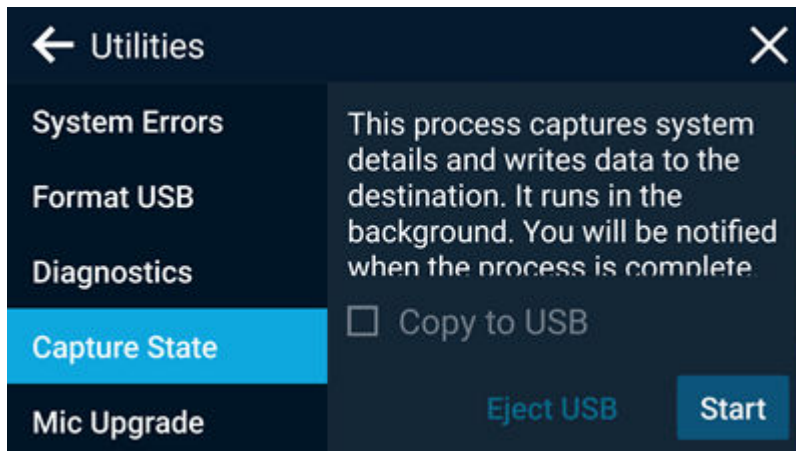
Generating a State Capture (Utilities)

The **Utilities Capture State** screen allows you to generate a state capture of the M500 system. The state capture automatically uploads to the agency evidence management system. You can also copy it to a USB drive.

Procedure:

1. From the **Utilities** screen, touch **Capture State**.

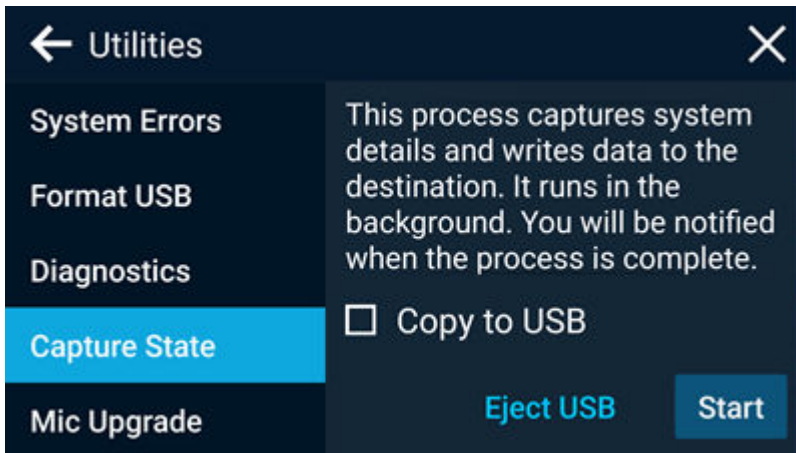
The **Utilities Capture State** screen appears.



2. (Optional) Insert a USB drive if you want to copy the state capture to a USB drive.

 **NOTE:** If you need to remove a USB drive before you insert one, touch **Eject USB** and remove it safely ([Ejecting the USB Drive Using Utilities on page 49](#)).

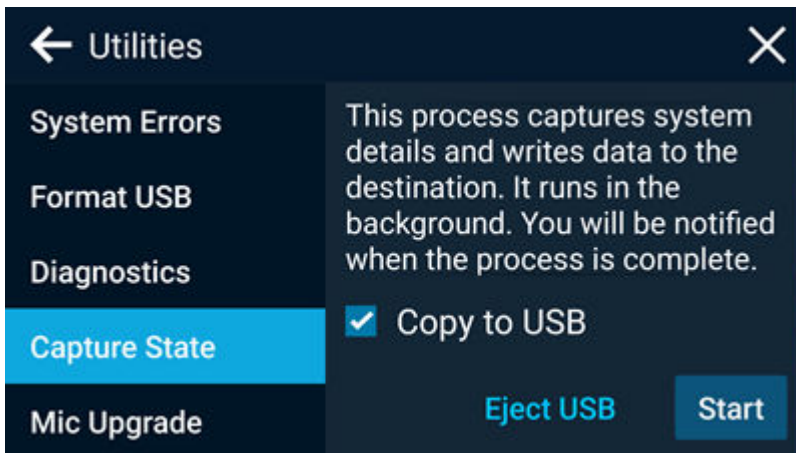
The **Copy to USB** check box becomes active.



3. (Optional) Touch the **Copy to USB** check box to check it.

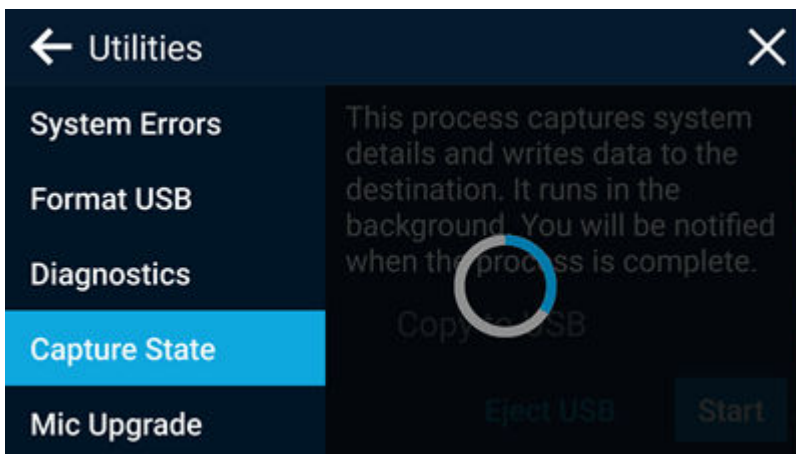


NOTE: A copy of the state capture is always automatically uploaded to your evidence management system location.



4. Touch **Start**.

The system begins to generate the state capture.

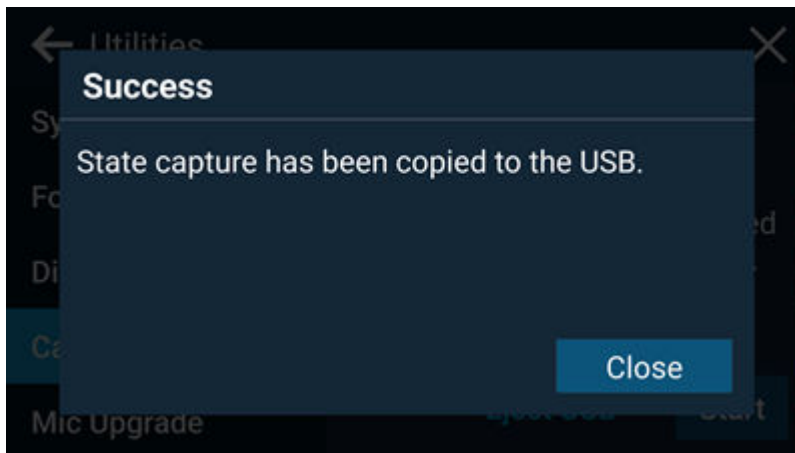


The process may take a few minutes.



NOTE: You can navigate to other parts of the system while the state capture is being generated.

When the state capture is finished, the **Success** message appears.



5. Touch **Close**.
6. If applicable, touch **Eject USB**, then remove the USB drive ([Ejecting the USB Drive Using Utilities on page 49](#)).

Result:

 **NOTE:** State captures can also be requested remotely through the agency evidence management system. See your evidence management system documentation for more information.

Detection

The M500 can detect the human form within a camera view if enabled by your administrator in the agency evidence management system. If enabled, the camera turns on automatically when a human form appears in the detection area.

You can set the detection area on the M500 if you have the correct permissions. See [Adjusting the Detection Area on page 89](#) for more information.

If **Detection** is enabled by your administrator and the camera recording state is set to **Off**, the camera state changes to **On** when a human form is detected during an event. The camera view is included in the recorded event upload.

If **Detection** is enabled by your administrator and the camera recording state is set to **Background**, the camera state remains **Background** when a human form is detected. The camera view upload status changes from excluded to **Include in upload** and the camera view is included in the recorded event upload.

See [Event Tags Cameras on page 30](#) and [Background Recording on page 99](#) for more information.

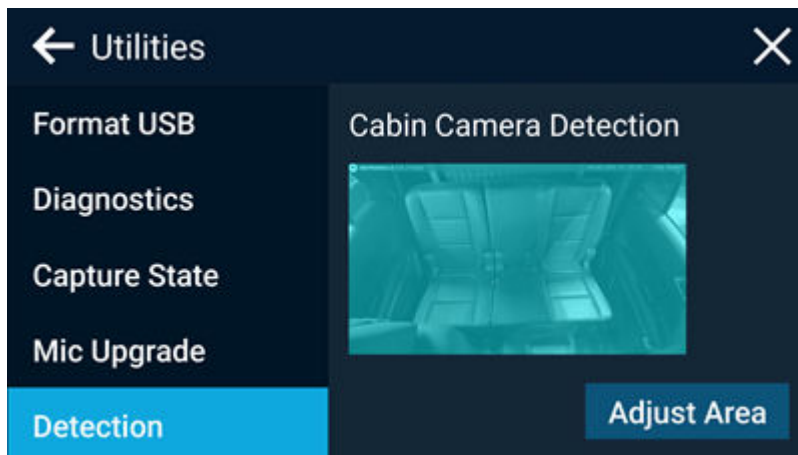
 **NOTE:** If you have permissions, you have the option to include a backgrounded camera view during event tagging. See [Event Tags Cameras on page 30](#) for more information.

Adjusting the Detection Area

The **Detection** screen allows you to view and adjust the detection area.

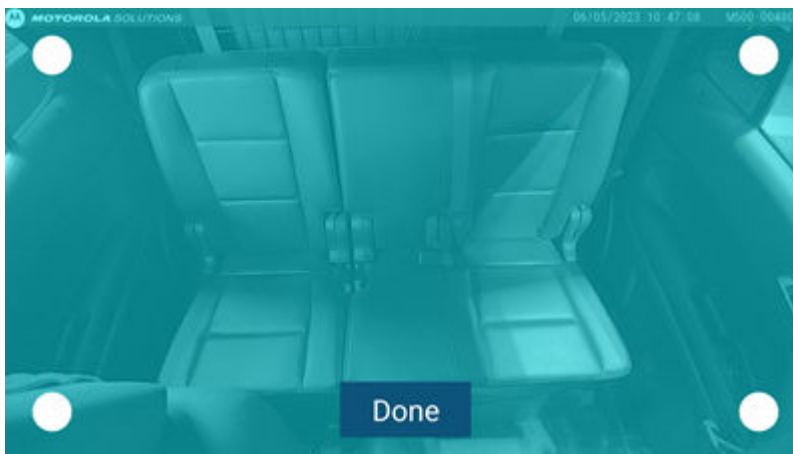
Procedure:

1. From the **Utilities** screen, touch **Detection**.
The **Utilities Detection** screen appears.

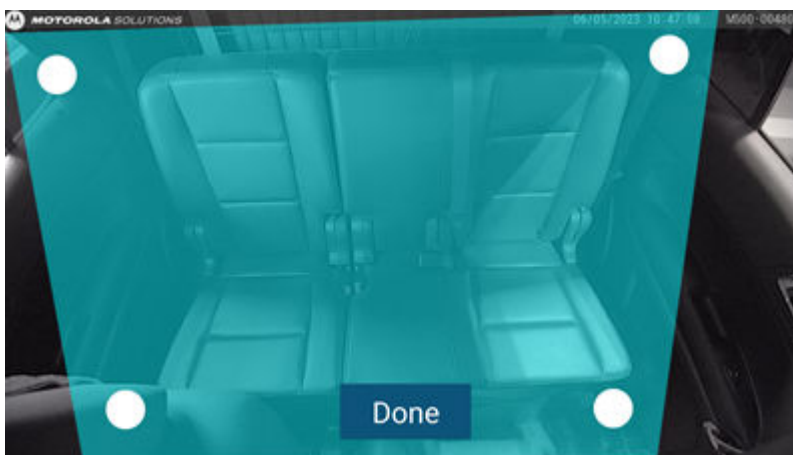


2. Touch **Adjust Area** to adjust the detection area.
3. Enter the **Administrator** password if prompted.

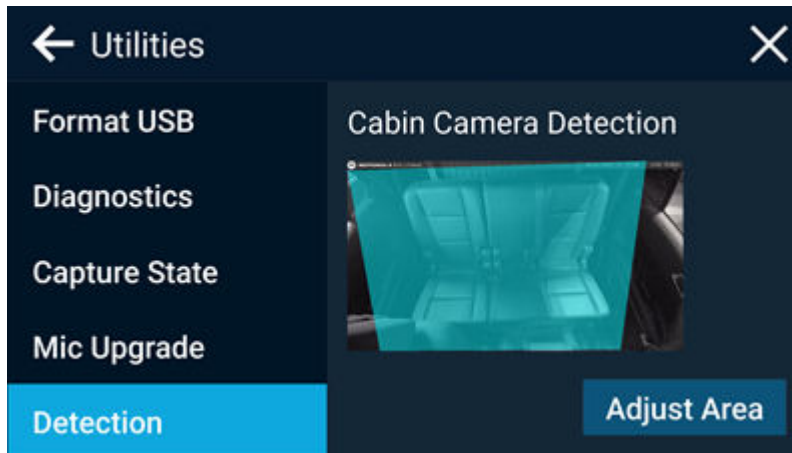
The adjust area screen appears.



4. Touch and drag the four white circular points on the screen to adjust the detection area.



5. Touch **Done**.
- The **Detection** screen appears showing the new detection area.



Re-Export Events to the USB Drive

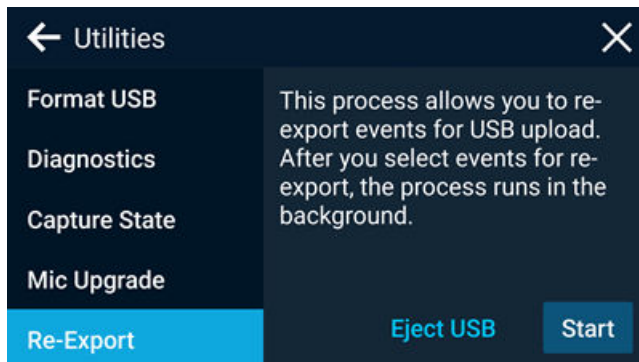
Your administrator can manually select one or more events on your M500 for re-export to the USB drive. Events that have already been exported successfully or unsuccessfully can be selected from the event list and then re-exported to the USB drive.

 **NOTE:** This feature is protected by an administrator password.

Manually Re-Exporting Events to the USB Drive

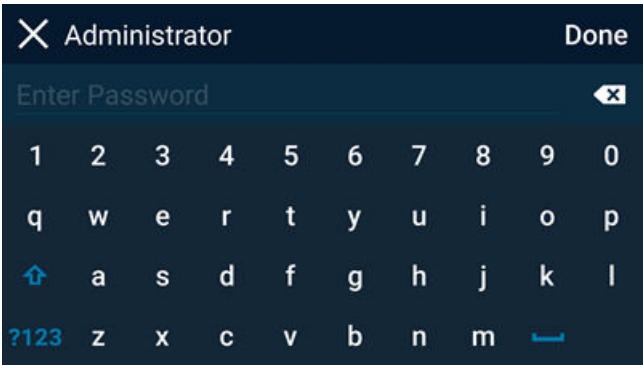
Procedure:

1. From the **Utilities** screen, touch **Re-Export**.
2. Touch **Start** to re-export events.



 **NOTE:** If applicable, you can touch **Eject USB** to remove the USB drive. If there is no USB drive present in the system, the **Eject USB** option is not available.

The **Administrator** screen appears.



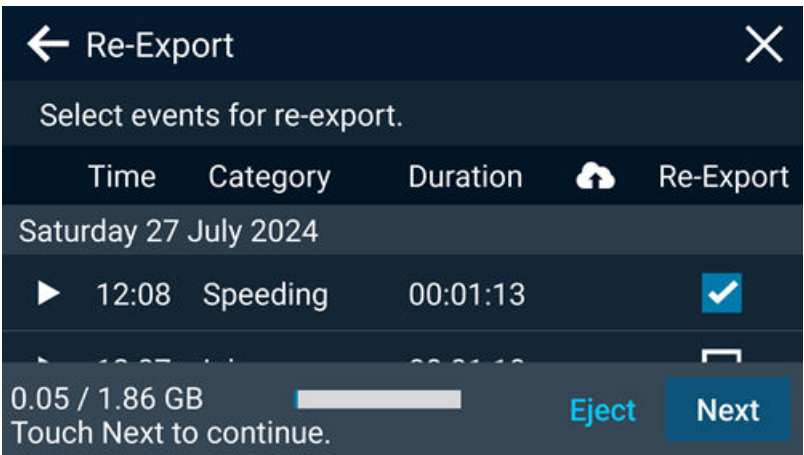
3. Enter the administrator password and touch **Done**.
The **Re-Export** screen appears.



 **NOTE:** To review the event before re-exporting it, touch the play icon on the left side of the screen.

4. Touch the check box next to the events you want to re-export.
The bottom action bar appears

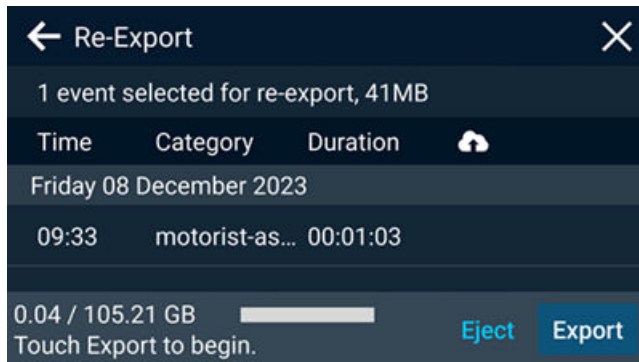
 **NOTE:** You can touch the play icon to review an event before you re-export it.



 **NOTE:** If a USB drive is inserted, a bar appears at the bottom of the screen showing storage space.

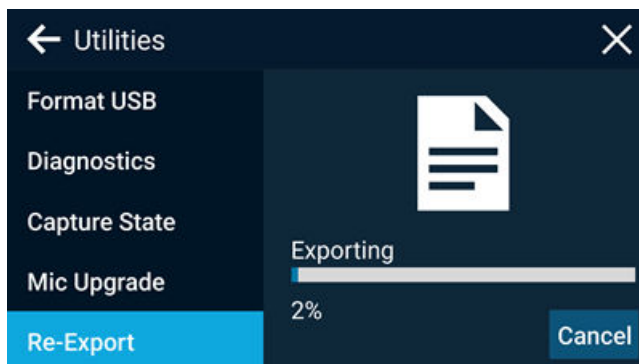
5. Touch **Next** to continue.

The **Re-Export** event review screen appears.



6. If applicable, insert the USB drive into the M500 to re-export the selected events.
7. Touch **Export**.

The Export Progress screen appears



8. After the system is done **Exporting**, touch **Done**.

Delete Events

Your administrator can manually select one or more events on your M500 to delete.

 **NOTE:** This feature is protected by an administrator password.

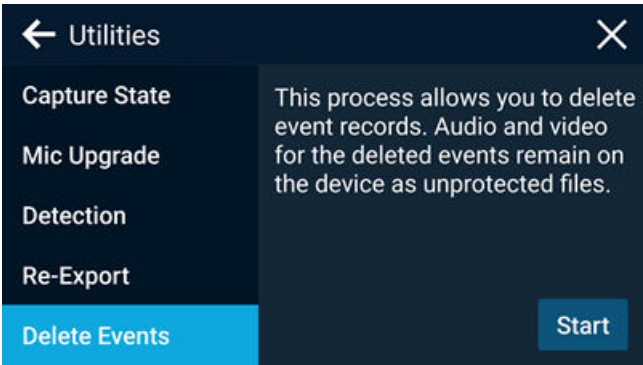
When your administrator deletes an event, the event record is removed from the event database, but it does not delete the audio and video files. The remaining audio and video files on the device are unprotected but can be used in an RATF (Record-After-the-Fact) event. Deleted events are reported in the metadata log for forensic tracking and auditing purposes.

 **IMPORTANT:** **Review** and **Re-Export** are not available during the event deletion process.

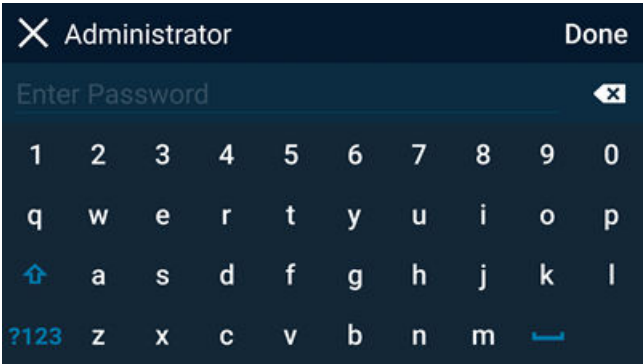
Manually Delete Events

Procedure:

1. From the **Utilities** screen, touch **Delete Events**.
2. Touch **Start**.

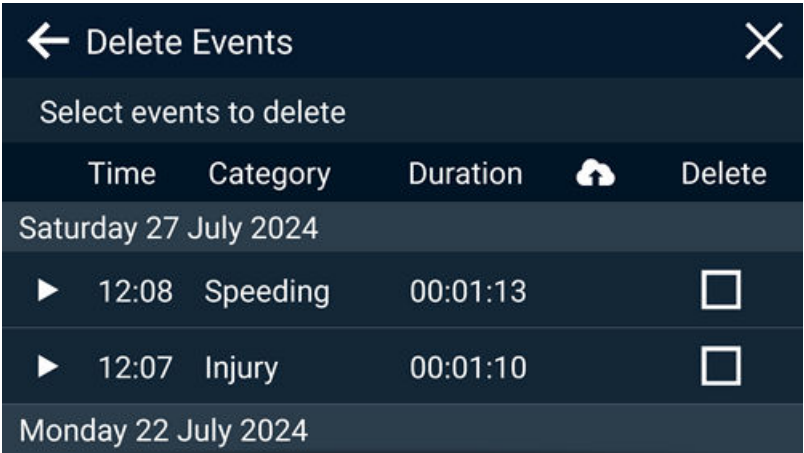


The **Administrator** screen appears.



3. Enter the administrator password and touch **Done**.

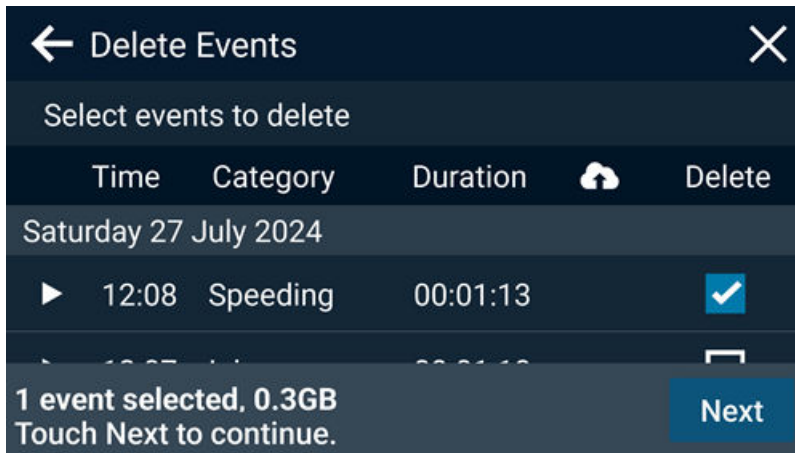
The **Delete Events** screen appears.



 **TIP:** To access event information, touch the event and the event information screen appears.

4. Touch the check box next to the events you want to delete.

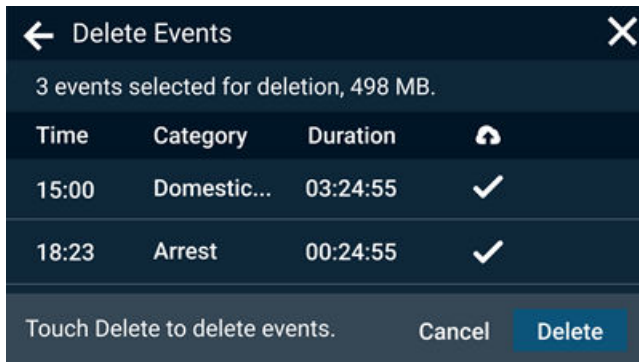
 **NOTE:** The number of events selected for deletion appears at the bottom of the screen.



 **NOTE:** To review the event before deleting it, touch the play icon on the left side of the screen.

5. Touch **Next**.

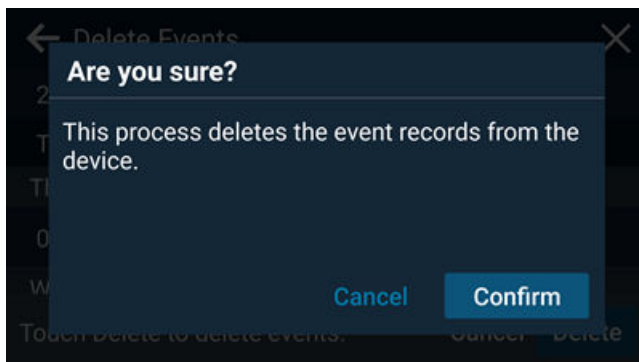
The **Delete Events** review screen appears.



6. Touch **Delete**.

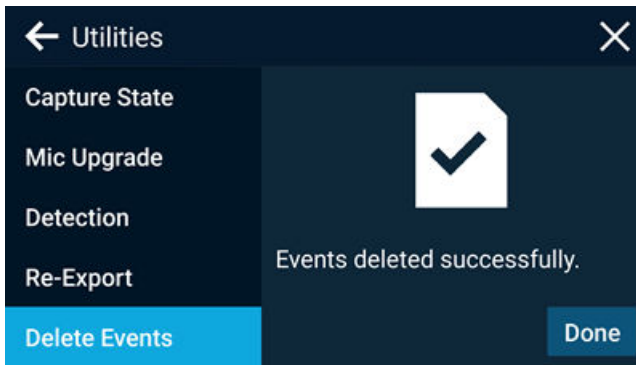
The **Are you sure?** dialog box appears.

7. Touch **Confirm** to delete the selected events.



The event deletion process starts.

8. Touch **Done**.



Cameras Button

You can access the system camera views and their recording states by pressing **Cameras**.



The **Cameras** button on the Display Control Panel allows you to:

- See the view from each system camera



NOTE: The M500 only shows views from the cameras that it has detected in the system.

- See the recording state of each system camera

The camera is either recording to disk, recording to disk in a background state, or not recording ([Camera States on page 98](#)).



IMPORTANT: There must be at least one camera on in the M500 system.

- Turn off recording from a camera, if you have permissions

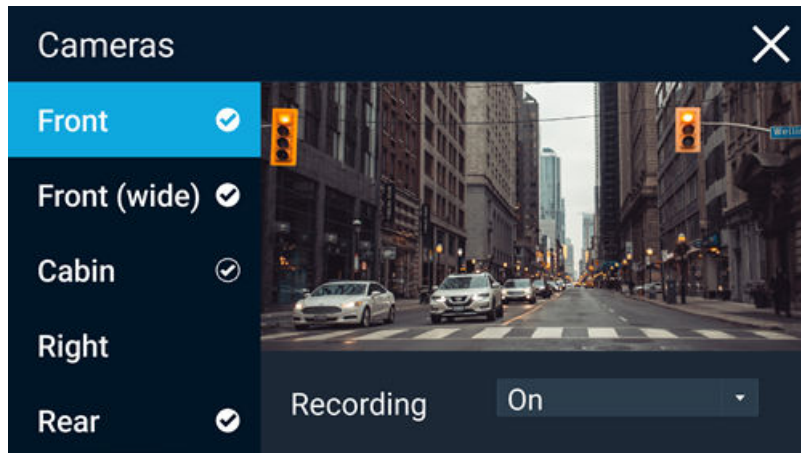


IMPORTANT: Depending on your configuration, you may not be able to change the state of a camera if human form detection is enabled during an event. See [AI Human Form Detection Override on page 99](#) for more information.

- Set a camera to a background recording state, if you have permissions ([Background Recording on page 99](#))

Camera Views

The available camera views are listed on the left side of the **Cameras** screen.



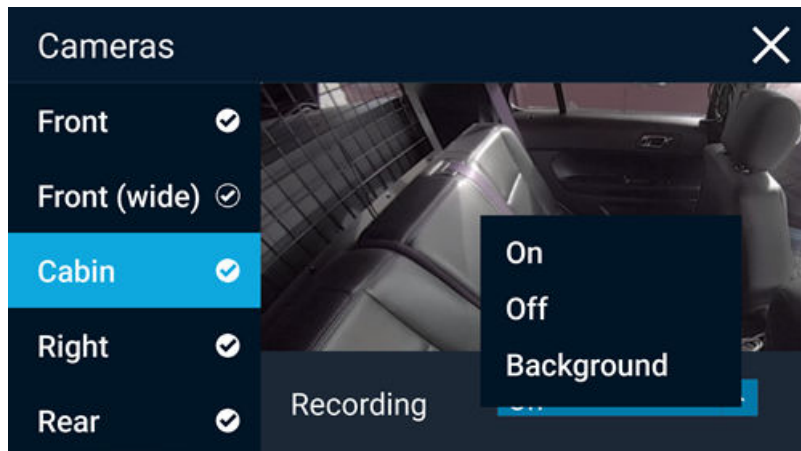
You access each camera view by touching its name on the list.

 **NOTE:** You may have a different list of camera views. The M500 shows a list of views from the cameras that it has detected in the system.

The **Cameras Front** and **Cameras Front (wide)** screens show a snapshot from the **Front** camera view of the M5F camera.

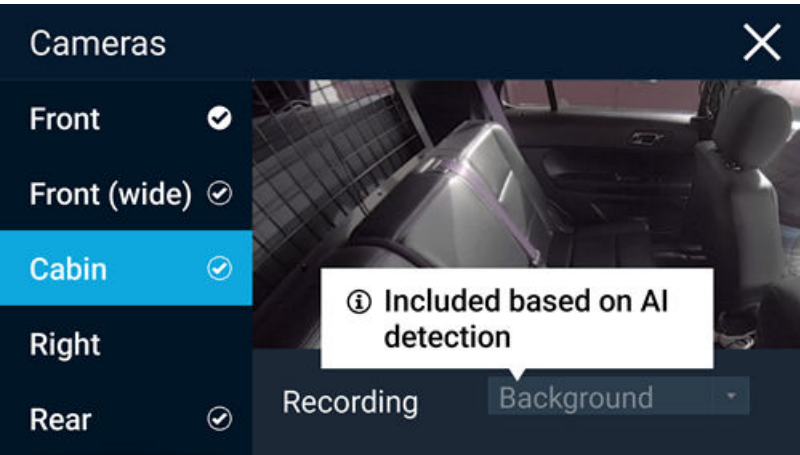
If an M5F front camera is installed in your vehicle, the **Front (wide)** camera view is a second view from the physical M5F front camera. The system treats it as a separate camera view from the **Front** camera view.

Each screen shows a snapshot from the camera view.



The recording state of the camera shows below the snapshot. See [Camera States on page 98](#) for more information.

Depending on your configuration, you may not be able to change the state of a camera if AI human form detection is enabled during an event.



 **NOTE:** See [AI Human Form Detection Override on page 99](#) for more information.

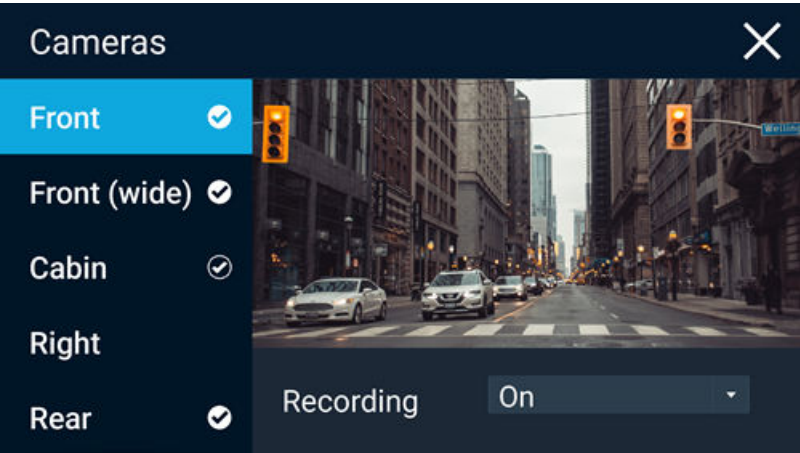
For more information, see:

[Display on page 60](#)

[Cameras on page 102](#)

Camera States

The recording state of the camera shows below the snapshot and an icon shows in the camera view list on the **Cameras** screen.



If you have permissions, you can change the camera recording state to one of the following options from the drop-down list:

 **IMPORTANT:** Depending on your configuration, you may not be able to change the state of a camera if human form detection is enabled during an event. See [AI Human Form Detection Override on page 99](#) for more information.

State	Icon
On	
Off	No icon appears

State	Icon
Background	

 **IMPORTANT:** There must be at least one camera on in the M500 system.

 **NOTE:** For more information, see [Background Recording on page 99](#).

Turning off recording for a camera view means that the camera view is no longer being recorded to disk. No video from that camera view is included in events while the camera recording state is off.

The recording state of the camera view remains off until:

- You turn it back on
- You start Covert Mode ([Entering Covert Mode on page 24](#))
- A human form is detected by AI, if **Detection** is enabled ([Detection on page 89](#))

The recording state you set for the camera view is saved to the agency evidence management system as one of your officer preferences. Each time you log in to an M500, your set preferences are restored.

 **NOTE:** Your administrator configures your permissions to change the recording state in the agency evidence management system.

See [Officer Preferences on page 19](#) for information about all your officer preference settings.

Background Recording

If you have permissions, you can change the recording state of a camera to **Background**. A camera that is set to a background recording state is on and recording.

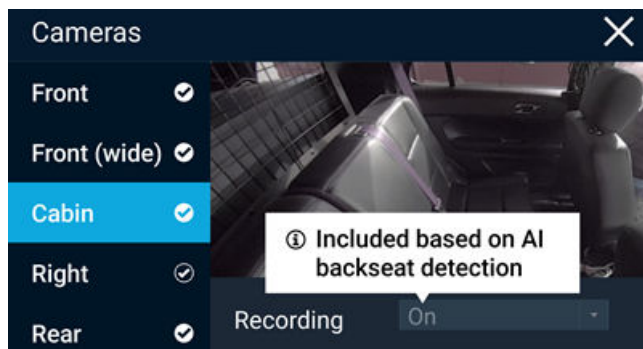
Depending on your configuration, a camera view set to a background recording state is either included or excluded by default. When you tag a recorded event, depending on your M500 configuration, you have the option to include or exclude the view from the camera in the event upload.

If you need the camera view, select the **Include in upload** check box when tagging the event.

 **NOTE:** See [Tagging an Event on page 31](#) and [Event Tags Cameras on page 30](#) for more information.

AI Human Form Detection Override

Your agency determines your ability to override AI human form detection during an event.



During an event, human form detection may be enabled. By default, the camera recording with detection enabled is included in the event upload. Depending on your configuration, you may not be able to change the

state of a camera if human form detection is enabled during an event. Your agency administrator determines your ability to change the state of a camera recording with detection enabled.

 **NOTE:** See [Detection on page 89](#) and [Camera States on page 98](#) and for more information.

On the **Cameras** screen, the list of camera views shows the cameras present in the system. If a camera is included by default based on AI human form detection that cannot be overridden, the camera state is **On** but the drop-down menu is disabled.

 **NOTE:** See [Camera Views on page 96](#) for more information.

On the **Event Tags Cameras** screen, you can see which camera view is included in an event upload based on AI human form detection.

 **NOTE:** See [Event Tags Cameras on page 30](#) for more information.

Display Button

The **Display** button on the Display Control Panel allows you to:

- Adjust the Display Screen brightness ([Using Screen Brightness on page 101](#))
- Turn off all Display illumination (screen and buttons) and camera LEDs ([Enabling Dark Mode on page 101](#))
- Hide the icons and information on Live View ([Enabling Hide Info on page 102](#))
- Adjust the Display Control Panel button brightness ([Using Button Brightness on page 102](#))

 **NOTE:** Your Display Screen may change to an M500 logo when your vehicle begins to move. You cannot interact with any Display Screen functions when this happens. Your administrator configures this behavior in the agency evidence management system.

Accessing Display Settings

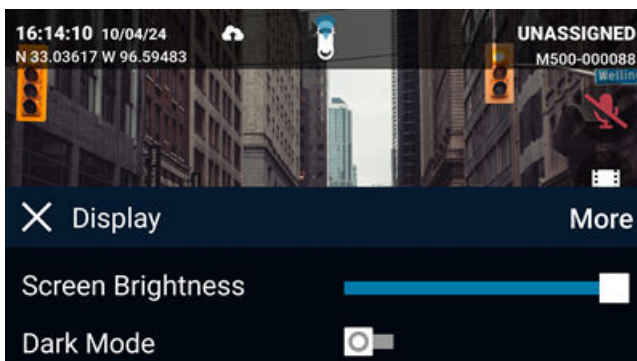
The **Display** screen has two versions, an initial and a full version.

Procedure:

1. To access the initial **Display** screen, press **Display**.



The initial **Display** screen opens over the bottom half of the Live View.



You can adjust screen brightness or enter **Dark Mode** from the initial **Display** screen.

2. To access the full **Display** screen from the initial **Display** screen, touch **More**.

The full **Display** screen opens showing more **Display** options.



Result: For more information, see [Display Settings on page 78](#).

Using Screen Brightness

You can make your Display Screen brighter or darker using the **Screen Brightness** slider.

Procedure:

To increase or decrease the Display Screen brightness, drag the slider to the right or to the left as needed.

Result:

The brightness level you set is saved to the agency evidence management system as one of your officer preferences. Each time you log in to an M500, your set preferences are restored.

See [Officer Preferences on page 19](#) for information about all your officer preference settings.

Enabling Dark Mode

Enabling **Dark Mode** turns off all Display illumination (the screen and buttons) and front and rear camera LEDs. The Display, the front camera, and the rear camera go completely dark.

Procedure:

To enable **Dark Mode**, touch the **Dark Mode** toggle switch to turn it on .

The system immediately goes dark.

Result:

It remains dark until you disable **Dark Mode**. See [Disabling Dark Mode on page 102](#) for instructions.



NOTE: Dark Mode and Covert Mode are not the same feature. Dark Mode turns off all illumination on the Display, M5F camera, and M5R camera. Covert Mode does the same. But Covert Mode also starts a recorded event, puts all cameras into an active recording state, and unmutes all microphones. See [Covert Mode on page 24](#) for more information.

Disabling Dark Mode

Disabling **Dark Mode** turns on all Display illumination (the screen and buttons) and the front and rear camera LEDs.

Procedure:

To disable **Dark Mode**, touch the Display Screen or press any button on the Display Control Panel.
The system fully illuminates. The Display Screen shows the Live View.

Enabling Hide Info

Enabling **Hide Info** hides any icons and information that show on the Live View. Everything that is laid over the live video stream is hidden.

Procedure:

To enable **Hide Info**, touch the **Hide Info** toggle switch to turn it on .

All the icons and information as well as the darker overlay bar at the top of the Live View screen disappear.

Result:

They remain hidden until you disable **Hide Info**. See [Disabling Hide Info on page 102](#) for instructions.
See [Live View on page 62](#) for information about what is available on the Live View.

Disabling Hide Info

Disabling **Hide Info** causes all icons and information to reappear on the Live View.

Procedure:

To disable **Hide Info**, touch the **Hide Info** toggle switch to turn it off .

All the icons, information, and the overlay bar reappear on the Live View.

Using Button Brightness

You can make your Display Control Panel buttons brighter or darker by using the **Button Brightness** slider.

Procedure:

To increase or decrease the button brightness, drag the slider to the right or to the left as needed.

Result:

The brightness level you set is saved to the agency evidence management system as one of your officer preferences. Each time you log in to an M500, your set preferences are restored.

See [Officer Preferences on page 19](#) for information about all your officer preference settings.

Cameras

The M500 in-car video system uses four cameras:

- M5F ([M5F Front Camera on page 103](#))
- M5P ([M5P Camera on page 104](#))
- M5R ([M5R Camera on page 104](#))

- M5E ([M5E Camera on page 105](#))

Each camera can be configured to record at least two streams of video.

 **NOTE:** New cameras will be added to the system as they are developed.

M5F Front Camera

The M5F front camera is a 4K high resolution camera with a wide field of view and a wide dynamic range. It has color night vision and produces best-in-class video in a low light environment.



The M5F front camera can record up to 4 streams of video. These streams are configured as two separate camera views, **Front** and **Front (wide)**, each with a primary and secondary stream. The M5F front camera is not required for the M500 system.

The M5F front camera has two LEDs:

- Power: LED is green and is on the left as you look at the front of the camera
The power LED turns on when the system is powered on, unless it has been disabled on the **Settings Camera LEDs** screen ([Disabling the Camera LEDs on page 77](#)).
- Record: LED is red and is on the right as you look at the front of the camera
The record LED turns on when an event is started, unless it has been disabled on the **Settings Camera LEDs** screen.

 **NOTE:** Changes made to a camera's LEDs on the **Settings Camera LEDs** screen are independent from other cameras.

The M5F is a small camera that mounts just behind the rear view mirror so it does not interfere with the officer's field of view.



 **NOTE:** Contact your Motorola Solutions representative for information about other possible M5F camera mount options.

M5P Camera

The M5P camera is a 1080p HD high resolution camera with a wide field of view and a wide dynamic range. The M5P can record up to two streams of video.

The M5P camera field of view covers the interior of the rear cabin with night vision providing 180 degree cabin illumination.



The M5P is a compact, non-obtrusive camera that can be mounted on the cage or the roof liner.



 **NOTE:** The M5P camera has an integrated microphone. See [M5P Camera Integrated Microphone on page 106](#) for more information.

The M5P can be connected to either the rear or cabin camera port on the back of the M500 DVR. However, only the cabin camera port supports audio streaming. The M5P connected to the cabin camera port creates the audio stream.

If you connect the M5P to the rear camera port, the camera does not create an audio stream.

M5R Camera

The M5R camera is a 1080p HD high resolution camera with a wide field of view (108-degrees) and a wide dynamic range. It performs well in low light environments. The M5R can record up to two streams of video.



The M5R camera has two LEDs:

- **Power:** LED is green and is on the left side as you look at the front of the camera
The power LED turns on when the system is powered on unless it has been disabled on the **Settings Camera LED** screen ([Disabling the Camera LEDs on page 77](#)).
- **Record:** LED is red and is on the right side as you look at the front of the camera
The record LED turns on when an event is started unless it has been disabled on the **Settings Camera LED** screen.

 **NOTE:** Changes made to a camera's LEDs on the **Settings Camera LEDs** screen are independent from other cameras.

Unlike the M5P, the M5R does not have a microphone, so it does not create an audio stream. The M5R can connect to both cabin and rear camera ports on the back of the M500 DVR; however, even when connected to the cabin camera port, it does not create an audio stream.

 **NOTE:** Because the rear camera port does not support audio, an M5P connected to the rear camera port does not create an audio stream.

The M5R is a small camera that mounts facing out of the rear windshield, typically in one of the following positions:

- Rear windshield ceiling center
- Passenger-side rear pillar
- Driver-side rear pillar



The appropriate mounting position depends on the setup of the vehicle and the visibility needs of the driver.

M5E Camera

The M5E is a 1080p HD high resolution camera that is weatherized and can be mounted on the exterior of a vehicle.



The M5E camera has two LEDs:

- Power: LED is green and is on the left side as you look at the front of the camera
The power LED turns on when the system is powered on unless it has been disabled on the **Settings Camera LED** screen ([Disabling the Camera LEDs on page 77](#)).
- Record: LED is red and is on the right side as you look at the front of the camera
The record LED turns on when an event is started unless it has been disabled on the **Settings Camera LED** screen.

 **NOTE:** Changes made to a camera's LEDs on the **Settings Camera LEDs** screen are independent from other cameras.

Typically, the M5E is an exterior camera facing away from the vehicle in the following locations:

- Left exterior side of the vehicle
- Right exterior side of the vehicle (pictured below)



The appropriate mounting position depends on the setup of the vehicle.

Microphones

The M500 in-car video system uses the following types of microphones:

- Stand-alone cabin microphone ([Stand-Alone Cabin Microphones on page 106](#))
- Integrated microphone (in the M5P camera) ([M5P Camera Integrated Microphone on page 106](#))

For more information, see:

[Audio on page 71](#)

[Muting Microphones on page 19](#)

Stand-Alone Cabin Microphones

The cabin microphones are omnidirectional, wired, in-car microphones that connect to the DVR with a mini stereo jack. Typically, there are two installed in the vehicle, one in each top corner of the windshield.



1	Cabin microphone, installed
2	Front windshield, top passenger corner

M5P Camera Integrated Microphone

The M5P camera has an integrated omnidirectional microphone. It is a compact camera/microphone combination that can be mounted on the cage or the roof liner.



The integrated microphone only works when the M5P is connected to the cabin camera port. Only the cabin camera port supports audio stream.

For more information, see [M5P Camera on page 104](#).

RFID Reader

The Radio-Frequency Identification (RFID) reader allows officers to log in to the M500 by tapping their ID card on the RFID reader. The RFID reader connects to the USB port on the back of the M500 DVR. See [DVR on page 59](#) for more information.



The RFID reader has an LED to indicate the status of the login process. See [RFID LEDs on page 107](#) for more information.

If RFID is enforced, the officer must use their RFID reader to log in to the M500.

 **NOTE:** See [Logging in with RFID on page 18](#) for more information.

If RFID is not enforced, the officer can either log in through live view or tap their RFID reader.

 **NOTE:** See [Logging In from Live View on page 17](#) for more information

If you attempt to log in and the system notifies you that another officer is currently logged in, you can log that officer out and continue to log in using the RFID reader.

 **IMPORTANT:** Officers cannot log in to an M500 using an RFID reader while a recorded event is in progress.

RFID LEDs

The RFID reader has two LEDs on the top of the reader to indicate the state of the login process.

LED	State
Green	Powered On/Officer is logged in

LED	State
Rapid Green	Login accepted
Red	Login not accepted

 **NOTE:** The RFID associates tones with each LED to indicate the state of the login process.

Peer Assisted Recording

Peer Assisted Recording allows devices such as the M500 and body-worn cameras to send to send and/or receive notifications when the devices are in range of each other depending on your agency configuration.

The M500 can be configured to notify or receive notifications from other peer devices:

- The M500 sends notifications to other peer devices
The M500 starts a recorded event and is in range of another device configured to receive notifications, a notification is sent. The device receiving the notification starts an event automatically.

 **NOTE:** The M500 must be configured by your administrator to send notifications to other peer devices.

- The M500 receives notifications from other peer devices
The M500 receives a notification from another peer device that a recorded event is in progress. The M500, when in range, receives the notification from the other peer device and automatically starts a recorded event. See [Start an Event Automatically on page 22](#) for more information.

 **NOTE:** The M500 must be configured by your administrator to receive notifications from other peer devices.

 **NOTE:** If the M500 already has a recorded event in progress, the notification is ignored and the in progress event continues.

The M500 can trigger a recorded event, but it cannot stop the event for other devices. Each device must stop their own recorded events.

 **NOTE:** See [Stopping an Event Manually on page 26](#) to stop an event on the M500.

Chapter 4

Administrator Functions

The M500 system provides multiple features and functions that an administrator can use for system setup and troubleshooting, including:

- Configuration deployment ([Configuration Deployment on page 109](#))
- Resetting the Default Network Configuration on the M500 ([Resetting the Default Network Configuration on the M500 on page 113](#))
- Firmware upgrade ([Firmware Upgrade on page 115](#))
- Maintenance Mode ([Maintenance Mode on page 118](#))



NOTE: New administrator features will be added to the system as they are developed.

Access to these administrator functions typically requires a password.



NOTE: Passwords (administrator and supervisor) are set in your evidence management system and pushed to the M500 during configuration deployment.

Configuration Deployment

An M500 configuration is a group of settings applied to an M500. When the configuration is applied, it automatically sets up the M500 to meet the needs of your agency.

You deploy the initial configuration manually using a USB drive ([Deploying Configurations Manually on page 111](#)). The initial configuration sets up the connection between the M500 and your agency evidence management system.



NOTE: You must have the correct permissions to deploy a configuration.

After the initial manual deployment, a configuration can be deployed:

- Automatically ([Automatic Configuration Deployment on page 109](#))
- Manually ([Manual Configuration Deployment on page 110](#))

Automatic Configuration Deployment

New configurations deploy automatically after the initial manual deployment, if your agency connects to the M500 through a network (wireless or wired).

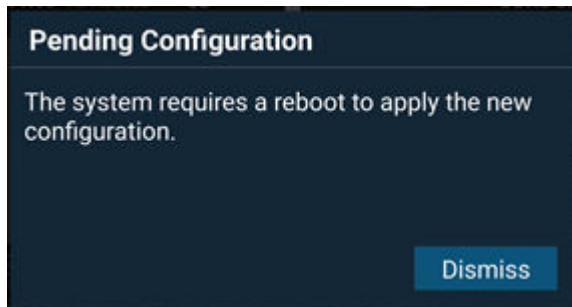
When an M500 connects to the agency network, the evidence management system automatically pushes an updated configuration, if applicable.

The configuration is staged on the M500, then applied the next time the DVR is rebooted.



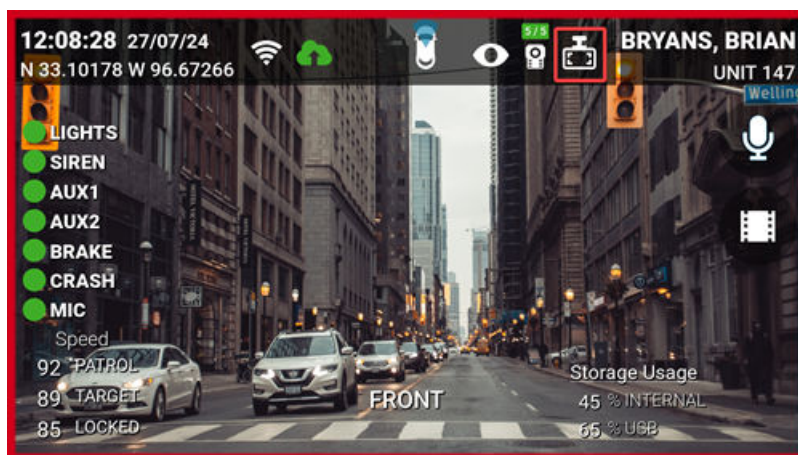
NOTE: If automatic configuration deployment is enabled, the evidence management system automatically pushes the new configuration to all applicable M500 DVRs either wirelessly or using a USB drive during event upload through the USB upload application.

The **Pending Configuration** message box appears on the screen when the configuration is pushed to the M500.



You can **Dismiss** the message box. The configuration remains staged on the M500 until the DVR is rebooted.

The pending update indicator  shows on the Live View when a configuration (or firmware) update is staged and waiting for a reboot. The indicator remains on the Live View until the M500 is rebooted.



Manual Configuration Deployment

You must deploy manually in the following two situations:

- First time you load a configuration on the M500
The initial configuration sets up the connection between the M500 and your agency evidence management system.
- Your agency does not connect to the M500 through a network (wireless or wired)
You must manually deploy any M500 configuration in this case, whether it is the initial configuration or an update.

You use a USB drive when you manually deploy a configuration ([Deploying Configurations Manually on page 111](#)).



IMPORTANT: The configuration file on the configuration USB drive must use the path structure
[Drive]:/Configurations/your_config_xml_here.xml.

The new configuration is copied from the USB drive and staged on the M500. It is applied when the M500 reboots. The M500 reboots automatically.

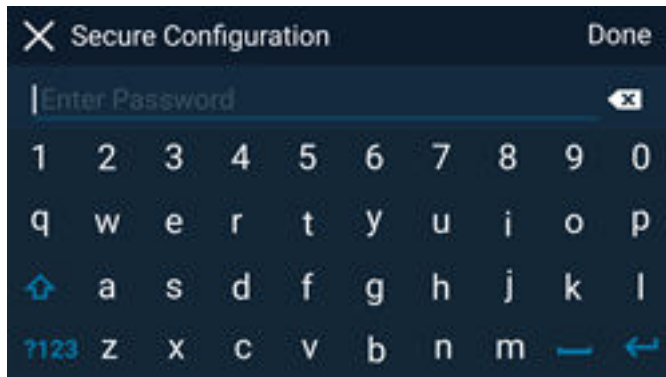


NOTE: You must have the correct permissions to deploy a configuration.

Secure Configurations

With the M500 firmware version 1.1.4, the configuration you manually deploy may need to be decrypted before it can be applied to the M500.

The password to decrypt the secure configuration is set up in the agency evidence management system during the process of preparing the configuration for deployment. You enter this password on the M500 when the system asks for it on the **Secure Configuration** screen.



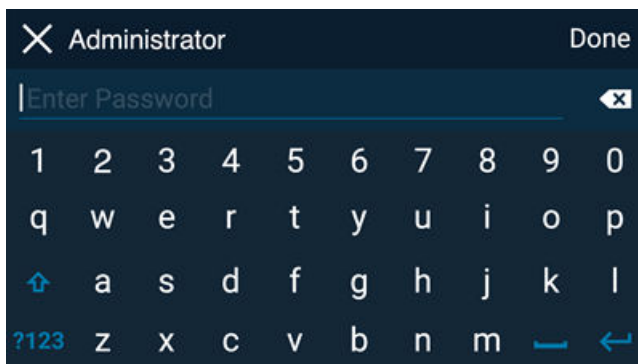
Deploying Configurations Manually

Procedure:

1. If a USB drive is already present in the M500, remove it safely ([Ejecting the USB Drive Using Utilities on page 49](#)).
2. Insert the USB drive containing the configuration into the M500.

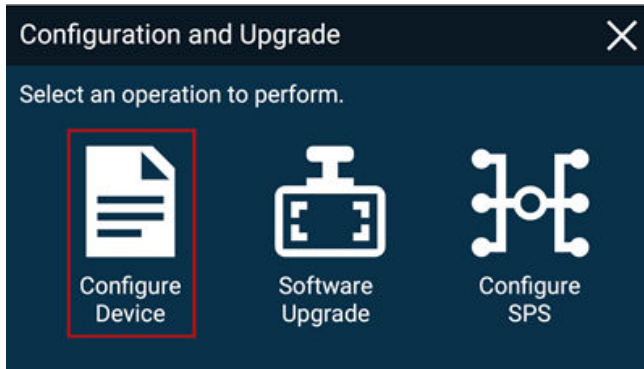
 **NOTE:** If this is the initial configuration of an M500 and there is only one configuration file on the USB drive, the configuration update process may start automatically.

3. Enter the **Administrator** password if prompted for it.

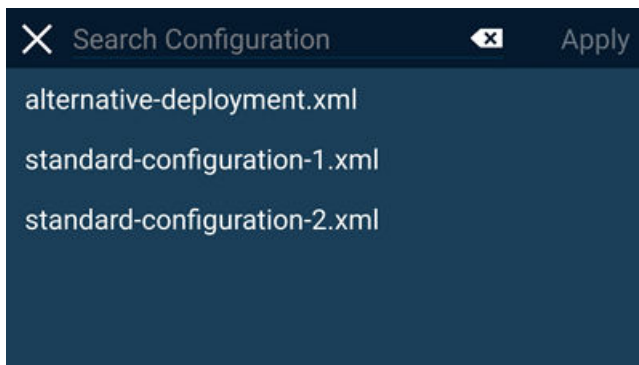


 **NOTE:** You must have the correct permissions to deploy a configuration.

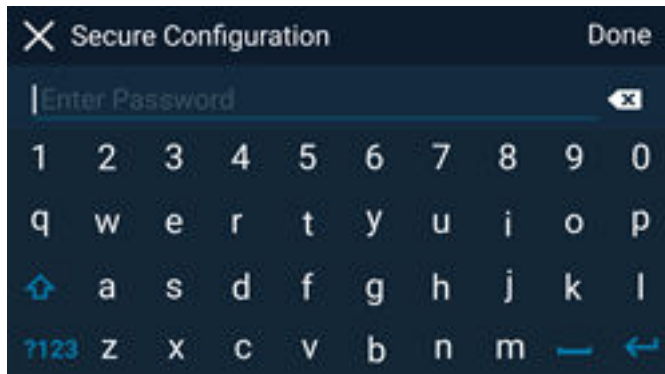
4. Select the configuration to apply if prompted for it:
 - a. If there is only one configuration file on the USB drive, the configuration update process may start automatically. Go to step 5.
 - b. If there are multiple file types on the USB drive, select **Configure Device** from the **Configuration and Upgrade** screen, then enter the **Administrator** password if prompted for it.



- c. If there are multiple configuration files on the USB drive, you can search for and then select the appropriate configuration from the **Search Configuration** list, then touch **Apply**.

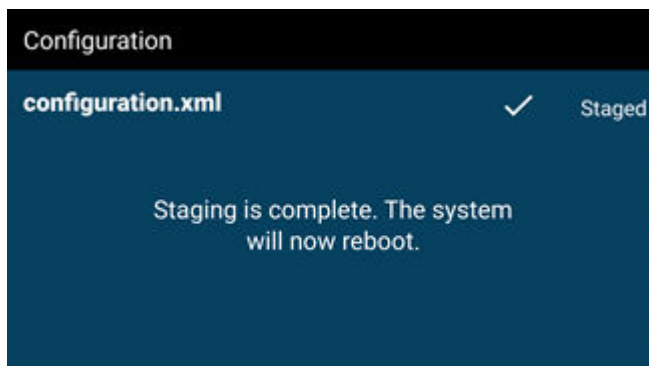


5. Enter the **Secure Configuration** password if prompted for it.



The configuration staging process starts.

The **Configuration** screen appears when the new configuration has finished staging.



The M500 ejects the USB drive when the staging process has finished.

6. Remove the configuration USB drive from the M500, if desired.

After a moment, the M500 automatically reboots to apply the configuration.



NOTE: If the M500 reboots before you remove the USB drive, you must eject the drive to remove it ([Ejecting the USB Drive Using Utilities on page 49](#)).

Resetting the Default Network Configuration on the M500

You can reset the default network configuration on the M500 using the USB drive that contains the configuration file.



WARNING: Resetting the default network configuration on the M500 can temporarily disrupt the M500 system network. This network disruption includes the in-car network and the connection to the agency evidence management system. The disruption should last about 2 minutes.

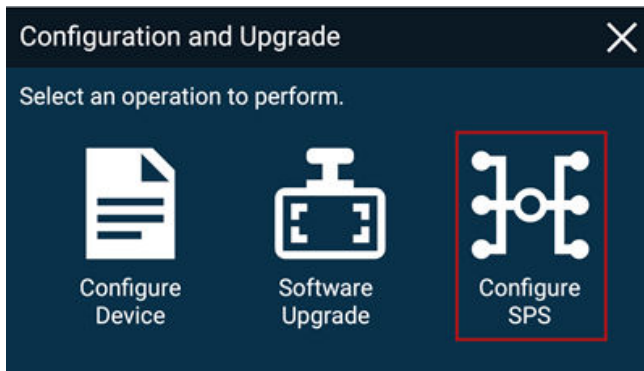
Procedure:

1. Load the new network configuration onto a USB drive:
 - a. Insert the USB drive into your PC.
 - b. Create a folder on the USB drive titled **SPS**.
 - c. Open the **Notepad** application on your PC.
 - d. Create a new **Notepad** file.
 - e. If you want to change the M500 network configuration, copy and paste the text from the network configuration text section ([Network Configuration Text on page 114](#)) to your **Notepad** text file.
 - f. Save the text file to the USB drive in the **SPS** folder as a JSON file.

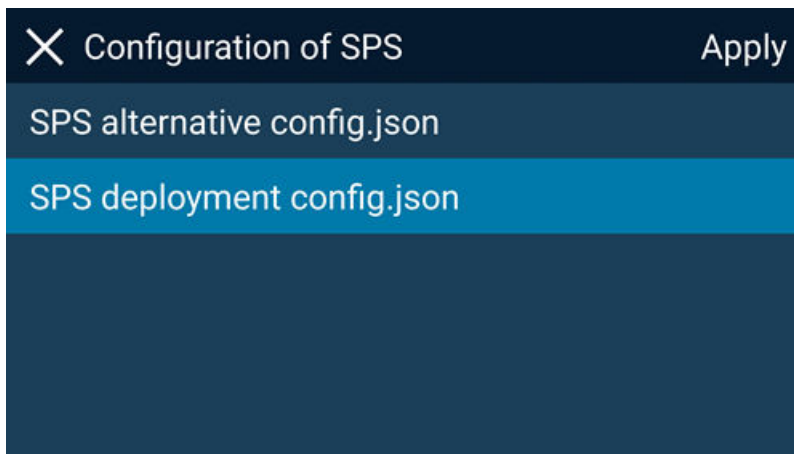


NOTE: For example, `sps_config.json`

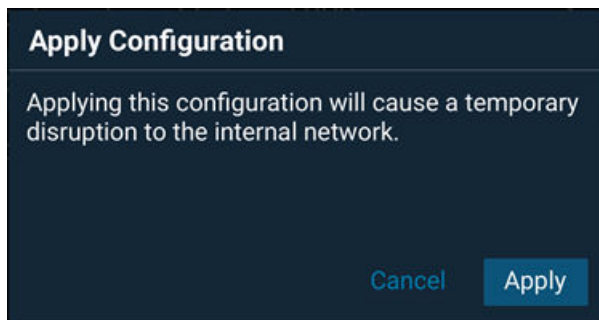
- g. Safely remove the USB drive from your PC.
2. Power on the M500.
3. Safely remove the USB drive from the M500 if there is one already inserted.
4. Insert the configuration USB drive into the M500.
5. Select the configuration to apply to the M500:
 - a. If there is only one configuration file on the USB drive, the configuration update process starts automatically.
 - b. If there are multiple file types on the USB drive, select **Configure SPS** from the **Configuration and Upgrade** screen.



- c. If there are multiple configuration files on the USB drive, select the appropriate configuration from the list, then touch **Apply**.



The **Apply Configuration** message appears to remind you of the network disruption.



6. Touch **Apply**.

The system applies the configuration file to the M500.

The **Internal network online** message appears when the system network is back online.

The USB drive automatically ejects from the M500.

7. After the configuration update finishes, reboot the M500.

Network Configuration Text

If you want to change the default network configuration, you can use the following configuration text:



CAUTION: Motorola Solutions recommends consulting your IT network expert for guidance.

```
{
  "sps_type": "internal",
  "network": {
    "ipaddr": "10.1.100.20",
    "netmask": "255.255.255.0"
  },
  "dhcp": {
    "enabled": true,
    "router": "10.1.100.1",
    "lease_start": "10.1.100.101",
    "lease_end": "10.1.100.120",
    "max_leases": 20,
    "lease_time": 86400,
    "static_lease": []
  }
}
```

Firmware Upgrade

Motorola Solutions regularly releases firmware upgrades for the M500.

You can upgrade the M500 firmware:

- Automatically ([Automatic Firmware Upgrade on page 115](#))
- Manually ([Manual Firmware Upgrade on page 116](#))



NOTE:

You must upgrade manually if your agency does not connect to the M500 through a network (wireless or wired).

When you apply a firmware upgrade to the M500, the firmware must be a later version than the one currently installed.

Automatic Firmware Upgrade

You can choose to upgrade M500 firmware automatically. You enable automatic firmware upgrade in your evidence management system.

If automatic upgrade is enabled, the evidence management system automatically pushes new firmware to all applicable M500 DVRs, either wirelessly or through an automatic distribution to a USB drive during event upload through the USB upload application.

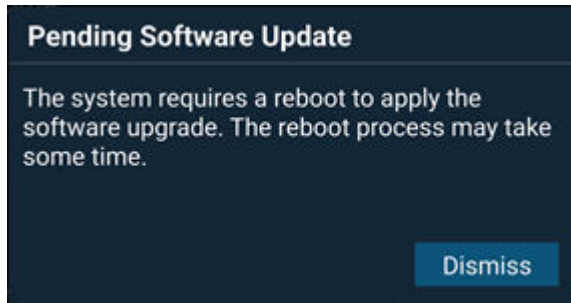


NOTE: If your agency uploads events over an LTE connection, your M500 may be configured to only upload critical events over LTE. If this is the case, new firmware versions are not downloaded from your agency evidence management system over an LTE connection. For more information about critical events, see [Critical Categories on page 29](#).

After the firmware upgrade is pushed wirelessly or the USB drive containing the firmware upgrade is inserted back into the M500, the new firmware is staged.

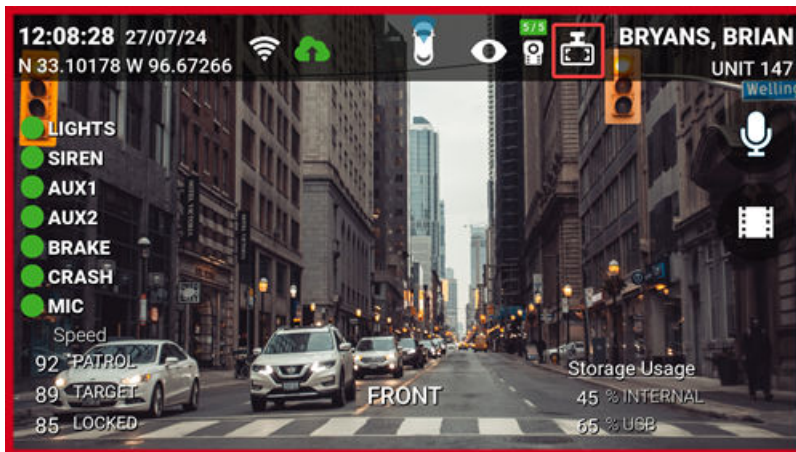
The new firmware is applied the next time the DVR is rebooted.

The **Pending Software Update** message box appears on the screen when the new firmware is pushed to the M500.



You can **Dismiss** the message box. The new firmware remains staged on the M500 until the DVR is rebooted.

The pending update indicator  shows on the Live View when a new firmware (or configuration) update is staged and waiting for a reboot. The indicator remains on the Live View until the M500 is rebooted.



 **NOTE:** You can verify the firmware version of the M500 in the **DVR Version** field on the **System Status** screen ([System on page 67](#)).

Manual Firmware Upgrade

You must upgrade firmware manually if your agency does not connect to the M500 through a network (wireless or wired). You use a USB drive when you manually upgrade firmware.

When you apply a firmware upgrade to the M500, the firmware must be a later version than the one currently installed.

 **NOTE:** The firmware file on the USB drive must use the path structure `[Drive]:/Upgrades/your_firmware_here.tar.gz`

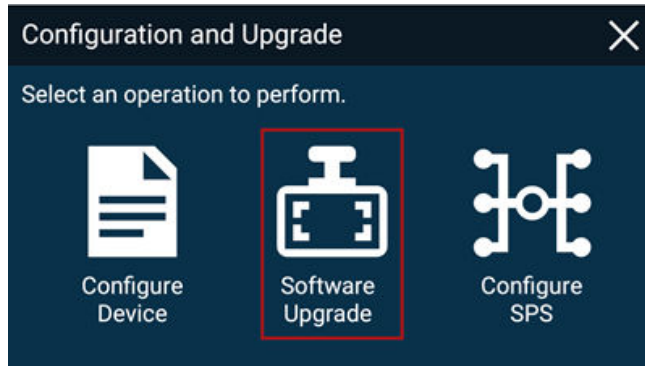
The new firmware is copied from the USB drive and staged on the M500. It is applied when the M500 reboots. The M500 reboots automatically.

Upgrading Firmware Manually

Procedure:

1. If a USB drive is already present in M500, remove it safely.
For more information, see [Ejecting the USB Drive Using Utilities on page 49](#).

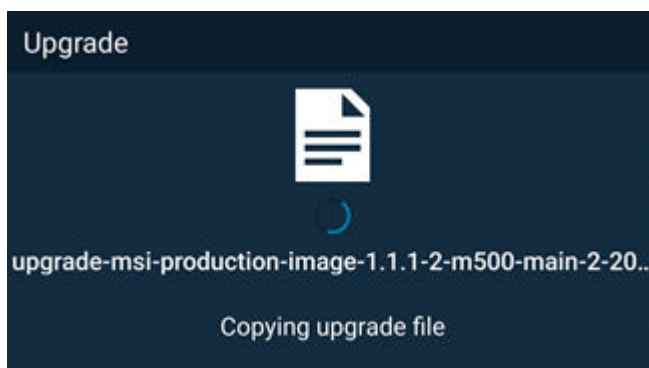
2. Insert the USB drive containing the firmware upgrade into the M500.
3. Select the firmware to apply if prompted for it:
 - a. If there is only one firmware upgrade file on the USB drive, the update process may start automatically. Go to [step 4](#).
 - b. If there are multiple file types on the USB drive, select **Software Upgrade** from the **Configuration and Upgrade** screen.



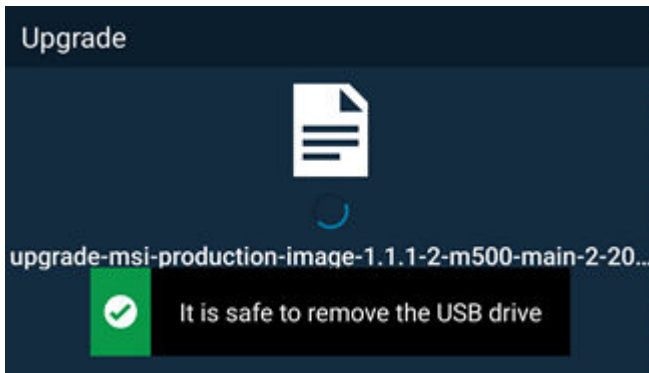
- c. If there are multiple firmware upgrade files on the USB drive, select the appropriate file from the **Upgrade** list, then touch **Apply**.



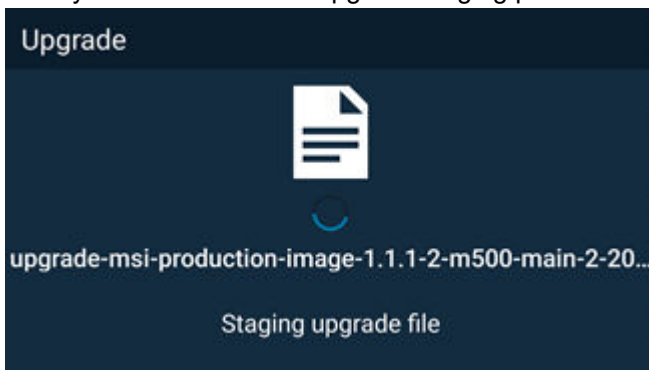
The upgrade file is copied from the USB drive to the M500.



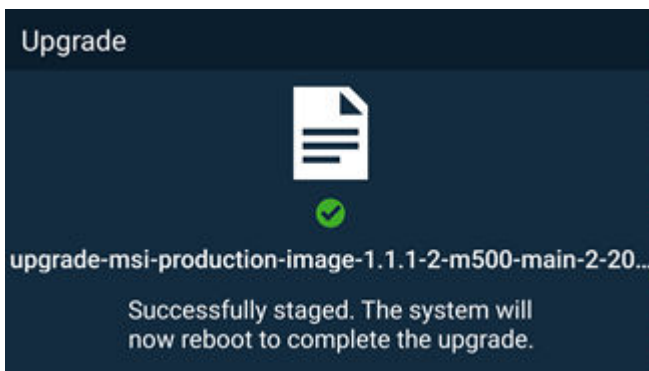
After the upgrade file has finished copying, the M500 ejects the USB drive.



4. Remove the firmware USB drive from the M500, if desired.
The system continues the upgrade staging process.



After the staging process has finished, the M500 reboots to apply the firmware upgrade.



-  **NOTE:** If the M500 reboots before you remove the USB drive, you must eject the thumb to remove it ([Ejecting the USB Drive Using Utilities on page 49](#)).
-  **NOTE:** You can verify the firmware version of the M500 in the **DVR Version** field on the **System Status** screen ([System on page 67](#)).

Maintenance Mode

Maintenance Mode provides tools to help you recover an M500 system that may be experiencing certain issues, for example, failing to boot.

You can perform the following operations on the M500 using Maintenance Mode:

- Generate a state capture ([Generating a State Capture \(Maintenance Mode\) on page 120](#))

- Reformat the internal storage ([Reformatting Internal Storage on page 121](#))
- Restore the default configuration ([Restoring the Default Configuration on page 122](#))
- Erase the event database ([Erasing the Event Database on page 124](#))
- Copy files securely to a USB ([Secure File Copy on page 125](#))
- Update the M500 network settings ([Network Settings on page 130](#))
- Reset the time source ([Resetting the Time Source on page 134](#))

 **NOTE:** New maintenance features will be added to Maintenance Mode as they are developed.

The M500 system automatically enters Maintenance Mode after multiple failed boots in a row.

You can manually enter Maintenance Mode by pressing **Stop** on the Display Control Panel when prompted during the boot sequence ([Entering Maintenance Mode on page 119](#)).

 **CAUTION:** Motorola Solutions recommends that you only manually enter Maintenance Mode when instructed to by customer service.

Maintenance Mode typically requires the administrator password.

Entering Maintenance Mode

Procedure:

1. Reboot the M500.

The M500 splash screen appears.



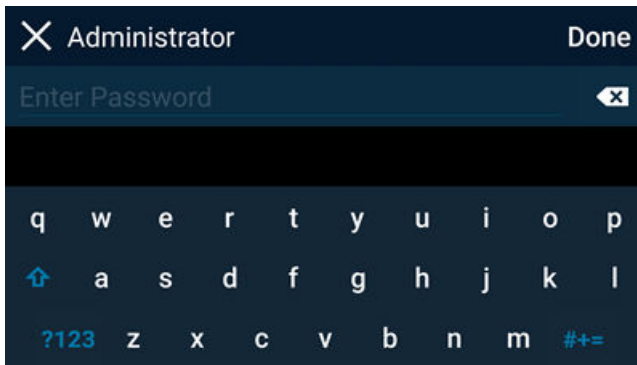
 **CAUTION:** Motorola Solutions recommends that you only manually enter Maintenance Mode when instructed to by customer service.

2. Press and hold **Stop** for 1 second on the Display Control Panel when **Checking for Maintenance Mode...** appears on the splash screen.

 **IMPORTANT:** You have 5 seconds to press **Stop** from the time you see **Checking for Maintenance Mode...** on the screen.

The **Administrator** password prompt screen appears.

 **NOTE:** If there is no administrator password set in the M500 configuration (initial out-of-the-box setup or after factory reset), the **Administrator** password prompt does not appear.



3. Enter the administrator password and touch **Done**.



NOTE: Touching the **X** on the **Administrator** password prompt in Maintenance Mode reboots the M500.

The **Maintenance Mode** screen opens.

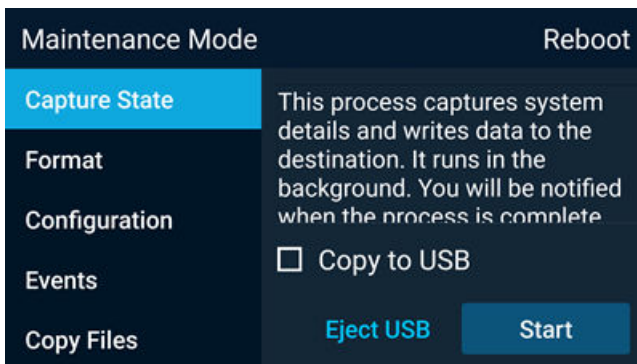
Generating a State Capture (Maintenance Mode)

The **Maintenance Mode Capture State** feature allows you to generate a state capture of the M500 system and copy it to a USB drive to help you troubleshoot system issues.

Procedure:

1. From the **Maintenance Mode** screen, touch **Capture State**.

The **Maintenance Mode Capture State** screen appears.



2. Insert a USB drive.

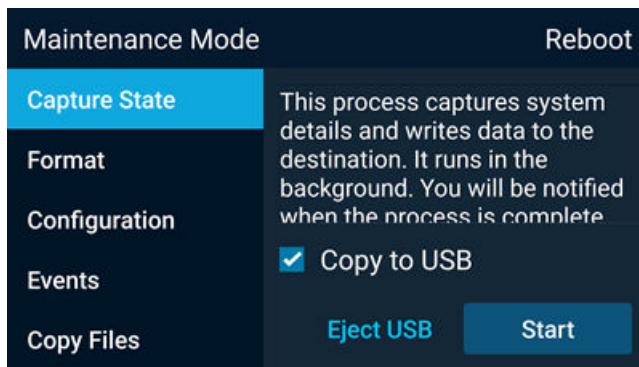


NOTE: If you need to remove a USB drive before you insert one, touch **Eject USB**, then remove the drive.

3. Touch the **Copy to USB** check box to check it.



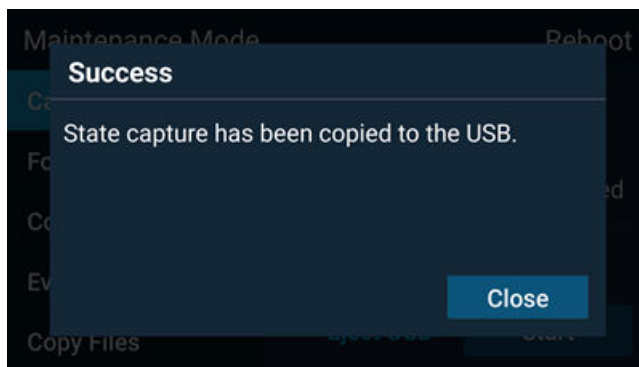
NOTE: A copy of the state capture is also automatically uploaded to your evidence management storage location after the M500 system is able to reconnect to the evidence management software.



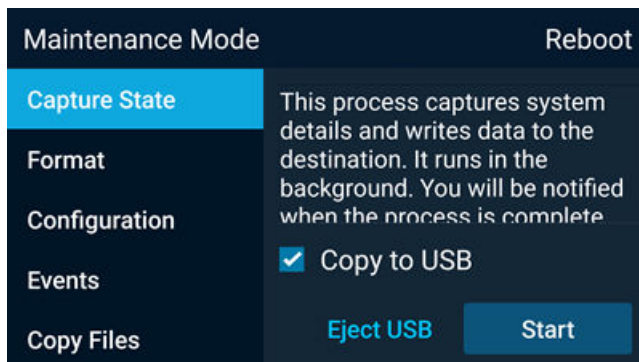
4. Touch **Start**.

The system begins to generate the state capture. The process may take a few minutes.

When the state capture is finished, the **Success** message appears.



5. Touch **Close**.



6. Touch **Eject USB**, then remove the USB drive ([Ejecting the USB Drive Using Utilities on page 49](#)).

Reformatting Internal Storage

The **Maintenance Mode Format Internal Storage** feature allows you to securely wipe the M500 internal storage SSD (solid-state drive) by doing a full reformat. The reformat can help you recover the system if the SSD becomes corrupted and does not allow the system to boot.

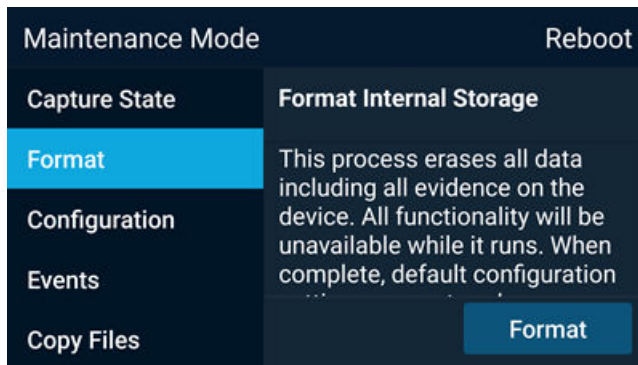


WARNING: The storage reformat erases all configurations, data, and evidence from the M500 system.

Procedure:

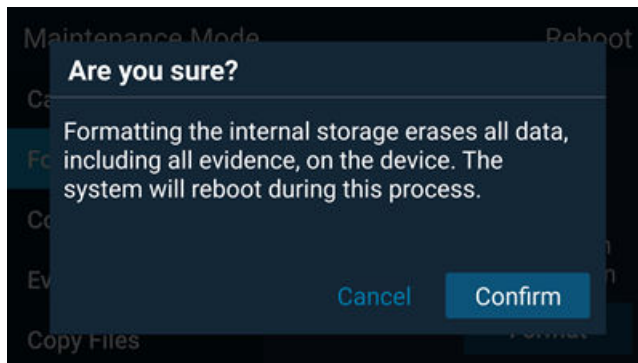
1. From the **Maintenance Mode** screen, touch **Format**.

The **Maintenance Mode Format Internal Storage** screen appears.



2. Touch **Format**.

The **Are you sure?** confirmation dialog box appears asking you to confirm that you want to reformat the internal storage.



3. Touch **Confirm**.

The system reboots to reformat the internal storage.



NOTE: The reboot process to reformat the internal storage may take a little longer than a normal reboot process.

When the reformat reboot has finished, the Display Screen shows the Live View streaming from the front camera.

Result:



IMPORTANT: The M500 loses its connection to the agency evidence management system when its storage is reformatted. You must redeploy a configuration from your agency evidence management system to re-establish the connection ([Manual Configuration Deployment on page 110](#)).

Restoring the Default Configuration

The **Maintenance Mode Restore Default Configuration** feature allows you to restore the M500 system to its factory default configuration. Restoring the default configuration can help recover a system that is malfunctioning.

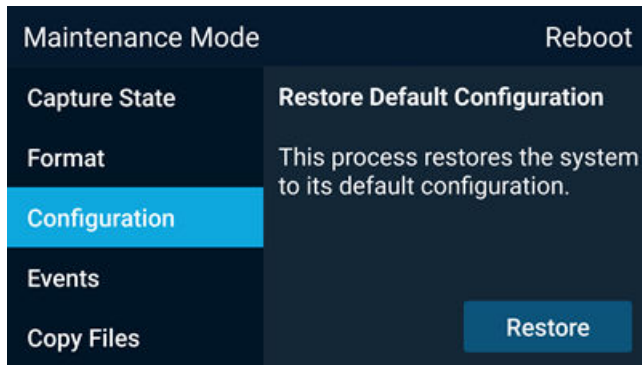


NOTE: Restoring the default configuration also resets the officer preferences to default settings, if applicable.

Procedure:

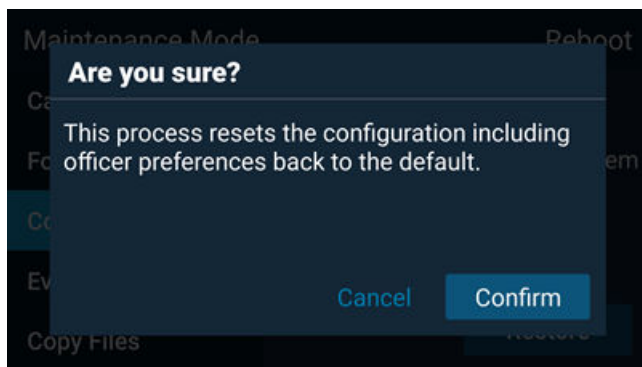
1. From the **Maintenance Mode** screen, touch **Configuration**.

The **Maintenance Mode Restore Default Configuration** screen appears.



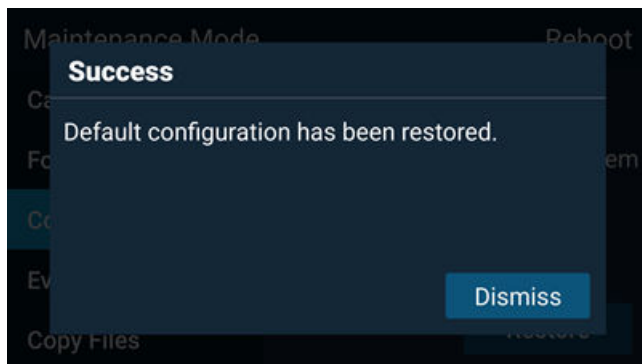
2. Touch **Restore**.

The **Are you sure?** confirmation dialog box appears asking you to confirm that you want to restore the default M500 configuration.



3. Touch **Confirm**.

The default configuration is staged on the M500 and the **Success** message box appears.



4. Touch **Dismiss**.
5. On the **Maintenance Mode Restore Default Configuration** screen, touch **Reboot**.

The default configuration is applied during the reboot process. When the reboot has finished, the Display Screen shows the Live View streaming from the front camera.

Erasing the Event Database

The **Maintenance Mode Erase Event Database** feature allows you to remove all event records (with corresponding event-specific files, for example, event metadata, manifests, VTTs) from the M500 system database. Erasing the event database may help you recover a system that is malfunctioning.

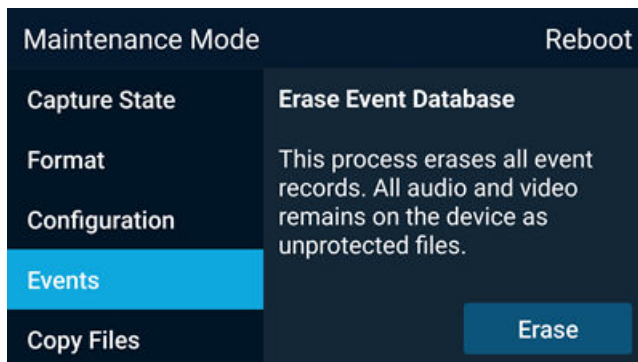


NOTE: All audio and video files remain on the M500 as unprotected files.

Procedure:

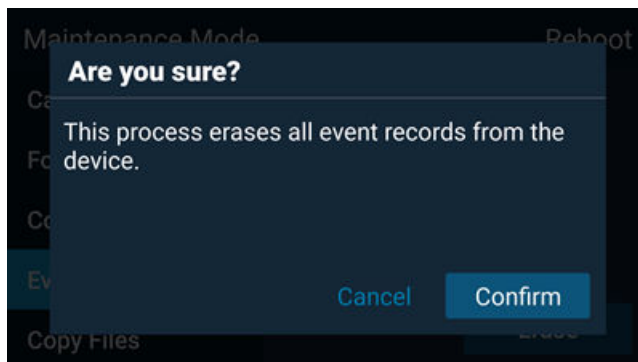
1. From the **Maintenance Mode** screen, touch **Events**.

The **Maintenance Mode Erase Event Database** screen appears.



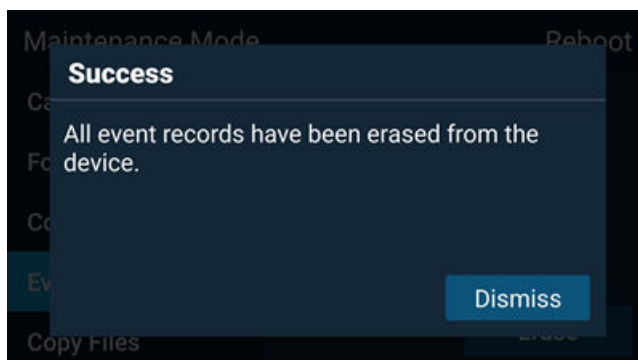
2. Touch **Erase**.

The **Are you sure?** confirmation dialog box appears asking you to confirm that you want to erase all event records from the M500.

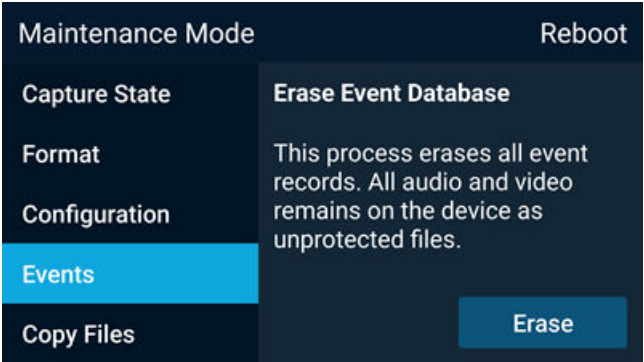


3. Touch **Confirm**.

The system erases all event records and the **Success** message box appears.



4. Touch **Dismiss**.
5. On the **Maintenance Mode** screen, touch **Reboot** to return to normal operating mode.

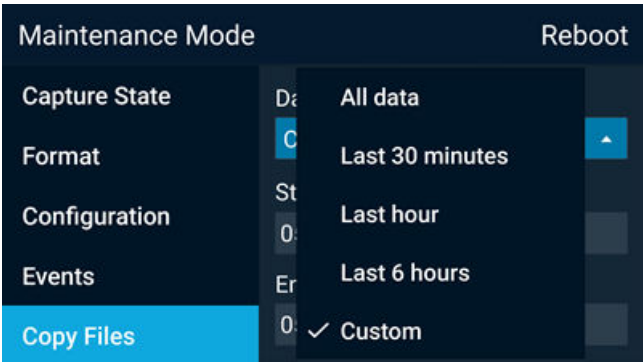


Secure File Copy

The **Maintenance Mode Secure File Copy** feature allows you to securely extract video files and other data from the M500 if the device is unable to upload events using normal methods.

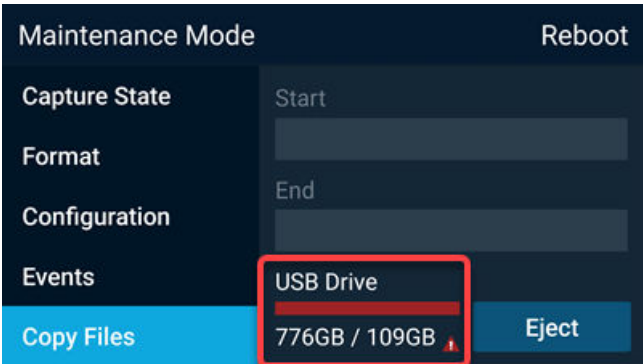
 **IMPORTANT:** The **Secure File Copy** feature provides immediate extraction of raw video (and other data) for investigative purposes. This video is not typically appropriate for evidentiary purposes.

You can choose to copy all the data stored on the M500 hard drive or a subset of the data, by time.



If you choose, for example, the **Last 30 minutes**, the system copies the last 30 minutes of evidence on the hard drive.

There must be enough space on the USB drive for all the files you want to copy. If there is not enough available space on the USB drive, the **Copy Files** screen shows you a red bar in the **USB Drive** section of the screen. It shows you the amount of data on the M500 hard drive for the time frame you selected versus the amount of space available on the USB drive.



In this case, you can either **Eject** the USB drive and replace it with a larger one or choose a smaller amount of data, by time, to copy to the USB drive.

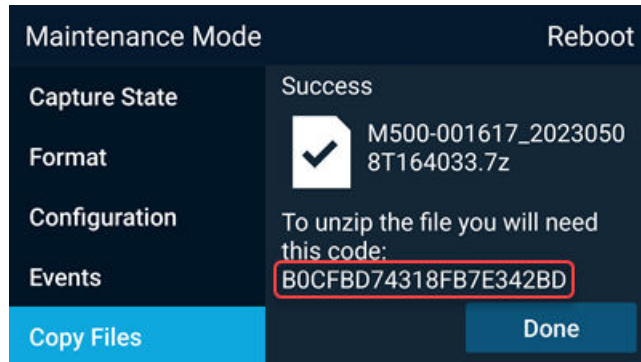
Secure Zip File

The **Secure File Copy** process produces a secure zip file that is loaded on the USB drive.

When the file copy process finishes, the **Success** screen shows you a code. You need to provide this code to Customer Service. They will give you the corresponding password you need to access the secure file.



NOTE: The code in the image below is an example. You receive your own code.



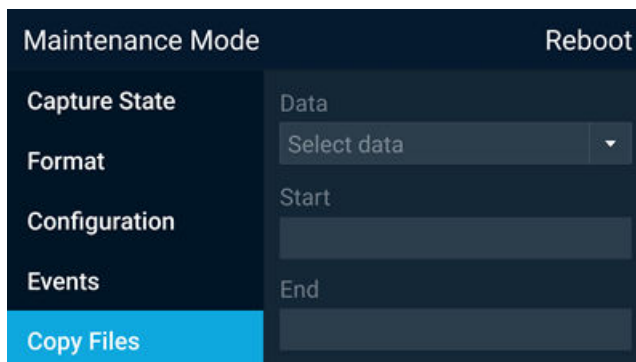
IMPORTANT: If you forget or misplace the code, it cannot be retrieved. In this case, you will need to re-run the **Secure File Copy** process to generate a new secure zip file and a new code to provide to Customer Service.

Copying Files

Procedure:

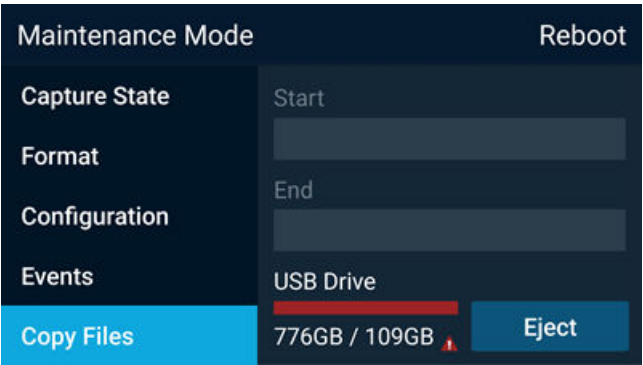
1. From the **Maintenance Mode** screen, touch **Copy Files**.

The **Maintenance Mode Copy Files** screen appears.



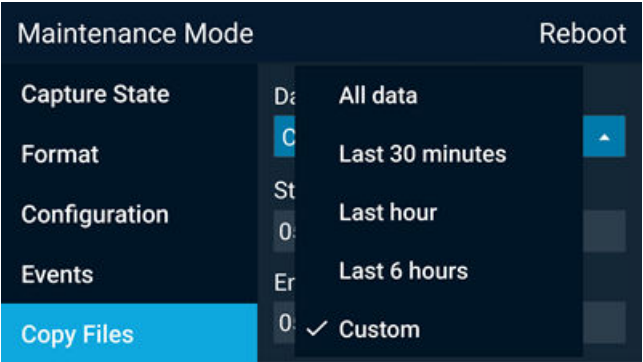
2. Insert the USB drive you want to use for copying the files.

The system defaults to copying **All data**.



 **NOTE:** If the USB drive does not have enough space available for all the data on the M500 hard drive, you must either replace the USB drive with a larger drive, or change the amount of data you want to copy.

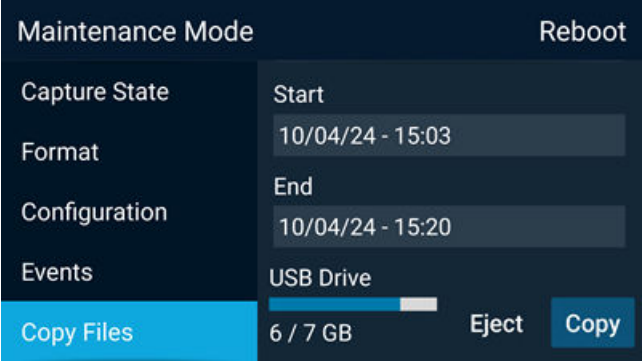
3. Touch the **Data** field and select the amount of data, in time, you want to copy to the USB drive.



If you select an option other than **Custom**, go to step 5.

 **NOTE:** If you select an amount of time/data that is greater than the available space on the USB drive, either replace the drive with a larger one or select a smaller amount of time/data.

4. If you selected **Custom** from the **Data** field:



- a. Touch the **Start** field to enter a start date and time for the custom data window.

The screenshot shows a 'Start' dialog box with a dark blue background. At the top left is a close button (X) and at the top right is a 'Done' button. Below the title bar, there are two sections: 'Date' and 'Time'. The 'Date' section has three input fields for day (dd), month (mm), and year (yy), containing the values 10, 04, and 24 respectively. The 'Time' section has two input fields for hour (h) and minute (m), containing the values 15 and 03 respectively. To the right of these fields is a numeric keypad with digits 1-9, 0, and a backspace icon.

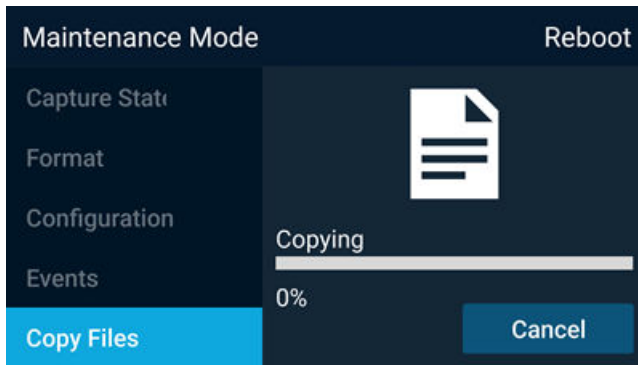
- b. Touch a field (**Date dd, mm, or yy** or **Time h or m**) then use the number pad to enter the **Start** date and time.
- c. Touch **Done**.
- d. Touch the **End** field to enter an end date and time for the custom data window.

The screenshot shows an 'End' dialog box, similar to the 'Start' dialog. It has a close button (X) and a 'Done' button. The 'Date' section has input fields for day (dd), month (mm), and year (yy) with values 10, 04, and 24. The 'Time' section has input fields for hour (h) and minute (m) with values 15 and 20. A numeric keypad is also present on the right.

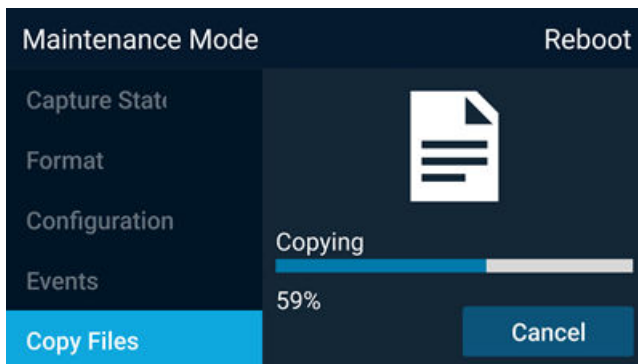
- e. Touch a field (**Date dd, mm, or yy** or **Time h or m**) then use the number pad to enter the **End** date and time.
 - f. Touch **Done**.
5. On the **Maintenance Mode Copy Files** screen, touch **Copy**.

The screenshot shows the 'Maintenance Mode' screen with a dark blue background. At the top right is a 'Reboot' button. On the left is a vertical menu with options: 'Capture State', 'Format', 'Configuration', 'Events', and 'Copy Files' (which is highlighted in blue). To the right of the menu, there are two text boxes: 'Start' containing '05/08/2023 - 07:28' and 'End' containing '05/08/2023 - 07:58'. Below these is a 'USB Drive' section with a progress bar and the text '0 / 115 GB'. At the bottom right are 'Eject' and 'Copy' buttons.

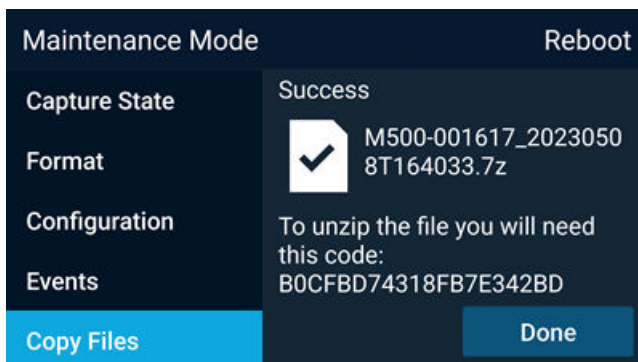
The **Copying** process begins.



The **Copying** process takes some time. A progress bar shows the percentage copied.

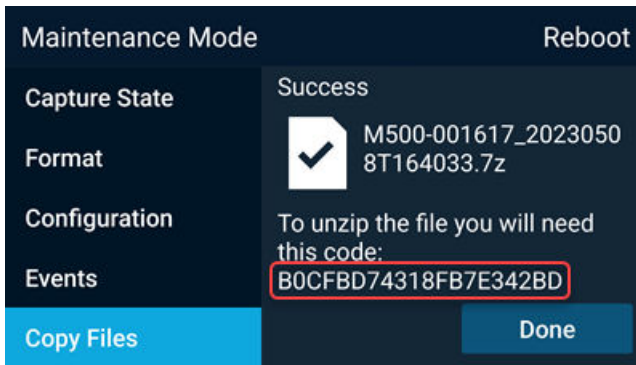


When the **Copying** process is finished, the **Success** message appears.



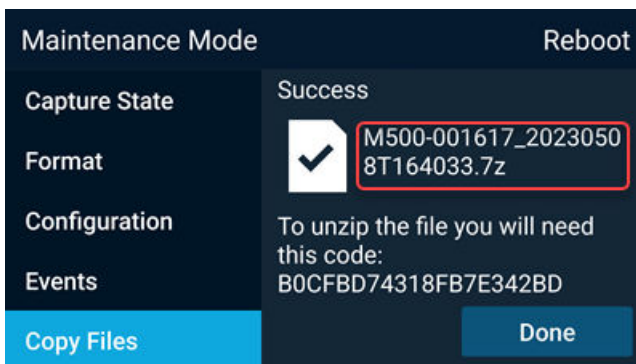
6. Make a note (or take a photo) of the code on the **Success** screen. You will need to provide it to Customer Service when you call in for the corresponding password to access the secure zip file.

 **NOTE:** The code in the image is an example. You receive your own code.



IMPORTANT: If you forget or misplace the code, it cannot be retrieved. In this case, you will need to re-run the **Secure File Copy** process to generate a new secure zip file and a new code to provide to Customer Service.

7. Make a note (or take a photo) of the secure file name on the **Success** screen so that you can identify it.



8. Touch **Done**.
The M500 ejects the USB drive automatically.
9. Remove the USB drive containing the secure zip file from the M500.
10. To access the secure zip file:
 - a. Call Customer Service and provide them the code from the **Success** screen.
Customer Service will provide you with a corresponding secure file password.
 - b. Insert the USB drive containing the secure zip file into a secure computer.
 - c. Enter the secure file password when prompted.

Network Settings

The **Maintenance Mode Network** screen shows information about the network settings and other utilities in the system.



IMPORTANT: Motorola Solutions recommends that you consult your IT administrator before you set up your network settings. The instructions in this Maintenance Mode section assumes that the user has administrator credentials and an understanding of IT concepts.

Maintenance Mode		Reboot
Format	Current Network Settings	
Configuration	IP address	192.168.254.171
Events	Netmask	255.255.255.0
Copy Files	Gateway	192.168.254.254
Network		

The **Maintenance Mode Network** screen shows the system **Current Network Settings**:

- **IP address**
- **Netmask**
- **Gateway**

Typically, agencies use the pre-configured DHCP settings provided for them. However, if you need to change the settings, you can disable the DHCP toggle. See [Disabling DHCP Network Settings on page 131](#) for more information.

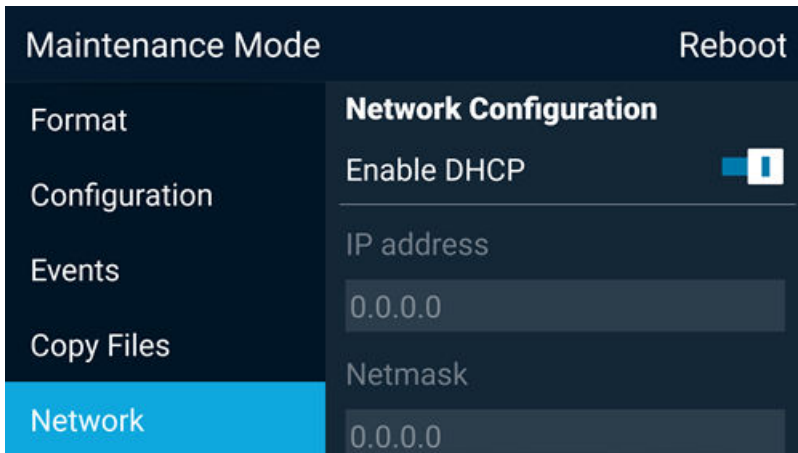
Disabling DHCP Network Settings

Procedure:

1. From the **Maintenance Mode Network** screen, select the **Config** button.

Maintenance Mode		Reboot
Format	IP address	192.168.254.171
Configuration	Netmask	255.255.255.0
Events	Gateway	192.168.254.254
Copy Files		
Network		
		Config

The **Maintenance Mode Network Configuration** screen appears.

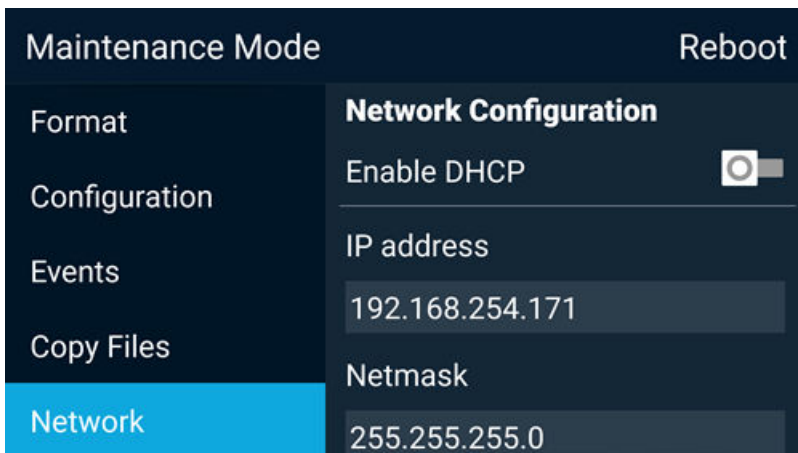


2. Touch the **Enable DHCP** toggle to disable the DHCP.

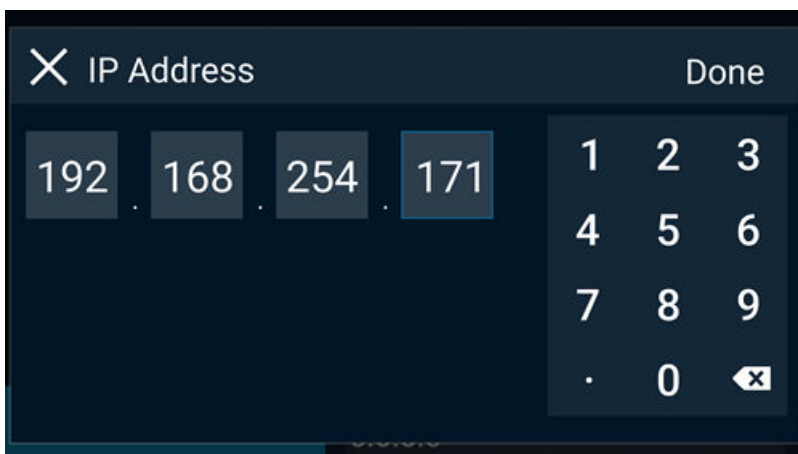


NOTE: Typically, agencies use the pre-configured DHCP settings provided for them. However, if you need to change the settings, you can disable the DHCP toggle

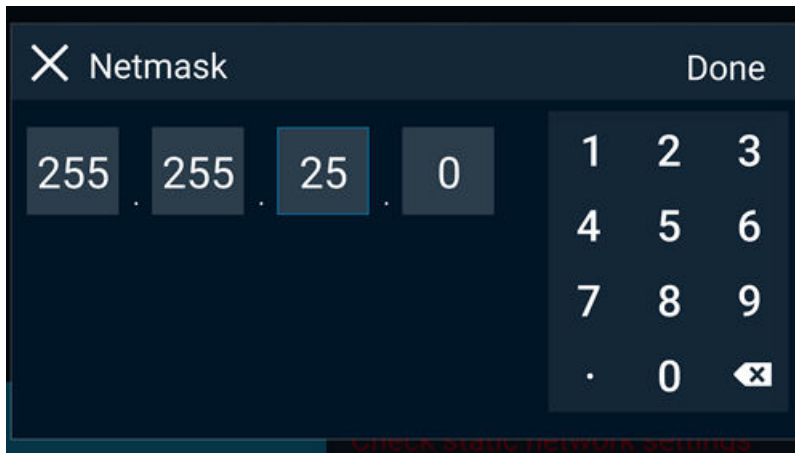
The **IP address**, **Netmask**, and **Gateway** fields are activated.



3. Touch the **IP Address** field to edit the **IP Address**.

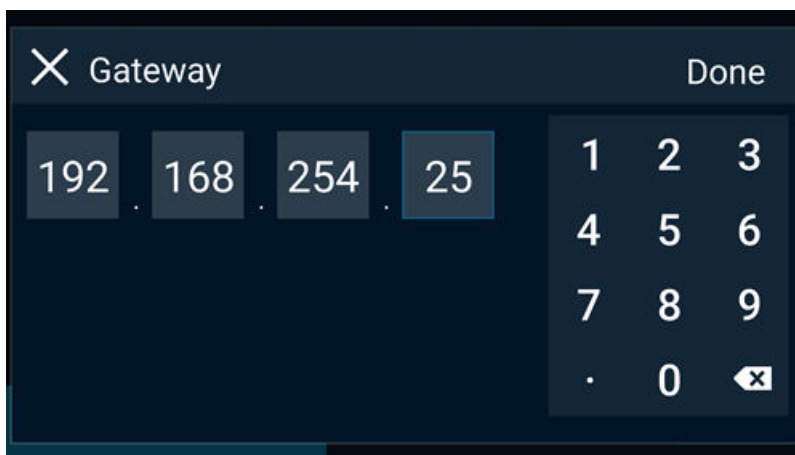


4. Touch the **Netmask** field to edit the **Netmask**.



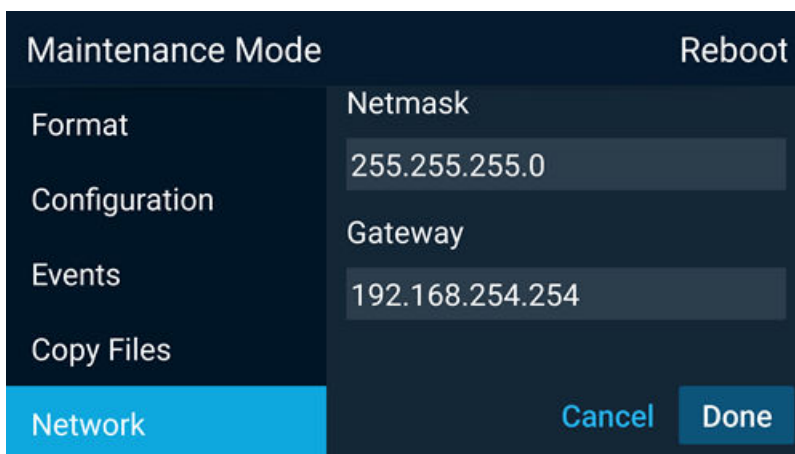
A screenshot of a 'Netmask' configuration screen. At the top left is a close button (X) and the title 'Netmask'. At the top right is a 'Done' button. Below the title, there are four input fields containing the values '255', '255', '25', and '0'. To the right of these fields is a numeric keypad with digits 1-9, 0, and a decimal point, along with a backspace button (X). At the bottom of the screen, there is a red text label that reads 'CHECK STATIC NETWORK SETTINGS'.

5. Touch the **Gateway** field to edit the **Gateway**.



A screenshot of a 'Gateway' configuration screen. At the top left is a close button (X) and the title 'Gateway'. At the top right is a 'Done' button. Below the title, there are four input fields containing the values '192', '168', '254', and '25'. To the right of these fields is a numeric keypad with digits 1-9, 0, and a decimal point, along with a backspace button (X).

6. When you finish editing the static network settings, touch **Done**.



A screenshot of the 'Maintenance Mode' screen. At the top left is the title 'Maintenance Mode' and at the top right is a 'Reboot' button. On the left side, there is a list of menu items: 'Format', 'Configuration', 'Events', 'Copy Files', and 'Network'. The 'Network' item is highlighted with a blue background. On the right side, there are two text input fields. The first field is labeled 'Netmask' and contains the value '255.255.255.0'. The second field is labeled 'Gateway' and contains the value '192.168.254.254'. At the bottom right, there are two buttons: 'Cancel' and 'Done'.

A success message appears confirming your network setting updates.

7. Touch **Reboot** on the **Current Network Settings** screen.
After the M500 reboots, the Live View screen appears.

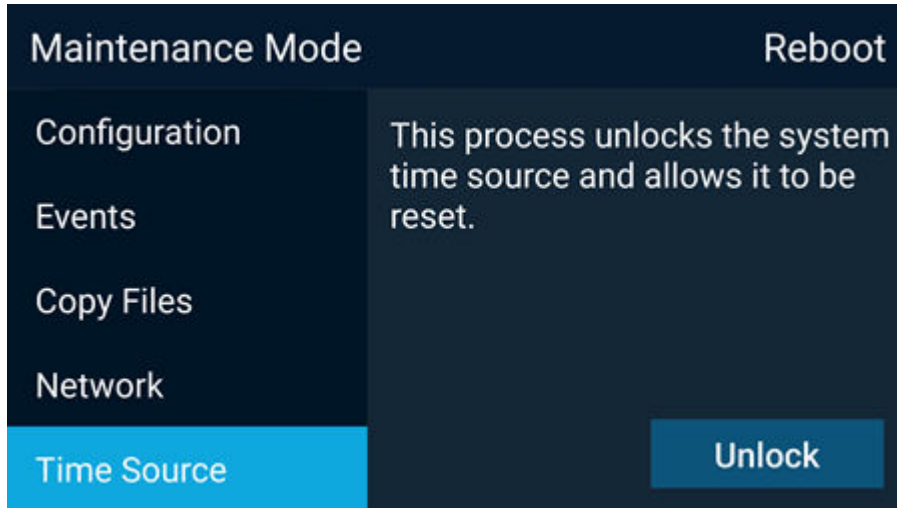
Resetting the Time Source

In **Maintenance Mode**, you can reset the system time source.

Procedure:

1. From the **Maintenance Mode** screen, touch **Time Source**.

The **Maintenance Mode Time Source** screen appears.



2. To unlock the time source, touch the **Unlock** button.



NOTE: If the **Unlock** button is disabled, the system does not currently have a time source it is locked to.